

A Summary of PDEs

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Abstract

This document provides a summary of partial differential equations (PDEs). The main reference is the lecture notes from École Polytechnique.

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1 Preliminaries

Unless otherwise stated, Ω denotes an open subset of $\mathbb{R}^n$ and ν denotes the unit outer normal vector at the boundary $\partial\Omega$.

Definition 1.1 (Regularity of boundary). We say that the boundary $\partial\Omega$ is sufficiently regular if for every point $x_0 \in \partial\Omega$, there exist a neighborhood U of x_0 , an open set V in $\mathbb{R}^{n-1}$, a Lipschitz continuous function $\varphi : V \rightarrow \mathbb{R}$ such that

$$\Omega \cap U = \{x = (x', x_n) \in U \mid x_n > \varphi(x')\}.$$

Moreover, if φ belongs to C^k for some $k \in \mathbb{N}$, we say that the boundary $\partial\Omega$ is of class C^k .

Remark. Since φ is Lipschitz continuous, by Rademacher's theorem, φ is differentiable almost everywhere.

1.1 Integral Theory

All omitted proofs are standard and can be found in any textbook.

Lemma 1.2 (du Bois-Reymond). *Let $f \in C(\Omega)$. If*

$$\int_{\Omega} f(x) \varphi(x) dx = 0, \quad \forall \varphi \in C_c^\infty(\Omega),$$

then $f \equiv 0$ on Ω .

Theorem 1.3 (Divergence Theorem). *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with sufficiently regular boundary. Then for all $F \in C^1(\bar{\Omega}; \mathbb{R}^n)$, we have*

$$\int_{\Omega} \operatorname{div} F(x) dx = \int_{\partial\Omega} F(x) \cdot \underline{\nu}(x) d\sigma,$$

where $\underline{\nu}(x)$ is the unit outer normal vector at $x \in \partial\Omega$.

Corollary 1.4.

1. *Let u be a C^1 function on $\bar{\Omega}$. Then*

$$\int_{\Omega} \frac{\partial u(x)}{\partial x_i} dx = \int_{\partial\Omega} u(x) \nu_i(x) d\sigma.$$

2. *Let u, v be C^1 functions on $\bar{\Omega}$. Then*

$$\int_{\Omega} \frac{\partial u(x)}{\partial x_i} v(x) dx + \int_{\Omega} u(x) \frac{\partial v(x)}{\partial x_i} dx = \int_{\partial\Omega} u(x) v(x) \nu_i(x) d\sigma.$$

3. *Let $u \in C^1(\bar{\Omega}; \mathbb{R}^n)$ and $v \in C^1(\bar{\Omega}; \mathbb{R})$. Then*

$$\int_{\Omega} v(x) \operatorname{div} u(x) dx + \int_{\Omega} \nabla v(x) \cdot u(x) dx = \int_{\partial\Omega} v(x) u(x) \cdot \underline{\nu}(x) d\sigma.$$

4. *Let $u \in C^2(\bar{\Omega}; \mathbb{R})$ and $v \in C^1(\bar{\Omega}; \mathbb{R})$. Then*

$$\int_{\Omega} v(x) \Delta u(x) dx + \int_{\Omega} \nabla v(x) \cdot \nabla u(x) dx = \int_{\partial\Omega} v(x) \frac{\partial u}{\partial \nu}(x) d\sigma.$$

We denote by $L^2(\Omega)$ the space of (equivalence classes of) square-integrable measurable functions on Ω , where two functions are identified if they are equal almost everywhere, i.e.,

$$L^2(\Omega) = \left\{ u : \Omega \rightarrow \mathbb{R} \text{ measurable} \mid \int_{\Omega} |u(x)|^2 dx < +\infty \right\} / \sim,$$

Proposition 1.5 (Density of $C_c^\infty(\Omega)$ in $L^2(\Omega)$). *The space $C_c^\infty(\Omega)$ is dense in $L^2(\Omega)$, i.e., for every $u \in L^2(\Omega)$, there exists a sequence $\{u_n\}$ in $C_c^\infty(\Omega)$ such that*

$$\|u_n - u\|_{L^2(\Omega)} \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Corollary 1.6. Let $f \in L^2(\Omega)$. If for every $\varphi \in C_c^\infty(\Omega)$, we have

$$\int_{\Omega} f(x)\varphi(x) \, dx = 0,$$

then $f = 0$ a.e. on Ω , i.e., $f = 0$ in $L^2(\Omega)$.

Definition 1.7 (Weak derivative). For $u \in L^2(\Omega)$ and $i \in \{1, \dots, n\}$, we call $f \in L^2(\Omega)$ the weak derivative of u with respect to x_i , denoted by $\frac{\partial u}{\partial x_i} = f$, if for every $\varphi \in C_c^\infty(\Omega)$,

$$\int_{\Omega} u(x) \frac{\partial \varphi(x)}{\partial x_i} \, dx = - \int_{\Omega} f(x) \varphi(x) \, dx.$$

Proposition 1.8. Let $u \in L^2(\Omega)$. If for $i \in \{1, 2, \dots, n\}$, there exists $C_i > 0$ such that

$$\left| \int_{\Omega} u(x) \frac{\partial \varphi(x)}{\partial x_i} \, dx \right| \leq C_i \|\varphi\|_{L^2(\Omega)}, \quad \forall \varphi \in C_c^\infty(\Omega),$$

then the weak derivative $\frac{\partial u}{\partial x_i}$ exists in $L^2(\Omega)$.

Proposition 1.9. Let Ω be a domain, i.e., a connected open set in $\mathbb{R}^n$. Let $u \in L^2(\Omega)$. If for every $i \in \{1, 2, \dots, n\}$, the weak derivative $\frac{\partial u}{\partial x_i}$ exists in $L^2(\Omega)$ and vanishes almost everywhere on Ω , then u is constant almost everywhere on Ω .

Definition 1.10 (Weak divergence). Let $\underline{u} \in L^2(\Omega; \mathbb{R}^n)$. We say that $f \in L^2(\Omega)$ is the weak divergence of $\underline{u}$, denoted by $\operatorname{div} \underline{u} = f$, if for every $\varphi \in C_c^\infty(\Omega)$, we have

$$\int_{\Omega} \underline{u}(x) \cdot \nabla \varphi(x) \, dx = - \int_{\Omega} f(x) \varphi(x) \, dx.$$

1.2 Hilbert Spaces

All omitted proofs are standard and can be found in any textbook of functional analysis.

Theorem 1.11 (Riesz Representation). Let H be a Hilbert space. Then for every bounded linear functional $f \in H'$, there exists a unique $u \in H$ such that

$$f(v) = (u, v)_H, \quad \forall v \in H.$$

Moreover, $\|f\|_{H'} = \|u\|_H$.

Corollary 1.12. Let H be a Hilbert space, and let $a : H \times H \rightarrow \mathbb{R}$ be a bilinear form which is continuous, i.e., there exists a constant $C > 0$ such that

$$|a(u, v)| \leq C \|u\|_H \|v\|_H, \quad \forall u, v \in H.$$

Moreover, there exists a unique element $A \in \mathcal{L}(H)$ such that

$$a(u, v) = (Au, v)_H, \quad \forall v \in H.$$

Definition 1.13 (Weak convergence). Let H be a Hilbert space. A sequence $\{u_n\}$ in H is said to converge weakly to $u \in H$, denoted by $u_n \rightharpoonup u$, if

$$(u_n, v)_H \rightarrow (u, v)_H, \quad \forall v \in H.$$

Remark. If $u_n \rightarrow u$ strongly in H , then $u_n \rightharpoonup u$ weakly in H . The converse is not true in general.

Proposition 1.14. Let H be a Hilbert space. If $u_n \rightharpoonup u$ weakly in H , then the sequence $\{u_n\}$ is bounded in H , i.e., there exists a constant $C > 0$ such that

$$\|u_n\|_H \leq C, \quad \forall n \in \mathbb{N}.$$

Proposition 1.15. Let H be a Hilbert space. If $u_n \rightharpoonup u$ weakly in H and $v_n \rightarrow v$ strongly in H , then

$$\lim_{n \rightarrow \infty} (u_n, v_n)_H = (u, v)_H.$$

Proposition 1.16. *Let H be a Hilbert space. If $u_n \rightharpoonup u$ weakly in H , then*

$$\|u\|_H \leq \liminf_{n \rightarrow \infty} \|u_n\|_H.$$

Moreover, if the equality holds, then $u_n \rightarrow u$ strongly in H .

Proposition 1.17. *Let H be a Hilbert space and f be a convex continuous function on H . If $u_n \rightharpoonup u$ weakly in H , then f is weakly lower semicontinuous, i.e.,*

$$f(u) \leq \liminf_{n \rightarrow \infty} f(u_n).$$

The proof of this proposition is given in the solution of Problem 17.

Theorem 1.18. *Let H be a Hilbert space. Then every bounded sequence $\{u_n\}$ in H has a subsequence which converges weakly to some $u \in H$.*

Theorem 1.19 (Lax-Milgram). *Let H be a Hilbert space, and let $a : H \times H \rightarrow \mathbb{R}$ be a bilinear form which is continuous and coercive, i.e., there exist constants $C > 0$ and $\alpha > 0$ such that*

$$|a(u, v)| \leq C \|u\|_H \|v\|_H, \quad \forall u, v \in H,$$

and

$$a(u, u) \geq \alpha \|u\|_H^2, \quad \forall u \in H.$$

Then for every bounded linear functional $f \in H'$, there exists a unique $u \in H$ such that

$$a(u, v) = f(v), \quad \forall v \in H.$$

Moreover, we have

$$\|u\|_H \leq \frac{1}{\alpha} \|f\|_{H'}.$$

Proof. By the Riesz representation theorem, there exists a unique $w \in H$ such that

$$f(v) = (w, v)_H, \quad \forall v \in H.$$

We define the operator $T : H \rightarrow H$ by

$$(Tu, v)_H = a(u, v), \quad \forall u, v \in H.$$

By the continuity of a , we have

$$|(Tu, v)_H| = |a(u, v)| \leq C \|u\|_H \|v\|_H.$$

Taking $v = Tu$, we get

$$\|Tu\|_H^2 = (Tu, Tu)_H = a(u, Tu) \leq C \|u\|_H \|Tu\|_H,$$

which implies that T is a bounded linear operator on H . Moreover, by the coercivity of a , we have

$$(Tu, u)_H = a(u, u) \geq \alpha \|u\|_H^2, \quad \forall u \in H,$$

which implies that T is injective. Since H is a Hilbert space and $\|Tu\|_H \geq \alpha \|u\|_H$, the range of T is closed. Hence, $H = \text{Im}(T) \oplus \text{Im}(T)^\perp$. Note that

$$\text{Im}(T)^\perp = \{v \in H \mid (Tu, v)_H = 0, \forall u \in H\} = \{0\}.$$

We conclude that $\text{Im}(T) = H$, i.e., T is surjective. Therefore, there exists a unique $u \in H$ such that

$$Tu = w,$$

which is equivalent to

$$a(u, v) = (Tu, v)_H = (w, v)_H = f(v), \quad \forall v \in H.$$

Moreover, we have

$$\alpha \|u\|_H^2 \leq a(u, u) = f(u) = (w, u)_H \leq \|w\|_H \|u\|_H = \|f\|_{H'} \|u\|_H,$$

which implies that

$$\|u\|_H \leq \frac{1}{\alpha} \|f\|_{H'}.$$

□

Corollary 1.20. *We use the same notations as in the theorem above. Let a be symmetric, i.e., $a(v, w) = a(w, v)$ for all $v, w \in H$. We define the energy functional $J : H \rightarrow \mathbb{R}$ by*

$$J(v) = \frac{1}{2}a(v, v) - f(v), \quad \forall v \in H.$$

Then J is strictly convex on H . Moreover, J has a unique minimizer $u \in H$ if and only if $a(u, v) = f(v)$ for all $v \in H$.

Proof. For every $v, w \in H$ with $v \neq w$ and $t \in (0, 1)$, we have

$$\begin{aligned} J(tv + (1-t)w) &= \frac{1}{2}a(tv + (1-t)w, tv + (1-t)w) - f(tv + (1-t)w) \\ &= \frac{t^2}{2}a(v, v) + \frac{(1-t)^2}{2}a(w, w) + t(1-t)a(v, w) - tf(v) - (1-t)f(w) \\ &< tJ(v) + (1-t)J(w). \end{aligned}$$

If $a(u, v) = f(v)$ for all $v \in H$, by the Lax-Milgram theorem, the solution u is unique. And we have

$$J(u + v) = J(u) + a(u, v) - f(v) + \frac{1}{2}a(v, v) = J(u) + \frac{1}{2}a(v, v) \geq J(u).$$

Hence u is the minimizer of J . Conversely, if J has a unique minimizer $u \in H$, we fix $v \in H$, then we have

$$J(u + tv) = J(u) + t(a(u, v) - f(v)) + \frac{t^2}{2}a(v, v) \geq J(u),$$

i.e., $t(a(u, v) - f(v)) + \frac{t^2}{2}a(v, v) \geq 0$. For $t > 0$, dividing by t and letting $t \rightarrow 0$, we get $a(u, v) - f(v) \geq 0$. For $t < 0$, dividing by t and letting $t \rightarrow 0$, we get $a(u, v) - f(v) \leq 0$. Hence $a(u, v) = f(v)$. This works for all $v \in H$. $\square$

Theorem 1.21. *Let H be a Hilbert space and let f be a strictly convex, continuous function on H such that*

$$f(v) \rightarrow +\infty, \text{ as } \|v\|_H \rightarrow +\infty.$$

Then there exists a unique $u \in H$ such that

$$f(u) = \min_{v \in H} f(v).$$

Proof. Suppose $m = \inf_{v \in H} f(v)$. Then there exists a sequence $\{u_n\}$ such that $f(u_n) \rightarrow m$. Since $f(v) \rightarrow +\infty$ as $\|v\|_H \rightarrow +\infty$, we may suppose that $u_n \in B(0, R)$, i.e., the sequence $\{u_n\}$ is bounded. By Theorem 1.18, there exists a subsequence $\{u_{n_k}\}$ such that

$$u_{n_k} \rightharpoonup u, \text{ as } k \rightarrow \infty,$$

for some $u \in H$. By Proposition 1.17, we know that f is weakly lower semicontinuous, i.e.,

$$f(u) \leq \liminf_{k \rightarrow \infty} f(u_{n_k}) = m.$$

Hence, $f(u) = m$, which implies that u is a minimizer of f . Moreover, if there exists another minimizer $v \neq u$, then by the convexity of f , we have

$$f\left(\frac{u+v}{2}\right) < \frac{f(u) + f(v)}{2} = m,$$

which is a contradiction. Hence, u is the unique minimizer of f . $\square$

Proposition 1.22 (Internal approximation). *Let H be a Hilbert space and H_N be a finite-dimensional subspace of H . Let $a : H \times H \rightarrow \mathbb{R}$ be a bilinear form which is continuous and coercive, and let $f \in H'$. Then there exists a unique $u_N \in H_N$ such that*

$$a(u_N, v_N) = f(v_N), \quad \forall v_N \in H_N.$$

Proof. Note that the subspace H_N is also a Hilbert space with the inner product inherited from H . Hence, it is a direct application of the Lax-Milgram theorem. $\square$

Lemma 1.23 (Céa). *We use the same notations as in the proposition above. Let $u \in H$ be the unique solution of*

$$a(u, v) = f(v), \quad \forall v \in H.$$

Then since $u_N - v_N \in H_N$, we have

$$\|u - u_N\|_H \leq \frac{C}{\alpha} \inf_{v_N \in H_N} \|u - v_N\|_H,$$

where C and α are the constants in the continuity and coercivity conditions.

Proof. For every $w_N \in H_N$, we have

$$a(u - u_N, w_N) = a(u, w_N) - a(u_N, w_N) = f(w_N) - f(w_N) = 0.$$

Taking $w_N = u_N - v_N$, we have $a(u - u_N, u_N - v_N) = 0$. Thus,

$$\alpha \|u - u_N\|_H^2 \leq a(u - u_N, u - u_N) = a(u - u_N, u - v_N) \leq C \|u - u_N\|_H \|u - v_N\|_H.$$

Dividing both sides by $\alpha \|u - u_N\|_H$, we get

$$\|u - u_N\|_H \leq \frac{C}{\alpha} \|u - v_N\|_H.$$

Since v_N is arbitrary in H_N , we have

$$\|u - u_N\|_H \leq \frac{C}{\alpha} \inf_{v_N \in H_N} \|u - v_N\|_H.$$

□

Remark. If a is symmetric, i.e., $a(u, v) = a(v, u)$ for all $u, v \in H$, then we have

$$\|u - u_N\|_H \leq \sqrt{\frac{C}{\alpha}} \inf_{v_N \in H_N} \|u - v_N\|_H,$$

Proof. Note that

$$\begin{aligned} \alpha \|u - u_N\|_H^2 &\leq a(u - u_N, u - u_N) = a(u - u_N, u - v_N) \\ &= a(u - v_N, u - v_N) + a(v_N - u_N, u - v_N) \\ &= a(u - v_N, u - v_N) - a(v_N - u_N, v_N - u_N) \\ &\leq a(u - v_N, u - v_N) \leq C \|u - v_N\|_H^2. \end{aligned}$$

□

Proposition 1.24 (Convergence of internal approximation). *We use the same notations as in the proposition and lemma above. If there exists a dense subspace W of H and a sequence of maps $(P_N : W \rightarrow H_N)_N$ satisfying*

$$\|w - P_N w\|_H \rightarrow 0, \quad \text{as } N \rightarrow \infty, \quad \forall w \in W,$$

we have

$$\|u - u_N\|_H \rightarrow 0, \quad \text{as } N \rightarrow \infty.$$

Proof. Let $\varepsilon > 0$. Since W is dense in H , there exists an element $w \in W$ such that $\|u - w\|_H < \varepsilon$. By the assumption, there exists $N_0 = N_0(\varepsilon) \in \mathbb{N}$ such that for every $N \geq N_0$, we have $\|w - P_N w\|_H < \varepsilon$. Hence, for every $N \geq N_0$, by Lemma 1.23, we have

$$\|u - u_N\|_H \leq \frac{C}{\alpha} \inf_{v_N \in H_N} \|u - v_N\|_H \leq \frac{C}{\alpha} \|u - P_N w\|_H \leq \frac{C}{\alpha} (\|u - w\|_H + \|w - P_N w\|_H) < \frac{2C}{\alpha} \varepsilon.$$

□

1.3 Sobolev Spaces

All omitted proofs are standard and can be found in [3]. In this section, unless otherwise stated, Ω denotes a bounded open set in $\mathbb{R}^n$.

Definition 1.25 (Sobolev space $H^1(\Omega)$). We define the Sobolev space $H^1(\Omega)$ as

$$H^1(\Omega) = \left\{ u \in L^2(\Omega) \mid \frac{\partial u}{\partial x_i} \in L^2(\Omega), \ i = 1, 2, \dots, n \right\}.$$

The space $H^1(\Omega)$ is a Hilbert space with the inner product

$$(u, v)_{H^1(\Omega)} = \int_{\Omega} \left(u(x)v(x) + \nabla u(x) \cdot \nabla v(x) \right) dx, \quad (1)$$

and the associated norm

$$\|u\|_{H^1(\Omega)} = \left(\|u\|_{L^2(\Omega)}^2 + \sum_{i=1}^n \left\| \frac{\partial u}{\partial x_i} \right\|_{L^2(\Omega)}^2 \right)^{1/2}.$$

Theorem 1.26 (Density of $C^\infty(\overline{\Omega})$ in $H^1(\Omega)$). *Let Ω be a bounded open set in $\mathbb{R}^n$ with sufficiently regular boundary. The space $C^\infty(\overline{\Omega})$ is dense in $H^1(\Omega)$, i.e., for every $u \in H^1(\Omega)$, there exists a sequence $\{u_n\}$ in $C^\infty(\overline{\Omega})$ such that*

$$\|u_n - u\|_{H^1(\Omega)} \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Definition 1.27 (Sobolev space $H_0^1(\Omega)$). We define the Sobolev space $H_0^1(\Omega)$ as the closure of $C_c^\infty(\Omega)$ in $H^1(\Omega)$, i.e.,

$$H_0^1(\Omega) = \overline{C_c^\infty(\Omega)}^{H^1(\Omega)}.$$

Equipped with the inner product (1), the space $H_0^1(\Omega)$ is also a Hilbert space.

Definition 1.28 (Sobolev space $H(\Omega, \text{div})$). We define the Sobolev space $H(\Omega, \text{div})$ as

$$H(\Omega, \text{div}) = \left\{ u \in L^2(\Omega; \mathbb{R}^n) \mid \text{div } \underline{u} \in L^2(\Omega) \right\}.$$

Equipped with the inner product

$$(\underline{u}, \underline{v})_{H(\Omega, \text{div})} := (\underline{u}, \underline{v})_{L^2(\Omega; \mathbb{R}^n)} + (\text{div } \underline{u}, \text{div } \underline{v})_{L^2(\Omega)},$$

the space $H(\Omega, \text{div})$ is also a Hilbert space.

Theorem 1.29 (Density of $C^\infty(\overline{\Omega}; \mathbb{R}^n)$ in $H(\Omega, \text{div})$). *Let Ω be a bounded open set in $\mathbb{R}^n$ with sufficiently regular boundary. The space $C^\infty(\overline{\Omega}; \mathbb{R}^n)$ is dense in $H(\Omega, \text{div})$.*

Definition 1.30 (Sobolev space $H^m(\Omega)$). We define the Sobolev space $H^m(\Omega)$ for $m \in \mathbb{N}$ as

$$H^m(\Omega) = \left\{ u \in L^2(\Omega) \mid D^\alpha u \in L^2(\Omega), \ \forall \alpha \in \mathbb{N}^n, |\alpha| \leq m \right\}.$$

Equipped with the inner product

$$(u, v)_{H^m(\Omega)} = \sum_{|\alpha| \leq m} (D^\alpha u, D^\alpha v)_{L^2(\Omega)}, \quad (2)$$

the space $H^m(\Omega)$ is also a Hilbert space.

Theorem 1.31 (Density of $C^\infty(\overline{\Omega})$ in $H^m(\Omega)$). *Let Ω be a bounded open set in $\mathbb{R}^n$ with sufficiently regular boundary. The space $C^\infty(\overline{\Omega})$ is dense in $H^m(\Omega)$.*

Definition 1.32 (Sobolev space $H_0^m(\Omega)$). We define the Sobolev space $H_0^m(\Omega)$ as the closure of $C_c^\infty(\Omega)$ in $H^m(\Omega)$, i.e.,

$$H_0^m(\Omega) = \overline{C_c^\infty(\Omega)}^{H^m(\Omega)}.$$

Equipped with the inner product (2), the space $H_0^m(\Omega)$ is also a Hilbert space.

Theorem 1.33 (Trace Theorem). *Let Ω be a bounded domain with sufficiently regular boundary. The continuous operator*

$$\gamma_0 : C^\infty(\overline{\Omega}) \rightarrow L^2(\partial\Omega), \quad \gamma_0(u) = u|_{\partial\Omega}$$

is well-defined and linear. Moreover, there exists a constant $C > 0$ such that

$$\|\gamma_0(u)\|_{L^2(\partial\Omega)} \leq C\|u\|_{H^1(\Omega)}, \quad \forall u \in C^\infty(\overline{\Omega}).$$

Since $C^\infty(\overline{\Omega})$ is dense in $H^1(\Omega)$, there exists a unique extension of γ_0 to $H^1(\Omega)$, denoted by γ , such that

$$\gamma : H^1(\Omega) \rightarrow L^2(\partial\Omega), \quad \gamma(u) = u|_{\partial\Omega}.$$

Moreover, we have

$$\|\gamma(u)\|_{L^2(\partial\Omega)} \leq C\|u\|_{H^1(\Omega)}, \quad \forall u \in H^1(\Omega).$$

We call γ the trace operator. We will often write $\gamma(u)$ simply as $u|_{\partial\Omega}$, referred to as the trace of u on $\partial\Omega$.

Proposition 1.34. *Let Ω be a bounded open set in $\mathbb{R}^n$ with sufficiently regular boundary. And $\overline{\Omega} = \overline{\Omega}_1 \cup \overline{\Omega}_2$ with Ω_1, Ω_2 being two disjoint open sets. We define $\Gamma = \overline{\Omega}_1 \cap \overline{\Omega}_2$. Let $v_i = v|_{\Omega_i} \in H^1(\Omega_i)$ for $i = 1, 2$, then*

$$v_1|_\Gamma = v_2|_\Gamma, \text{ i.e., } \gamma_\Gamma(v_1) = \gamma_\Gamma(v_2) \iff v \in H^1(\Omega).$$

Moreover, we have

$$\nabla v = \nabla v_1 \mathbf{1}_{\Omega_1} + \nabla v_2 \mathbf{1}_{\Omega_2}.$$

Proposition 1.35. *The trace operator is not injective. The kernel of the trace operator γ is $H_0^1(\Omega)$, i.e.,*

$$\ker(\gamma) = H_0^1(\Omega).$$

Hence, we have

$$H_0^1(\Omega) = \{u \in H^1(\Omega) \mid u|_{\partial\Omega} = 0\}.$$

Proposition 1.36. *The trace operator is not surjective. The image of the trace operator γ is $H^{1/2}(\partial\Omega)$, i.e.,*

$$\text{Im}(\gamma) = H^{1/2}(\partial\Omega) := \left\{ \varphi \in L^2(\partial\Omega) : \int_{\partial\Omega \times \partial\Omega} \frac{|\varphi(x) - \varphi(y)|^2}{|x - y|^n} d\sigma_x d\sigma_y < \infty \right\}.$$

Equipped with the norm

$$\|\varphi\|_{\text{Im}(\gamma)} = \inf \{ \|u\|_{H^1(\Omega)} \mid u \in H^1(\Omega), u|_{\partial\Omega} = \varphi \},$$

the space $\text{Im}(\gamma)$ is a Banach space which is dense in $L^2(\partial\Omega)$. Moreover, for every $\varphi \in \text{Im}(\gamma)$, there exists $u \in H^1(\Omega)$ such that

$$u|_{\partial\Omega} = \varphi, \quad \|u\|_{H^1(\Omega)} \leq C\|\varphi\|_{\text{Im}(\gamma)}.$$

We call such a function u an extension of φ (The extension is not unique).

Theorem 1.37 (Trace Theorem for $H^2(\Omega)$). *Let Ω be a bounded domain with sufficiently regular boundary. We have the continuous operator*

$$\gamma_1 : H^2(\Omega) \rightarrow L^2(\partial\Omega), \quad \gamma_1(u) = \gamma(\nabla u \cdot \underline{\nu}).$$

Moreover, we have

$$\|\gamma_1(u)\|_{L^2(\partial\Omega)} \leq C\|u\|_{H^2(\Omega)}, \quad \forall u \in H^2(\Omega).$$

Theorem 1.38 (Trace Theorem for $H(\Omega, \text{div})$). *Let Ω be a bounded domain with sufficiently regular boundary. We have the continuous operator*

$$\gamma_\nu : (H^1(\Omega))^n \rightarrow L^2(\partial\Omega), \quad \gamma_\nu(\underline{u}) = \gamma(\underline{u} \cdot \underline{\nu}).$$

This operator can be extended to a continuous linear operator on $H(\Omega, \text{div})$, i.e.,

$$\gamma_\nu : H(\Omega, \text{div}) \rightarrow (\text{Im}(\gamma))', \quad \gamma_\nu(\underline{u}) = \gamma(\underline{u} \cdot \underline{\nu}).$$

where for all $\varphi \in \text{Im}(\gamma)$, we have

$$\langle \gamma_\nu(\underline{u}), \varphi \rangle = \int_\Omega \underline{u}(x) \cdot \nabla \tilde{\varphi}(x) dx + \int_\Omega \text{div } \underline{u}(x) \tilde{\varphi}(x) dx,$$

where $\tilde{\varphi} \in H^1(\Omega)$ is any extension of φ , i.e., $\tilde{\varphi}|_{\partial\Omega} = \varphi$.

Theorem 1.39 (Green formula). *Let Ω be a bounded domain with sufficiently regular boundary. Let $u \in H^1(\Omega)$, then for every $i \in \{1, 2, \dots, n\}$, we have*

$$\int_{\Omega} \frac{\partial u(x)}{\partial x_i} dx = \int_{\partial\Omega} u|_{\partial\Omega} \nu_i d\sigma.$$

Corollary 1.40. *Let Ω be a bounded domain with sufficiently regular boundary and let $u \in H^2(\Omega)$, $v \in H^1(\Omega)$. Then we have*

$$\int_{\Omega} v(x) \Delta u(x) dx + \int_{\Omega} \nabla v(x) \cdot \nabla u(x) dx = \int_{\partial\Omega} v|_{\partial\Omega} \frac{\partial u}{\partial \nu} d\sigma.$$

Remark. By applying Theorem 1.38, we can extend Corollary 1.40 to the case where $u \in H^1(\Omega)$ with $\Delta u \in L^2(\Omega)$ and $v \in H^1(\Omega)$:

$$\int_{\Omega} v(x) \Delta u(x) dx + \int_{\Omega} \nabla v(x) \cdot \nabla u(x) dx = \langle \gamma_{\nu}(\nabla u), v|_{\partial\Omega} \rangle \quad (3)$$

Theorem 1.41 (Rellich-Kondrachov Compactness Theorem). *Let Ω be a bounded open set in $\mathbb{R}^n$ with sufficiently regular boundary. Then the embedding*

$$H^1(\Omega) \hookrightarrow L^2(\Omega)$$

is compact, i.e., every bounded sequence in $H^1(\Omega)$ has a subsequence which converges strongly in $L^2(\Omega)$.

Corollary 1.42. *Let $\{u_k\}$ be a bounded sequence in $H^1(\Omega)$. Then there exists a subsequence $\{u_{k_j}\}$ and a function $u \in H^1(\Omega)$ such that*

$$\begin{aligned} u_{k_j} &\rightarrow u \text{ strongly in } L^2(\Omega) \text{ as } j \rightarrow \infty, \\ u_{k_j} &\rightharpoonup u \text{ weakly in } H^1(\Omega) \text{ as } j \rightarrow \infty. \end{aligned}$$

Proof. Since $\{u_k\}$ is a bounded sequence in $H^1(\Omega)$, by Theorem 1.18, there exists a subsequence $\{u_{k_j}\}$ and a function $u \in H^1(\Omega)$ such that

$$u_{k_j} \rightharpoonup u \text{ weakly in } H^1(\Omega) \text{ as } j \rightarrow \infty.$$

By Theorem 1.41, there exists a subsequence of $\{u_{k_j}\}$, still denoted by $\{u_{k_j}\}$, such that

$$u_{k_j} \rightarrow v \text{ strongly in } L^2(\Omega) \text{ as } j \rightarrow \infty.$$

Now we show that $u = v$. For every $\phi \in L^2(\Omega)$, we can define the continuous linear functional $\ell_{\phi} \in (H^1(\Omega))'$ by

$$\ell_{\phi}(w) = \int_{\Omega} w(x) \phi(x) dx, \quad \forall w \in H^1(\Omega).$$

Since $H^1(\Omega)$ is a Hilbert space, by Riesz representation theorem, there exists a unique element $z \in H^1(\Omega)$ such that for every $w \in H^1(\Omega)$, we have

$$\ell_{\phi}(w) = (z, w)_{H^1(\Omega)}.$$

Since $u_{k_j} \rightharpoonup u$ weakly in $H^1(\Omega)$, we have

$$\ell_{\phi}(u_{k_j}) = (z, u_{k_j})_{H^1(\Omega)} \rightarrow (z, u)_{H^1(\Omega)} = \ell_{\phi}(u).$$

On the other hand, since $u_{k_j} \rightarrow v$ strongly in $L^2(\Omega)$, we have

$$\ell_{\phi}(u_{k_j}) = \int_{\Omega} \phi(x) u_{k_j}(x) dx \rightarrow \int_{\Omega} \phi(x) v(x) dx = \ell_{\phi}(v).$$

Therefore, we have $\ell_{\phi}(u) = \ell_{\phi}(v)$ for every $\phi \in L^2(\Omega)$. This implies that $v = u \in L^2(\Omega)$. $\square$

Theorem 1.43 (Poincaré Inequality). *Let Ω be a bounded open set in $\mathbb{R}^n$ with sufficiently regular boundary. Then there exists a constant $C > 0$ such that*

$$\|u\|_{L^2(\Omega)} \leq C \|\nabla u\|_{(L^2(\Omega))^n}, \quad \forall u \in H_0^1(\Omega).$$

Proof. If the inequality does not hold, then there exists a sequence $\{u_n\}$ in $H_0^1(\Omega)$ such that

$$\|u_n\|_{L^2(\Omega)} > n \|\nabla u_n\|_{(L^2(\Omega))^n}, \quad \forall n \in \mathbb{N}.$$

We define $v_n = \frac{u_n}{\|u_n\|_{L^2(\Omega)}}$. Then we have $\|v_n\|_{L^2(\Omega)} = 1$ and $\|\nabla v_n\|_{(L^2(\Omega))^n} < \frac{1}{n}$ for every $n \in \mathbb{N}$. By Corollary 1.42, there exists a subsequence $\{v_{n_j}\}$ and a function $v \in H_0^1(\Omega)$ such that

$$v_{n_j} \rightharpoonup v \text{ weakly in } H_0^1(\Omega) \text{ and } v_{n_j} \rightarrow v \text{ strongly in } L^2(\Omega) \text{ as } j \rightarrow \infty.$$

By Proposition 1.16, we have

$$\|\nabla v\|_{(L^2(\Omega))^n} \leq \liminf_{j \rightarrow \infty} \|\nabla v_{n_j}\|_{(L^2(\Omega))^n} = 0.$$

Hence, we have $\nabla v = 0$ almost everywhere on Ω . Since $v \in H_0^1(\Omega)$, we have $v|_{\partial\Omega} = 0$. Hence $v = 0$ in $L^2(\Omega)$. On the other hand, since $\|v_{n_j}\|_{L^2(\Omega)} = 1$ for every $j \in \mathbb{N}$, we have $\|v\|_{L^2(\Omega)} = 1$. This is a contradiction. $\square$

Theorem 1.44 (Poincaré-Wirtinger Inequality). *Let Ω be a bounded domain in $\mathbb{R}^n$ with sufficiently regular boundary. Then there exists a constant $C > 0$ such that*

$$\|u - \bar{u}\|_{L^2(\Omega)} \leq C \|\nabla u\|_{(L^2(\Omega))^n}, \quad \forall u \in H^1(\Omega),$$

where

$$\bar{u} = \frac{1}{|\Omega|} \int_{\Omega} u(x) \, dx.$$

Proof. We can use the same method as in the proof of the Poincaré inequality. $\square$

We define the seminorm

$$|u|_{H^1(\Omega)} := \|\nabla u\|_{(L^2(\Omega))^n}, \quad u \in H^1(\Omega).$$

Corollary 1.45. *The seminorm $|\cdot|_{H^1(\Omega)}$ is a norm on $H_0^1(\Omega)$ which is equivalent to the norm $\|\cdot\|_{H^1(\Omega)}$.*

Proof. It is the consequence of the Poincaré inequality. $\square$

Remark. Equipped with the seminorm $|\cdot|_{H^1(\Omega)}$, the space $H_0^1(\Omega)$ is also a Hilbert space.

Proposition 1.46. *There exists a constant $C > 0$ such that*

$$\|u\|_{L^2(\Omega)} \leq C[|u|_{H^1(\Omega)} + \|u|_{\partial\Omega}\|_{L^2(\partial\Omega)}], \quad \forall u \in H^1(\Omega).$$

Proof. We can use the same method as in the proof of the Poincaré inequality. $\square$

Theorem 1.47. *Let Ω be a bounded open set in $\mathbb{R}^n$ with sufficiently regular boundary and let $\ell : H^1(\Omega) \rightarrow V$ be a continuous linear operator, where V is a normed space such that*

$$\|\ell(v)\|_V^2 + |v|_{H^1(\Omega)}^2 = 0 \Rightarrow v = 0.$$

Then there exists a constant $C > 0$ such that

$$\|v\|_{L^2(\Omega)} \leq C[|v|_{H^1(\Omega)} + \|\ell(v)\|_V], \quad \forall v \in H^1(\Omega).$$

Proof. We can use the same method as in the proof of the Poincaré inequality. $\square$

2 Elliptic Equations

Unless otherwise stated, Ω denotes a bounded domain of $\mathbb{R}^n$ with sufficiently regular boundary. Now we consider the following equation:

$$-\Delta u + \alpha u = f, \text{ in } \Omega, \quad (4)$$

where $\alpha \geq 0$ and $f \in L^2(\Omega)$ are given.

To ensure that the elliptic equation (4) gives rise to a well-posed boundary value problem, appropriate constraints must be imposed on the boundary $\partial\Omega$. Here are some examples of boundary value problems:

1. Homogeneous Dirichlet boundary value problem:

$$u = 0 \text{ on } \partial\Omega. \quad (5)$$

2. Inhomogeneous Dirichlet boundary value problem:

$$u = g \text{ on } \partial\Omega. \quad (6)$$

3. Inhomogeneous Neumann boundary value problem:

$$\frac{\partial u}{\partial \nu} = g \text{ on } \partial\Omega. \quad (7)$$

4. Inhomogeneous Robin boundary value problem:

$$\frac{\partial u}{\partial \nu} + \lambda u = g \text{ on } \partial\Omega. \quad (8)$$

2.1 Homogeneous Dirichlet Boundary Value Problem

We seek a function $u \in H^1(\Omega)$ satisfying (4) and (5). However, in order to apply Corollary 1.40, let us temporarily assume $u \in H^2(\Omega)$ and $v \in H^1(\Omega)$. Then we have

$$\begin{aligned} \int_{\Omega} f v \, dx &= \int_{\Omega} (-\Delta u + \alpha u) v \, dx \\ &= \int_{\Omega} (\nabla u \cdot \nabla v + \alpha u v) \, dx - \int_{\partial\Omega} \frac{\partial u}{\partial \nu} v|_{\partial\Omega} \, d\sigma. \end{aligned}$$

Since we have no information about $\frac{\partial u}{\partial \nu}$, we may suppose that $v \in H_0^1(\Omega)$ to simplify this equation. Thus,

$$\int_{\Omega} f v \, dx = \int_{\Omega} (\nabla u \cdot \nabla v + \alpha u v) \, dx, \quad \forall v \in H_0^1(\Omega).$$

To avoid assuming that $u \in H^2(\Omega)$, we can use the extended Green's formula (3) since $\Delta u = \alpha u - f \in L^2(\Omega)$. Now we have the variational problem:

$$\text{Find } u \in H_0^1(\Omega) \text{ such that } \int_{\Omega} (\nabla u \cdot \nabla v + \alpha u v) \, dx = \int_{\Omega} f v \, dx, \quad \forall v \in H_0^1(\Omega). \quad (9)$$

An alternative approach is to consider $v \in C_c^\infty(\Omega)$, then we have

$$\int_{\Omega} -\Delta u v \, dx = \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad \forall v \in C_c^\infty(\Omega).$$

Since $C_c^\infty(\Omega)$ is dense in $H_0^1(\Omega)$, we can extend this equation to $H_0^1(\Omega)$. Hence we obtain the same variational formulation.

Now we prove the equivalence between the boundary value problem and the variational formulation (9).

Theorem 2.1. *Let $f \in L^2(\Omega)$ and $\alpha \geq 0$ be given. If $u \in H_0^1(\Omega)$ is a solution of the variational formulation (9), then u is a weak solution of the boundary value problem (4)-(5).*

Proof. Assume that u is a solution of the variational formulation.

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} (f - \alpha u) v \, dx, \quad \forall v \in C_c^\infty(\Omega).$$

It follows that $\Delta u \in L^2(\Omega)$ exists in the weak sense (∇u has a weak divergence). Moreover $-\Delta u + \alpha u = f$ almost everywhere. Hence $-\Delta u + \alpha u = f$ in $L^2(\Omega)$. $\square$

Theorem 2.2. Suppose $\alpha > 0$. The variational formulation (9) has a unique solution $u \in H_0^1(\Omega)$.

Proof. Define the bilinear form

$$a(u, v) = \int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) \, dx,$$

and the linear form

$$\ell(v) = \int_{\Omega} f v \, dx.$$

$a(\cdot, \cdot)$ is continuous and $\ell(\cdot)$ is continuous on $H_0^1(\Omega)$ since

$$\begin{aligned} \forall v, w \in H_0^1(\Omega), \quad |a(v, w)| &\leq \|\nabla v\|_{L^2(\Omega)} \|\nabla w\|_{(L^2(\Omega))^n} + \alpha \|v\|_{L^2(\Omega)} \|w\|_{L^2(\Omega)} \leq (1 + \alpha) \|v\|_{H^1(\Omega)} \|w\|_{H^1(\Omega)}; \\ \forall v \in H_0^1(\Omega), \quad |\ell(v)| &\leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)}. \end{aligned}$$

Note that $a(\cdot, \cdot)$ is coercive since $\alpha > 0$ and for all $v \in H_0^1(\Omega)$

$$|a(v, v)| \geq \min(1, \alpha) \|v\|_{H^1(\Omega)}^2.$$

By the Lax-Milgram theorem, there exists a unique $u \in H_0^1(\Omega)$ such that

$$a(u, v) = \ell(v), \quad \forall v \in H_0^1(\Omega).$$

□

Remark. Note that we have also $a(u, u) = \ell(u)$. So we have

$$\|u\|_{H^1(\Omega)} \leq \frac{\|f\|_{L^2(\Omega)}}{\min(1, \alpha)}.$$

Theorem 2.3. Suppose $\alpha = 0$. The variational formulation (9) has a unique solution $u \in H_0^1(\Omega)$.

Proof. Following the method of Theorem 2.2, we can conclude that $a(\cdot, \cdot)$ and $\ell(\cdot)$ are continuous on $H_0^1(\Omega)$. However, since $\alpha = 0$, the coercivity of $a(\cdot, \cdot)$ needs to be established differently. By applying the Poincaré inequality, we have

$$\|v\|_{L^2(\Omega)} \leq C |v|_{H^1(\Omega)}, \quad \forall v \in H_0^1(\Omega).$$

Note that

$$\|v\|_{H^1(\Omega)}^2 = \|v\|_{L^2(\Omega)}^2 + |v|_{H^1(\Omega)}^2 \leq (C^2 + 1) |v|_{H^1(\Omega)}^2.$$

Hence, we obtain

$$|a(v, v)| = |v|_{H^1(\Omega)}^2 \geq \frac{1}{(C^2 + 1)} \|v\|_{H^1(\Omega)}^2, \quad \forall v \in H_0^1(\Omega).$$

By the Lax-Milgram theorem, there exists a unique $u \in H_0^1(\Omega)$ such that

$$a(u, v) = \ell(v), \quad \forall v \in H_0^1(\Omega).$$

□

Remark. Note that we have also $a(u, u) = \ell(u)$. Denote by C the constant in Poincaré inequality, it follows that

$$\|u\|_{H^1(\Omega)} \leq (C^2 + 1) \|f\|_{L^2(\Omega)}.$$

At the end of this section, we give some theorems about the regularity of the solution without proof.

Theorem 2.4. Let Ω be a bounded open set in $\mathbb{R}^n$ with C^2 boundary. Let $f \in L^2(\Omega)$ and $\alpha \geq 0$ be given. If $u \in H_0^1(\Omega)$ is a solution, then $u \in H^2(\Omega)$ and there exists a constant $C > 0$ such that

$$\|u\|_{H^2(\Omega)} \leq C \|f\|_{L^2(\Omega)}.$$

Theorem 2.5. Let Ω be a bounded open set in $\mathbb{R}^n$ with Lipschitz boundary. Let $f \in L^2(\Omega)$ and $\alpha \geq 0$ be given. If Ω is convex and $u \in H_0^1(\Omega)$ is a solution, then $u \in H^2(\Omega)$ and there exists a constant $C > 0$ such that

$$\|u\|_{H^2(\Omega)} \leq C \|f\|_{L^2(\Omega)}.$$

2.2 Inhomogeneous Dirichlet Boundary Value Problem

We seek a function $u \in H^1(\Omega)$ satisfying (4) and (6). Suppose $g \in \text{Im}(\gamma)$, as it is the trace of some function in $H^1(\Omega)$.

Using the method in the previous section, we can obtain the variational formulation:

$$\text{Find } u \in H^1(\Omega) \text{ such that } u|_{\partial\Omega} = g \text{ and } \int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) dx = \int_{\Omega} f v dx, \quad \forall v \in H_0^1(\Omega). \quad (10)$$

The equivalence between the boundary value problem and the variational formulation can be established similarly as in the previous section. We omit the proof here.

Theorem 2.6. *The variational formulation (10) has a unique solution $u \in H^1(\Omega)$.*

Proof. Let $u_g \in H^1(\Omega)$ be the extension of g given by a bounded lifting of the trace operator, i.e., $u_g|_{\partial\Omega} = g$ and $\|u_g\|_{H^1(\Omega)} \leq C\|g\|_{\text{Im}(\gamma)}$. Define $u_0 = u - u_g$. Then we need to find $u_0 \in H_0^1(\Omega)$ such that

$$\int_{\Omega} (\nabla u_0 \cdot \nabla v + \alpha u_0 v) dx = \int_{\Omega} (f - \alpha u_g) v dx - \int_{\Omega} \nabla u_g \cdot \nabla v dx, \quad \forall v \in H_0^1(\Omega).$$

Define the bilinear form

$$a(v, w) = \int_{\Omega} (\nabla v \cdot \nabla w + \alpha vw) dx,$$

and the linear form

$$\ell(v) = \int_{\Omega} (f - \alpha u_g) v dx - \int_{\Omega} \nabla u_g \cdot \nabla v dx.$$

By the same method as in Theorem 2.2 and Theorem 2.3, we can prove that $a(\cdot, \cdot)$ is continuous and coercive on $H_0^1(\Omega)$. Now it suffices to prove that $\ell(\cdot)$ is continuous on $H_0^1(\Omega)$. Note that

$$\begin{aligned} |\ell(v)| &\leq (\|f - \alpha u_g\|_{L^2(\Omega)} + \|\nabla u_g\|_{(L^2(\Omega))^n}) \|v\|_{H^1(\Omega)} \\ &\leq (\|f - \alpha u_g\|_{L^2(\Omega)} + C\|g\|_{\text{Im}(\gamma)}) \|v\|_{H^1(\Omega)}, \quad \forall v \in H_0^1(\Omega). \end{aligned}$$

By the Lax-Milgram theorem, there exists a unique $u_0 \in H_0^1(\Omega)$ such that

$$a(u_0, v) = \ell(v), \quad \forall v \in H_0^1(\Omega).$$

Then it suffices to check the solution doesn't depend on the choice of u_g . □

2.3 Neumann Boundary Value Problem

We seek a function $u \in H^1(\Omega)$ satisfying (4) and (7), where $g \in L^2(\partial\Omega)$. Supposing $u \in H^2(\Omega)$, by the same method in Section 2.1, we have

$$\begin{aligned} \int_{\Omega} f v dx &= \int_{\Omega} (-\Delta u + \alpha u) v dx \\ &= \int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) dx - \int_{\partial\Omega} \frac{\partial u}{\partial \nu} v|_{\partial\Omega} d\sigma, \quad \forall v \in H^1(\Omega). \end{aligned}$$

Using the boundary condition (7), we get the following variational formulation:

$$\text{Find } u \in H^1(\Omega) \text{ such that } \int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) dx = \int_{\Omega} f v dx + \int_{\partial\Omega} g v|_{\partial\Omega} d\sigma, \quad \forall v \in H^1(\Omega). \quad (11)$$

We can drop the assumption $u \in H^2(\Omega)$ and obtain the same result.

Theorem 2.7. *Let $f \in L^2(\Omega)$, $g \in L^2(\partial\Omega)$ and $\alpha \geq 0$ be given. If $u \in H^1(\Omega)$ is a solution of the variational formulation (11), then u is a weak solution of the boundary value problem (4)-(7).*

Proof. Assume that u is a solution of the variational formulation (11).

$$\int_{\Omega} \nabla u \cdot \nabla v dx = \int_{\Omega} (f - \alpha u) v dx + \int_{\partial\Omega} g v|_{\partial\Omega} d\sigma = \int_{\Omega} (f - \alpha u) v dx, \quad \forall v \in C_c^\infty(\Omega).$$

We see that $\Delta u \in L^2(\Omega)$ exists in the weak sense (∇u has a weak divergence). Moreover $-\Delta u + \alpha u = f$ almost everywhere. Hence $-\Delta u + \alpha u = f$ in $L^2(\Omega)$. Since Δu exists in the weak sense, by the extended Green's formula (3), we have

$$\int_{\Omega} v \Delta u dx + \int_{\Omega} \nabla v \cdot \nabla u dx = \langle \gamma_\nu(\nabla u), v|_{\partial\Omega} \rangle = \int_{\partial\Omega} (\nabla u \cdot \underline{\nu})|_{\partial\Omega} v|_{\partial\Omega} d\sigma, \quad \forall v \in H^1(\Omega).$$

Since $\Delta u = \alpha u - f$ in $L^2(\Omega)$, we obtain

$$\int_{\partial\Omega} (\nabla u \cdot \underline{\nu})|_{\partial\Omega} v|_{\partial\Omega} d\sigma = \int_{\partial\Omega} g v|_{\partial\Omega} d\sigma, \quad \forall v \in H^1(\Omega).$$

Hence, for every $\phi \in \text{Im}(\gamma)$, we have

$$\int_{\partial\Omega} [(\nabla u \cdot \underline{\nu})|_{\partial\Omega} - g] \phi d\sigma = 0.$$

Therefore, we have $(\nabla u \cdot \underline{\nu})|_{\partial\Omega} - g \in \text{Im}(\gamma)^\perp$. Since $\text{Im}(\gamma)$ is dense in $L^2(\partial\Omega)$, we conclude that $(\nabla u \cdot \underline{\nu})|_{\partial\Omega} = g$ almost everywhere on $\partial\Omega$. $\square$

Theorem 2.8 (Neumann Boundary Value Problem). *Suppose $\alpha > 0$. The variational formulation (11) has a unique solution $u \in H^1(\Omega)$.*

Proof. Define the bilinear form

$$a(u, v) = \int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) dx,$$

and the linear form

$$\ell(v) = \int_{\Omega} f v dx + \int_{\partial\Omega} g v|_{\partial\Omega} d\sigma.$$

$a(\cdot, \cdot)$ is continuous and $\ell(\cdot)$ is continuous on $H^1(\Omega)$ since

$$\begin{aligned} \forall v, w \in H^1(\Omega), \quad |a(v, w)| &\leq \|\nabla v\|_{L^2(\Omega)} \|\nabla w\|_{L^2(\Omega)} + \alpha \|v\|_{L^2(\Omega)} \|w\|_{L^2(\Omega)} \leq (1 + \alpha) \|v\|_{H^1(\Omega)} \|w\|_{H^1(\Omega)}; \\ \forall v \in H^1(\Omega), \quad |\ell(v)| &\leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \|v|_{\partial\Omega}\|_{L^2(\partial\Omega)} \leq (\|f\|_{L^2(\Omega)} + C \|g\|_{L^2(\partial\Omega)}) \|v\|_{H^1(\Omega)}. \end{aligned}$$

Note that $a(\cdot, \cdot)$ is coercive since $\alpha > 0$ and for all $v \in H^1(\Omega)$

$$|a(v, v)| \geq \min(1, \alpha) \|v\|_{H^1(\Omega)}^2.$$

By the Lax-Milgram theorem, there exists a unique $u \in H^1(\Omega)$ such that

$$a(u, v) = \ell(v), \quad \forall v \in H^1(\Omega).$$

$\square$

Remark. Note that we have also $a(u, u) = \ell(u)$. So we have

$$\|u\|_{H^1(\Omega)} \leq \frac{\|f\|_{L^2(\Omega)} + C \|g\|_{L^2(\partial\Omega)}}{\min(1, \alpha)}.$$

For the case $\alpha = 0$, the Neumann boundary value problem is not always well-posed. At first, we see that if u is a solution then $u + c$ is also a solution for any constant $c \in \mathbb{R}$. Hence, we can only expect uniqueness up to an additive constant. Moreover, by taking $v = 1$, a necessary condition for the existence of solutions is

$$\int_{\Omega} f(x) dx + \int_{\partial\Omega} g(x) d\sigma = 0.$$

This condition is called the **compatibility condition**.

Why does uniqueness fail? Note that the bilinear form

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v dx,$$

is not coercive on $H^1(\Omega)$ since for every constant function c , we have $a(c, c) = 0$.

Now, we assume that the compatibility condition is satisfied and try to find a solution in

$$V = \left\{ v \in H^1(\Omega) \mid \int_{\Omega} v(x) dx = 0 \right\}.$$

Consider the variational formulation:

$$\text{Find } u \in V \text{ such that } \int_{\Omega} \nabla u \cdot \nabla v dx = \int_{\Omega} f v dx + \int_{\partial\Omega} g v|_{\partial\Omega} d\sigma, \quad \forall v \in V. \quad (12)$$

Theorem 2.9 (Neumann Boundary Value Problem). *Suppose $\alpha = 0$. The variational formulation (12) has a unique solution $u \in V$.*

Proof. By the Lax-Milgram theorem, it suffices to prove that the bilinear form

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx,$$

is coercive on V . By applying the Poincaré inequality, we have

$$\|v\|_{L^2(\Omega)} \leq C|v|_{H^1(\Omega)}, \quad \forall v \in V.$$

Hence, we obtain

$$|a(v, v)| = |v|_{H^1(\Omega)}^2 \geq \frac{1}{(C^2 + 1)} \|v\|_{H^1(\Omega)}^2, \quad \forall v \in V.$$

□

2.4 Robin Boundary Value Problem

We seek a function $u \in H^1(\Omega)$ satisfying (4) and (8), where $g \in L^2(\partial\Omega)$ and $\lambda > 0$ is a constant. Supposing $u \in H^2(\Omega)$, by the same method in Section 2.1, we have

$$\begin{aligned} \int_{\Omega} f v \, dx &= \int_{\Omega} (-\Delta u + \alpha u) v \, dx \\ &= \int_{\Omega} (\nabla u \cdot \nabla v + \alpha u v) \, dx - \int_{\partial\Omega} \frac{\partial u}{\partial \nu} v|_{\partial\Omega} \, d\sigma, \quad \forall v \in H^1(\Omega). \end{aligned}$$

Using the boundary condition (8), we get the following variational formulation:

$$\text{Find } u \in H^1(\Omega) \text{ such that } \int_{\Omega} (\nabla u \cdot \nabla v + \alpha u v) \, dx + \int_{\partial\Omega} \lambda u v|_{\partial\Omega} \, d\sigma = \int_{\Omega} f v \, dx + \int_{\partial\Omega} g v|_{\partial\Omega} \, d\sigma, \quad \forall v \in H^1(\Omega). \quad (13)$$

We can drop the assumption $u \in H^2(\Omega)$ and obtain the same result.

Theorem 2.10. *Let $f \in L^2(\Omega)$, $g \in L^2(\partial\Omega)$ and $\alpha \geq 0, \lambda > 0$ be given. If $u \in H^1(\Omega)$ is a solution of the variational formulation (13), then u is a weak solution of the boundary value problem (4)-(8).*

Proof. Assume that u is a solution of the variational formulation (13). For every $v \in C_c^\infty(\Omega)$, we have

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} (f - \alpha u) v \, dx, \quad \forall v \in C_c^\infty(\Omega).$$

We see that $\Delta u \in L^2(\Omega)$ exists in the weak sense (∇u has a weak divergence). Moreover $-\Delta u + \alpha u = f$ almost everywhere. Hence $-\Delta u + \alpha u = f$ in $L^2(\Omega)$. Since Δu exists in the weak sense, by the extended Green's formula (3), we have

$$\int_{\Omega} v \Delta u \, dx + \int_{\Omega} \nabla v \cdot \nabla u \, dx = \langle \gamma_\nu(\nabla u), v|_{\partial\Omega} \rangle = \int_{\partial\Omega} (\nabla u \cdot \underline{\nu})|_{\partial\Omega} v|_{\partial\Omega} \, d\sigma, \quad \forall v \in H^1(\Omega).$$

Since $\Delta u = \alpha u - f$ in $L^2(\Omega)$, we obtain

$$\int_{\partial\Omega} (\nabla u \cdot \underline{\nu})|_{\partial\Omega} v|_{\partial\Omega} \, d\sigma = \int_{\partial\Omega} (\alpha u - f) v|_{\partial\Omega} \, d\sigma + \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad \forall v \in H^1(\Omega).$$

By the variational formulation (13), we have

$$\int_{\partial\Omega} (\nabla u \cdot \underline{\nu})|_{\partial\Omega} v|_{\partial\Omega} \, d\sigma = \int_{\partial\Omega} g v|_{\partial\Omega} \, d\sigma - \int_{\partial\Omega} \lambda u v|_{\partial\Omega} \, d\sigma, \quad \forall v \in H^1(\Omega).$$

Simplifying, we get

$$\int_{\partial\Omega} [(\nabla u \cdot \underline{\nu})|_{\partial\Omega} + \lambda u - g] v|_{\partial\Omega} \, d\sigma = 0, \quad \forall v \in H^1(\Omega).$$

Hence, for every $\phi \in \text{Im}(\gamma)$, we have

$$\int_{\partial\Omega} [(\nabla u \cdot \underline{\nu})|_{\partial\Omega} + \lambda u - g] \phi \, d\sigma = 0.$$

Therefore, we have $(\nabla u \cdot \underline{\nu})|_{\partial\Omega} + \lambda u - g \in \text{Im}(\gamma)^\perp$. Since $\text{Im}(\gamma)$ is dense in $L^2(\partial\Omega)$, we conclude that $(\nabla u \cdot \underline{\nu})|_{\partial\Omega} + \lambda u = g$ almost everywhere on $\partial\Omega$. □

Theorem 2.11 (Robin Boundary Value Problem). *Suppose $\alpha \geq 0$ and $\lambda > 0$. The variational formulation (13) has a unique solution $u \in H^1(\Omega)$.*

Proof. Define the bilinear form

$$a(u, v) = \int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) dx + \int_{\partial\Omega} \lambda u v|_{\partial\Omega} d\sigma,$$

and the linear form

$$\ell(v) = \int_{\Omega} f v dx + \int_{\partial\Omega} g v|_{\partial\Omega} d\sigma.$$

$a(\cdot, \cdot)$ is continuous and $\ell(\cdot)$ is continuous on $H^1(\Omega)$. We first show the continuity of $a(\cdot, \cdot)$. Since $\lambda > 0$, by the trace theorem, we have

$$\begin{aligned} |a(v, w)| &\leq \|\nabla v\|_{L^2(\Omega)} \|\nabla w\|_{L^2(\Omega)} + \alpha \|v\|_{L^2(\Omega)} \|w\|_{L^2(\Omega)} + \lambda \|v\|_{\partial\Omega} \|w\|_{\partial\Omega} \\ &\leq (1 + \alpha + \lambda C^2) \|v\|_{H^1(\Omega)} \|w\|_{H^1(\Omega)}, \quad \forall v, w \in H^1(\Omega). \end{aligned}$$

Next, we show the continuity of $\ell(\cdot)$. By the trace theorem, we have

$$\begin{aligned} |\ell(v)| &\leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \|v\|_{\partial\Omega} \\ &\leq (\|f\|_{L^2(\Omega)} + C\|g\|_{L^2(\partial\Omega)}) \|v\|_{H^1(\Omega)}, \quad \forall v \in H^1(\Omega). \end{aligned}$$

Now we show that $a(\cdot, \cdot)$ is coercive on $H^1(\Omega)$.

By applying the Poincaré inequality, we have

$$\|v\|_{L^2(\Omega)} \leq C(\|v\|_{H^1(\Omega)} + \|v\|_{\partial\Omega}), \quad \forall v \in H^1(\Omega).$$

Therefore, we have

$$\|v\|_{H^1(\Omega)} \leq (C + 1)(\|v\|_{H^1(\Omega)} + \|v\|_{\partial\Omega}), \quad \forall v \in H^1(\Omega).$$

Since $\lambda > 0$, for all $v \in H^1(\Omega)$

$$\begin{aligned} a(v, v) &= \|v\|_{H^1(\Omega)}^2 + \alpha \|v\|_{L^2(\Omega)}^2 + \lambda \|v\|_{\partial\Omega}^2 \\ &\geq \min(1, \lambda)(\|v\|_{H^1(\Omega)}^2 + \|v\|_{\partial\Omega}^2) \\ &\geq \frac{\min(1, \lambda)}{2}(\|v\|_{H^1(\Omega)} + \|v\|_{\partial\Omega})^2 \\ &\geq \frac{\min(1, \lambda)}{2(C + 1)^2} \|v\|_{H^1(\Omega)}^2. \end{aligned}$$

By the Lax-Milgram theorem, there exists a unique $u \in H^1(\Omega)$ such that

$$a(u, v) = \ell(v), \quad \forall v \in H^1(\Omega).$$

□

2.5 Mixed Boundary Value Problem

We can also consider the mixed boundary value problem, for example, Dirichlet boundary condition on part of the boundary and Neumann boundary condition on the other part.

$$\begin{cases} -\Delta u + \alpha u = f, & \text{in } \Omega, \\ u = g, & \text{on } \Gamma_D, \\ \frac{\partial u}{\partial \nu} = h, & \text{on } \Gamma_N, \end{cases} \quad (14)$$

where Γ_D and Γ_N are two disjoint nonempty open subsets of $\partial\Omega$ such that $\partial\Omega = \bar{\Gamma}_D \cup \bar{\Gamma}_N$. The variational formulation can be established similarly: Find $u \in V := \{u \in H^1(\Omega) : u|_{\Gamma_D} = g\}$ such that

$$\int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) dx = \int_{\Omega} f v dx + \int_{\Gamma_N} h v|_{\Gamma_N} d\sigma, \quad \forall v \in H_{\Gamma_D}^1(\Omega) := \{u \in H^1(\Omega) : u|_{\Gamma_D} = 0\}. \quad (15)$$

We omit the details here.

A method is to reduce this problem to the inhomogeneous Dirichlet boundary value problem. We take any extension $u_g \in H^1(\Omega)$ of g , i.e., $u_g|_{\Gamma_D} = g$. Define $u_0 = u - u_g$. Then we need to find $u_0 \in H_{\Gamma_D}^1(\Omega)$ such that

$$\int_{\Omega} (\nabla u_0 \cdot \nabla v + \alpha u_0 v) dx = \int_{\Omega} (f - \alpha u_g) v dx - \int_{\Omega} \nabla u_g \cdot \nabla v dx + \int_{\Gamma_N} h v|_{\Gamma_N} d\sigma, \quad \forall v \in H_{\Gamma_D}^1(\Omega).$$

Theorem 2.12. Suppose $\alpha > 0$. The variational formulation (15) has a unique solution $u \in H^1(\Omega)$.

Proof. By the same method as in Theorem 2.2 and Theorem 2.8, we can still apply the Lax-Milgram theorem. Then it suffices to check the solution doesn't depend on the choice of u_g . $\square$

Theorem 2.13. Suppose $\alpha = 0$. The variational formulation (15) has a unique solution $u \in H^1(\Omega)$.

Proof. By the same method as in Theorem 2.3 and Theorem 2.9, we can still apply the Lax-Milgram theorem. Then it suffices to check the solution doesn't depend on the choice of u_g . $\square$

2.6 A Direct Method of Computation

Consider the functional:

$$\mathcal{F}(v) = \frac{1}{2} \int_{\Omega} |\nabla v|^2 dx - \int_{\Omega} f v dx, \quad \forall v \in H_0^1(\Omega). \quad (16)$$

We want to find a minimizer $u \in H_0^1(\Omega)$. The proof consists of 3 steps:

1. Show that $\mathcal{F}$ has a lower bound, which implies that $m = \inf_{v \in H_0^1(\Omega)} \mathcal{F}(v)$ exists;
2. Find a minimizing sequence $\{u_n\} \subset H_0^1(\Omega)$ such that $\lim_{n \rightarrow \infty} \mathcal{F}(u_n) = m$, and extract a weakly convergent subsequence (we only need to show that $\{u_n\}$ is bounded in $H_0^1(\Omega)$);
3. Show that $\mathcal{F}(u) \leq \liminf_{n \rightarrow \infty} \mathcal{F}(u_{\phi(n)}) = m$, which implies that u is a minimizer.

Now we give the details.

1. By the Poincaré inequality, there exists a constant $C > 0$ such that

$$\|v\|_{L^2(\Omega)} \leq C \|v\|_{H^1(\Omega)}, \quad \forall v \in H_0^1(\Omega).$$

Hence, we have

$$\left| \int_{\Omega} f v dx \right| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq C \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)}.$$

Therefore, we obtain

$$\mathcal{F}(v) \geq \frac{1}{2} \|v\|_{H^1(\Omega)}^2 - C \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)} \geq -\frac{C^2}{2} \|f\|_{L^2(\Omega)}^2.$$

2. If $\mathcal{F}(u_n) < M$, then we have

$$\begin{aligned} M > \mathcal{F}(u_n) &\geq \frac{1}{2} \|u_n\|_{H^1(\Omega)}^2 - C \|f\|_{L^2(\Omega)} \|u_n\|_{H^1(\Omega)} \\ &\geq \frac{1}{4} \|u_n\|_{H^1(\Omega)}^2 - C^2 \|f\|_{L^2(\Omega)}^2. \end{aligned}$$

Hence, $\{u_n\}$ is bounded in $H_0^1(\Omega)$. By Theorem 1.18, there exists a subsequence $\{u_{\phi(n)}\}$ and $u \in H_0^1(\Omega)$ such that $u_{\phi(n)} \rightharpoonup u$ in $H_0^1(\Omega)$.

3. By Corollary 1.42, we may suppose the sequence $\{u_{\phi(n)}\}$ converges strongly to u in $L^2(\Omega)$. By Proposition 1.16, we have

$$\|u\|_{H^1(\Omega)} \leq \liminf_{n \rightarrow \infty} \|u_{\phi(n)}\|_{H^1(\Omega)}.$$

Since $u_{\phi(n)} \rightarrow u$ in $L^2(\Omega)$, we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} f u_{\phi(n)} dx = \int_{\Omega} f u dx.$$

Therefore, we obtain

$$\begin{aligned} \mathcal{F}(u) &= \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx - \int_{\Omega} f u dx \\ &\leq \liminf_{n \rightarrow \infty} \frac{1}{2} \int_{\Omega} |\nabla u_{\phi(n)}|^2 dx - \lim_{n \rightarrow \infty} \int_{\Omega} f u_{\phi(n)} dx \\ &= \liminf_{n \rightarrow \infty} \mathcal{F}(u_{\phi(n)}) = m. \end{aligned}$$

Therefore, u is a minimizer of $\mathcal{F}$.

This method allows us to solve some problems without applying the Lax-Milgram theorem.

Actually, this method can be applied to more general problems, for example, nonlinear energy functionals:

$$\mathcal{F}(v) = \frac{1}{2} \int_{\Omega} |\nabla v|^2 dx + \int_{\Omega} F(v) dx, \quad \forall v \in H_0^1(\Omega), \quad (17)$$

where $F : \mathbb{R} \rightarrow \mathbb{R}$ is a “nice” function satisfying the critical growth condition: $F(t) \leq C(1 + |t|^p)$, where $1 \leq p < \tilde{p} = \frac{2n}{n-2}$ ($n \geq 3$). Note that the injection $H^1(\Omega) \hookrightarrow L^p(\Omega)$ is compact for all $1 \leq p < \tilde{p}$ (generalization of the Rellich-Kondrachov compactness theorem) and there exists a constant $C_p > 0$ such that $\|v\|_{L^p(\Omega)} \leq C_p \|v\|_{H^1(\Omega)}$ for all $v \in H^1(\Omega)$ (Sobolev inequality). Hence, we can still apply the same method to find a minimizer of $\mathcal{F}$.

Example 2.14. Fix $n = 3$. Consider the following boundary value problem:

$$\begin{aligned} -\Delta u_{\epsilon} + \frac{1}{\epsilon}(u_{\epsilon}^2 - 1)u_{\epsilon} &= 0, & \text{in } \Omega, \\ \nabla u_{\epsilon} \cdot \underline{n} &= 0, & \text{on } \partial\Omega, \end{aligned}$$

where $\epsilon > 0$ is a small parameter. The energy functional is

$$\mathcal{F}_{\epsilon}(v) = \frac{1}{2} \int_{\Omega} |\nabla v|^2 dx + \frac{1}{4\epsilon} \int_{\Omega} (v^2 - 1)^2 dx, \quad \forall v \in H^1(\Omega).$$

We see that for all $\epsilon > 0$, $\mathcal{F}_{\epsilon}$ has a minimizer $u_{\epsilon} \in H^1(\Omega)$, which means that the equation has a solution. However, we cannot apply the Lax-Milgram theorem to get the existence of solutions since the equation is nonlinear.

2.7 Some Properties of Weak Solutions

We consider the inhomogeneous Dirichlet boundary value problem (4) and (6) with $f \in L^2(\Omega)$, $\alpha \geq 0$ and $g \in \text{Im}(\gamma)$. Let $u \in H^1(\Omega)$ be the weak solution.

Proposition 2.15 (Maximum Principle). *Suppose that $f \geq 0$ almost everywhere in Ω and $g \geq 0$ almost everywhere on $\partial\Omega$. Then the weak solution u satisfies $u \geq 0$ almost everywhere in Ω .*

Proof. Define $u^- = \max(-u, 0)$. Note that $u^- \in H_0^1(\Omega)$ since $u^-|_{\partial\Omega} = \max(-g, 0) = 0$. By the variational formulation (10), we have

$$\int_{\Omega} (\nabla u \cdot \nabla u^- + \alpha u u^-) dx = \int_{\Omega} f u^- dx.$$

Since $f \geq 0$ almost everywhere in Ω , we have

$$\int_{\Omega} (\nabla u \cdot \nabla u^- + \alpha u u^-) dx \geq 0.$$

Note that $\nabla u \cdot \nabla u^- = -|\nabla u^-|^2$ and $u u^- = -|u^-|^2$. Hence, we obtain

$$-\int_{\Omega} (|\nabla u^-|^2 + \alpha |u^-|^2) dx \geq 0.$$

Therefore, we have $u^- = 0$ almost everywhere in Ω , which implies that $u \geq 0$ almost everywhere in Ω . $\square$

Theorem 2.16 (Weak Maximum Principle). *If $\alpha > 0$, then the weak solution u satisfies*

$$\min \left(\alpha^{-1} \inf_{\Omega} f, \inf_{\partial\Omega} g \right) \leq u(x) \leq \max \left(\alpha^{-1} \sup_{\Omega} f, \sup_{\partial\Omega} g \right), \quad \text{for almost every } x \in \Omega.$$

If $\alpha = 0$, then the weak solution u satisfies

$$\begin{aligned} f \geq 0 &\Rightarrow \inf_{\partial\Omega} g \leq u(x), & \text{for almost every } x \in \Omega; \\ f \leq 0 &\Rightarrow u(x) \leq \sup_{\partial\Omega} g, & \text{for almost every } x \in \Omega; \\ f = 0 &\Rightarrow \inf_{\partial\Omega} g \leq u(x) \leq \sup_{\partial\Omega} g, & \text{for almost every } x \in \Omega. \end{aligned}$$

Proof. If $\alpha > 0$, define $M = \max(\alpha^{-1} \sup_{\Omega} f, \sup_{\partial\Omega} g)$ and $v = (u - M)^+ \geq 0$. Note that $v \in H_0^1(\Omega)$ since $v|_{\partial\Omega} = (g - M)^+ = 0$. By the variational formulation (10), we have

$$\int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) \, dx = \int_{\Omega} f v \, dx.$$

Since $f \leq \alpha M$, we have

$$\int_{\Omega} (\nabla u \cdot \nabla v + \alpha uv) \, dx \leq \int_{\Omega} \alpha M v \, dx,$$

which implies that

$$\int_{\Omega} \nabla u \cdot \nabla v + \alpha(u - M)v \, dx = \int_{\text{supp}(v)} |\nabla v|^2 + \alpha|v|^2 \, dx \leq 0.$$

Therefore, we have $v = 0$ almost everywhere in Ω , which implies that $u(x) \leq M$ for almost every $x \in \Omega$. The lower bound can be proved similarly. The case $\alpha = 0$ can be treated in the same way. $\square$

3 Some Examples

3.1 Linearized Elasticity System

Suppose $\Omega \subset \mathbb{R}^n$ is a bounded domain with Lipschitz boundary. Consider the following system:

$$\begin{cases} -\nabla \cdot \underline{\sigma} = \underline{f}, & \text{in } \Omega, \\ \underline{\sigma} \cdot \underline{n} = \underline{g}, & \text{on } \partial\Omega, \end{cases} \quad (18)$$

where $\underline{\sigma} : \Omega \rightarrow M_n(\mathbb{R})$ is the unknown matrix-valued function, $\underline{f} : \Omega \rightarrow \mathbb{R}^n$ is a given vector-valued function, $\underline{g} : \partial\Omega \rightarrow \mathbb{R}^n$ is a given vector-valued function and $\underline{n}$ is the unit outer normal vector on $\partial\Omega$.

The system is called the linearized elasticity system since it describes the deformation of an elastic body under the action of external forces.

From (18), we see that

$$\int_{\Omega} \underline{f} \, dx + \int_{\partial\Omega} \underline{g} \, d\sigma = \int_{\Omega} -\nabla \cdot \underline{\sigma} \, dx + \int_{\partial\Omega} \underline{\sigma} \cdot \underline{n} \, d\sigma \stackrel{\text{Divergence Theorem}}{=} 0.$$

We define

$$\underline{\varepsilon}(\underline{u}) = \frac{1}{2}(\nabla \underline{u} + (\nabla \underline{u})^T) = \left(\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right)_{1 \leq i, j \leq n}.$$

By Hooke's law, we have

$$\underline{\sigma}(\underline{u}) := 2\mu \underline{\varepsilon}(\underline{u}) + \lambda \text{Tr}(\underline{\varepsilon}(\underline{u})) I_n.$$

Hence, we can rewrite the first equation in (18) as

$$-\nabla \cdot (2\mu \underline{\varepsilon}(\underline{u}) + \lambda \text{Tr}(\underline{\varepsilon}(\underline{u})) I_n) = \underline{f}, \quad \text{in } \Omega. \quad (19)$$

where $\underline{u} : \Omega \rightarrow \mathbb{R}^n$ is the unknown vector-valued function, μ and λ are two constants called the Lamé parameters which satisfy $\mu > 0$ and $2\mu + \lambda n > 0$. This formulation naturally leads to another type of boundary condition: $\underline{u} = 0$ on $\partial\Omega$.

Consider the following mixed boundary value problem:

$$\begin{cases} -\nabla \cdot (2\mu \underline{\varepsilon}(\underline{u}) + \lambda \text{Tr}(\underline{\varepsilon}(\underline{u})) I_n) = \underline{f}, & \text{in } \Omega, \\ \underline{u} = 0, & \text{on } \Gamma_D, \\ (2\mu \underline{\varepsilon}(\underline{u}) + \lambda \text{Tr}(\underline{\varepsilon}(\underline{u})) I_n) \cdot \underline{n} = \underline{g}, & \text{on } \Gamma_N, \end{cases} \quad (20)$$

where Γ_D and Γ_N are two disjoint nonempty open subsets of $\partial\Omega$ such that $\partial\Omega = \bar{\Gamma}_D \cup \bar{\Gamma}_N$. And the variational formulation of this mixed boundary value problem is as follows:

$$\text{Find } \underline{u} \in V = \{ \underline{v} \in (H^1(\Omega))^n : \gamma_{\Gamma_D}(\underline{v}) = 0 \text{ on } \Gamma_D \} \text{ such that } a(\underline{u}, \underline{v}) = \ell(\underline{v}), \quad \forall \underline{v} \in V, \quad (21)$$

where

$$\begin{aligned} a(\underline{u}, \underline{v}) &= \int_{\Omega} (2\mu \underline{\varepsilon}(\underline{u}) : \underline{\varepsilon}(\underline{v}) + \lambda (\nabla \cdot \underline{u})(\nabla \cdot \underline{v})) \, dx, \quad \forall \underline{u}, \underline{v} \in V, \\ \ell(\underline{v}) &= \int_{\Omega} \underline{f} \cdot \underline{v} \, dx + \int_{\Gamma_N} \underline{g} \cdot \gamma_{\Gamma_N}(\underline{v}) \, d\sigma, \quad \forall \underline{v} \in V. \end{aligned}$$

Here $A : B = \sum_{i,j=1}^n A_{ij} B_{ij}$ denotes the Frobenius inner product of two matrices $A, B \in M_n(\mathbb{R})$.

Theorem 3.1. *The variational formulation (21) has a unique solution $\underline{u} \in V$ which is characterized by the following minimization problem:*

$$\underline{u} = \argmin \left\{ J(\underline{w}) = \int_{\Omega} W(\underline{\varepsilon}(\underline{w})) \, dx - \int_{\Omega} \underline{f} \cdot \underline{w} \, dx - \int_{\Gamma_N} \underline{g} \cdot \gamma_{\Gamma_N}(\underline{w}) \, d\sigma \right\},$$

where $W(\underline{\varepsilon}(\underline{w})) = \mu \underline{\varepsilon}(\underline{w}) : \underline{\varepsilon}(\underline{w}) + \frac{\lambda}{2} (\text{Tr}(\underline{\varepsilon}(\underline{w})))^2$.

Before proving Theorem 3.1, we need the following lemmas.

Lemma 3.2. *Set $\alpha = \min\{2\mu, 2\mu + \lambda n\} > 0$. Then we have*

$$\alpha \int_{\Omega} |\underline{\varepsilon}(\underline{v})|^2 \, dx \leq a(\underline{v}, \underline{v}), \quad \forall \underline{v} \in V.$$

Proof. Since

$$\underline{\underline{\varepsilon}}(\underline{v}) = \left(\underline{\underline{\varepsilon}}(\underline{v}) - \frac{1}{n}(\text{Tr}(\underline{\underline{\varepsilon}}(\underline{v})))I_n \right) + \frac{1}{n}(\text{Tr}(\underline{\underline{\varepsilon}}(\underline{v})))I_n,$$

we have

$$\begin{aligned} a(\underline{v}, \underline{v}) &= \int_{\Omega} (2\mu \underline{\underline{\varepsilon}}(\underline{v}) : \underline{\underline{\varepsilon}}(\underline{v}) + \lambda(\nabla \cdot \underline{v})^2) \, dx \\ &= \int_{\Omega} \left(2\mu \left| \underline{\underline{\varepsilon}}(\underline{v}) - \frac{1}{n}(\text{Tr}(\underline{\underline{\varepsilon}}(\underline{v})))I_n \right|^2 + \frac{2\mu + \lambda n}{n}(\text{Tr}(\underline{\underline{\varepsilon}}(\underline{v})))^2 \right) \, dx \\ &\geq \alpha \int_{\Omega} |\underline{\underline{\varepsilon}}(\underline{v})|^2 \, dx. \end{aligned}$$

□

Lemma 3.3. $\forall \underline{v} \in (H_0^1(\Omega))^n$, we have

$$\|\nabla \underline{v}\|_{L^2} \leq \sqrt{2} \|\underline{\underline{\varepsilon}}(\underline{v})\|_{L^2}.$$

Proof. For all $\underline{v} \in (C_c^\infty(\Omega))^n$, we have

$$\begin{aligned} 2\|\underline{\underline{\varepsilon}}(\underline{v})\|_{L^2}^2 &= \frac{1}{2} \int_{\Omega} |\nabla \underline{v} + \nabla \underline{v}^T|^2 \, dx = \int_{\Omega} (|\nabla \underline{v}|^2 + \nabla \underline{v} : \nabla \underline{v}^T) \, dx \\ &= \int_{\Omega} |\nabla \underline{v}|^2 \, dx + \sum_{i,j=1}^n \int_{\Omega} \frac{\partial v_i}{\partial x_j} \frac{\partial v_j}{\partial x_i} \, dx = \int_{\Omega} |\nabla \underline{v}|^2 \, dx - \sum_{i,j=1}^n \int_{\Omega} v_i \frac{\partial^2 v_j}{\partial x_j \partial x_i} \, dx \\ &= \int_{\Omega} |\nabla \underline{v}|^2 \, dx - \sum_{i=1}^n \int_{\Omega} v_i \frac{\partial}{\partial x_i} (\nabla \cdot \underline{v}) \, dx = \int_{\Omega} |\nabla \underline{v}|^2 \, dx + \int_{\Omega} |\nabla \cdot \underline{v}|^2 \, dx \\ &\geq \int_{\Omega} |\nabla \underline{v}|^2 \, dx = \|\nabla \underline{v}\|_{L^2}^2. \end{aligned}$$

Since $(C_c^\infty(\Omega))^n$ is dense in $(H_0^1(\Omega))^n$, we can obtain the desired result. □

When $\Gamma_D = \partial\Omega$, Lemma 3.2 and Lemma 3.3 imply that $a(\cdot, \cdot)$ is coercive on $V = (H_0^1(\Omega))^n$. For the general case, we need Korn's inequality to show that $a(\cdot, \cdot)$ is coercive on V .

Lemma 3.4 (Korn's inequality). *There exists a constant $C > 0$ such that*

$$\|\underline{v}\|_{(H^1(\Omega))^n} \leq C \left(\|\underline{\underline{\varepsilon}}(\underline{v})\|_{L^2}^2 + \|\underline{v}\|_{L^2}^2 \right)^{\frac{1}{2}}, \quad \forall \underline{v} \in (H^1(\Omega))^n.$$

Proof. We argue by contradiction. Suppose for every $k \in \mathbb{N}$, there exists a function $\underline{v}_k \in (H^1(\Omega))^n$ such that

$$\|\underline{v}_k\|_{(H^1(\Omega))^n} = 1,$$

but

$$\|\underline{\underline{\varepsilon}}(\underline{v}_k)\|_{L^2(\Omega)}^2 + \|\underline{v}_k\|_{L^2(\Omega)}^2 < \frac{1}{k}.$$

Since the sequence $\{\underline{v}_k\}$ is bounded in $(H^1(\Omega))^n$, the Rellich–Kondrachov compactness theorem guarantees the existence of a subsequence, still denoted by $\{\underline{v}_k\}$, and a function $\underline{v} \in (H^1(\Omega))^n$ such that

$$\underline{v}_k \rightharpoonup \underline{v} \text{ in } (H^1(\Omega))^n, \quad \underline{v}_k \rightarrow \underline{v} \text{ in } (L^2(\Omega))^n.$$

Since the linear operator $\underline{\underline{\varepsilon}}(\cdot)$ is continuous, passing to the weak limit in $\|\underline{\underline{\varepsilon}}(\underline{v}_k)\|_{L^2} \rightarrow 0$ yields

$$\underline{\underline{\varepsilon}}(\underline{v}) = 0 \quad \text{a.e. in } \Omega.$$

We now characterize all functions satisfying $\underline{\underline{\varepsilon}}(\underline{v}) = 0$. In component form, this reads

$$\partial_j v_i + \partial_i v_j = 0, \quad \forall i, j = 1, \dots, n.$$

Differentiating this equation with respect to x_k (valid in the sense of distributions), we obtain

$$\partial_{kj} v_i + \partial_{ki} v_j = 0, \quad \forall i, j, k = 1, \dots, n.$$

By cyclic permutation of the indices i, j, k , we also have

$$\partial_{ij} v_k + \partial_{ik} v_j = 0, \quad \partial_{ji} v_k + \partial_{jk} v_i = 0.$$

Since partial derivatives commute (Schwarz's theorem), we have

$$\partial_{kj}v_i = -\partial_{ki}v_j = -\partial_{ik}v_j = \partial_{ij}v_k = \partial_{ji}v_k = -\partial_{jk}v_i = -\partial_{kj}v_i,$$

i.e., all second-order derivatives of $\underline{v}$ vanish. Hence, $\partial_j v_i = W_{ij}$ for some constant matrix $W = (W_{ij}) \in M_n(\mathbb{R})$. Substituting this back to $\partial_j v_i + \partial_i v_j = 0$ yields

$$W_{ij} + W_{ji} = 0, \quad \forall i, j = 1, \dots, n,$$

which means W is skew-symmetric, i.e., $W^T = -W$. Integration of $\partial_j v_i = W_{ij}$ then gives

$$\underline{v}(x) = \underline{a} + Wx,$$

for some constant vector $\underline{a} \in \mathbb{R}^n$.

From the supposition, we have $\|\underline{v}_k\|_{L^2(\Omega)} \rightarrow 0$. Together with the strong convergence in $L^2(\Omega)$, this implies $\|\underline{v}\|_{L^2(\Omega)} = 0$, and hence $\underline{v} \equiv 0$ in Ω .

It remains to upgrade the convergence $\underline{v}_k \rightharpoonup \underline{v}$ to a strong one in $(H^1(\Omega))^n$. From the identity (valid in the sense of distributions)

$$\partial_k \partial_j v_i = \partial_j \varepsilon_{ik}(\underline{v}) + \partial_k \varepsilon_{ij}(\underline{v}) - \partial_i \varepsilon_{jk}(\underline{v}),$$

the second-order derivatives of $\underline{v}_m - \underline{v}_k$ are controlled in $H^{-1}(\Omega)$ by $\|\underline{\varepsilon}(\underline{v}_m) - \underline{\varepsilon}(\underline{v}_k)\|_{L^2(\Omega)}$, while $\underline{v}_m - \underline{v}_k \rightarrow 0$ in $(L^2(\Omega))^n$. By J.L. Lions' lemma (if w and all its first-order derivatives belong to $H^{-1}(\Omega)$, then $w \in L^2(\Omega)$ with $\|w\|_{L^2} \leq C(\|w\|_{H^{-1}} + \|\nabla w\|_{H^{-1}})$), applied to each $\partial_j(v_{m,i} - v_{k,i})$, the sequence $\{\nabla \underline{v}_k\}$ is Cauchy in $(L^2(\Omega))^{n \times n}$. Hence $\underline{v}_k \rightarrow \underline{v}$ strongly in $(H^1(\Omega))^n$, so that $\|\underline{v}\|_{(H^1(\Omega))^n} = 1$, which contradicts $\underline{v} \equiv 0$. Therefore, the assumption was false, and the inequality holds. $\square$

Remark. Some proofs of J.L. Lions' lemma can be found in [1].

Proof of Theorem 3.1. First, we show that there exists a constant $C > 0$ such that

$$\|\underline{v}\|_{(H^1(\Omega))^n} \leq C \|\underline{\varepsilon}(\underline{v})\|_{L^2(\Omega)}, \quad \forall \underline{v} \in V. \quad (22)$$

We argue by contradiction. Suppose for every $k \in \mathbb{N}$, there exists a function $\underline{v}_k \in V$ such that

$$\|\underline{v}_k\|_{(H^1(\Omega))^n} = 1, \quad \text{and} \quad \|\underline{\varepsilon}(\underline{v}_k)\|_{L^2(\Omega)} < \frac{1}{k}.$$

Since the sequence $\{\underline{v}_k\}$ is bounded in the Hilbert space $(H^1(\Omega))^n$, the Rellich–Kondrachov compactness theorem guarantees the existence of a subsequence, still denoted by $\{\underline{v}_k\}$, and a function $\underline{v} \in (H^1(\Omega))^n$ such that

$$\underline{v}_k \rightharpoonup \underline{v} \quad \text{weakly in } (H^1(\Omega))^n, \quad \underline{v}_k \rightarrow \underline{v} \quad \text{strongly in } (L^2(\Omega))^n.$$

Since the linear operator $\underline{\varepsilon}(\cdot)$ is continuous, passing to the weak limit in $\|\underline{\varepsilon}(\underline{v}_k)\|_{L^2} \rightarrow 0$ yields

$$\underline{\varepsilon}(\underline{v}) = 0 \quad \text{a.e. in } \Omega.$$

The proof of Lemma 3.4 shows that there exist a constant vector $\underline{a} \in \mathbb{R}^n$ and a skew-symmetric matrix $W \in M_n(\mathbb{R})$ such that

$$\underline{v}(x) = \underline{a} + Wx, \quad x \in \Omega.$$

Since each $\underline{v}_k$ belongs to V , we have $\gamma_{\Gamma_D}(\underline{v}_k) = 0$ on Γ_D . The trace operator $\gamma : (H^1(\Omega))^n \rightarrow (L^2(\partial\Omega))^n$ is continuous, so taking the limit gives

$$\gamma_{\Gamma_D}(\underline{v}) = 0 \quad \text{on } \Gamma_D.$$

Thus,

$$\underline{v} \equiv 0 \quad \text{in } \Omega.$$

Consequently, the strong convergence in $L^2(\Omega)$ implies

$$\|\underline{v}_k\|_{L^2(\Omega)} \rightarrow 0.$$

Together with the hypothesis $\|\underline{\varepsilon}(\underline{v}_k)\|_{L^2(\Omega)} \rightarrow 0$, we now invoke Korn's inequality to obtain:

$$\|\underline{v}_k\|_{(H^1(\Omega))^n} \leq C \left(\|\underline{\varepsilon}(\underline{v}_k)\|_{L^2(\Omega)}^2 + \|\underline{v}_k\|_{L^2(\Omega)}^2 \right)^{\frac{1}{2}}.$$

Letting $k \rightarrow +\infty$, the right-hand side tends to 0. Therefore,

$$\|\underline{v}_k\|_{(H^1(\Omega))^n} \rightarrow 0,$$

which directly contradicts the normalization $\|\underline{v}_k\|_{(H^1(\Omega))^n} = 1$ for all k . Thus, our initial assumption was false. Consequently, there exists a constant $C > 0$ such that

$$\|\underline{v}\|_{(H^1(\Omega))^n} \leq C \|\underline{\varepsilon}(\underline{v})\|_{L^2(\Omega)}, \quad \forall \underline{v} \in V.$$

Hence Lemma 3.2 implies that $a(\cdot, \cdot)$ is coercive on V . Hence, by the Lax-Milgram theorem, there exists a unique $\underline{u} \in V$ such that (21) holds. By Corollary 1.20, Theorem 3.1 holds. $\square$

3.2 Stokes Problem

Consider the following problem:

$$\begin{cases} -\mu\Delta \underline{u} + \nabla p = \underline{f}, & \text{in } \Omega, \\ \nabla \cdot \underline{u} = 0, & \text{in } \Omega, \\ \underline{u} = 0, & \text{on } \partial\Omega, \end{cases} \quad (23)$$

where $\underline{u} : \Omega \rightarrow \mathbb{R}^n$ is the unknown vector-valued function, $p : \Omega \rightarrow \mathbb{R}$ is the unknown scalar-valued function, $\underline{f} : \Omega \rightarrow \mathbb{R}^n$ is a given vector-valued function. And the variational formulation of this problem is as follows:

$$\text{Find } \underline{u} \in V \text{ such that } \int_{\Omega} \mu \nabla \underline{u} : \nabla \underline{v} \, dx = \int_{\Omega} \underline{f} \cdot \underline{v} \, dx, \quad \forall \underline{v} \in V, \quad (24)$$

where $V = \{\underline{v} \in (H_0^1(\Omega))^n : \nabla \cdot \underline{v} = 0\}$. Restricting to divergence-free test functions eliminates the pressure p from the variational formulation, since $\int_{\Omega} p \nabla \cdot \underline{v} \, dx = 0$ for all $\underline{v} \in V$.

The energy functional is given by

$$\mathcal{F}(\underline{v}) = \frac{\mu}{2} \int_{\Omega} |\nabla \underline{v}|^2 \, dx - \int_{\Omega} \underline{f} \cdot \underline{v} \, dx, \quad \forall \underline{v} \in V.$$

Theorem 3.5 (De Rham Theorem). *Suppose $\ell \in \mathcal{L}((H_0^1(\Omega))^n; \mathbb{R})$. Then there exists $p \in L^2(\Omega)$ such that*

$$\ell(\underline{v}) = \int_{\Omega} p \nabla \cdot \underline{v} \, dx, \quad \forall \underline{v} \in (H_0^1(\Omega))^n,$$

if and only if $\ell(\underline{v}) = 0$ for all $\underline{v} \in V$. p is unique up to a constant.

Proof. The “only if” part is trivial. Now we prove the “if” part. Consider the operator:

$$\Psi : (H_0^1(\Omega))^n \rightarrow L^2(\Omega), \quad \underline{v} \mapsto \nabla \cdot \underline{v}.$$

We see that Ψ is a bounded linear operator and $\text{Ker}(\Psi) = V$. Moreover, we can show that $\text{Im}(\Psi) = L_0^2(\Omega) := \{f \in L^2(\Omega) \mid \int_{\Omega} f \, dx = 0\}$. By divergence theorem, we have

$$\int_{\Omega} \nabla \cdot \underline{v} \, dx = 0$$

for all $\underline{v} \in (H_0^1(\Omega))^n$, which implies that $\text{Im}(\Psi) \subset L_0^2(\Omega)$. On the other hand, for all $f \in L_0^2(\Omega)$, there exists $\underline{u} \in (H_0^1(\Omega))^n$ such that $\nabla \cdot \underline{u} = f$ in Ω . The existence of such $\underline{u}$ is assured by the Bogovskii's lemma [2].

Hence, we obtain an isomorphism as follows:

$$\tilde{\Psi} : (H_0^1(\Omega))^n / V \rightarrow L_0^2(\Omega), \quad [\underline{v}] \mapsto \nabla \cdot \underline{v}.$$

Since ℓ is trivial on V , we can define $\tilde{\ell} : (H_0^1(\Omega))^n / V \rightarrow \mathbb{R}$ by $\tilde{\ell}([\underline{v}]) = \ell(\underline{v})$. Note that $\tilde{\ell}$ is a bounded linear functional on $(H_0^1(\Omega))^n / V$. Hence, the composition $\tilde{\ell} \circ \tilde{\Psi}^{-1} : L_0^2(\Omega) \rightarrow \mathbb{R}$ is also a bounded linear functional on $L_0^2(\Omega)$. By Riesz representation theorem, there exists $p \in L_0^2(\Omega)$ such that $\tilde{\ell} \circ \tilde{\Psi}^{-1}(w) = \int_{\Omega} p w \, dx$ for all $w \in L_0^2(\Omega)$. Therefore, we have

$$\ell(\underline{v}) = (\tilde{\ell} \circ \tilde{\Psi}^{-1})(\Psi(\underline{v})) = \int_{\Omega} p \nabla \cdot \underline{v} \, dx, \quad \forall \underline{v} \in (H_0^1(\Omega))^n.$$

However, if we take $p \in L^2(\Omega)$ instead of $p \in L_0^2(\Omega)$, then p is unique up to a constant since $\int_{\Omega} c \nabla \cdot \underline{v} \, dx = 0$ for all $\underline{v} \in (H_0^1(\Omega))^n$ and $c \in \mathbb{R}$. $\square$

Then we can apply the Lax-Milgram theorem to obtain the existence and uniqueness of the solution of (24).

4 Finite Element Method (FEM)

4.1 1D Case

Consider the following boundary value problem:

$$\begin{cases} -ku''(x) + u(x) = f(x), & \text{in } (0, L), \\ u(0) = u(L) = 0, \end{cases} \quad (25)$$

where $k > 0$ is a constant, $f \in L^2(0, L)$ is a given function

Then the variational formulation is: find $u \in H_0^1(0, L)$ such that

$$a(u, v) = \ell(v), \quad \forall v \in H_0^1(0, L), \quad (26)$$

where

$$\begin{aligned} a(u, v) &= \int_0^L (ku'(x)v'(x) + u(x)v(x)) \, dx, \quad \forall u, v \in H_0^1(0, L), \\ \ell(v) &= \int_0^L f(x)v(x) \, dx, \quad \forall v \in H_0^1(0, L). \end{aligned}$$

Note that $a(\cdot, \cdot)$ is continuous and coercive on $H_0^1(0, L)$, and $\ell(\cdot)$ is continuous on $H_0^1(0, L)$. By the Lax-Milgram theorem, there exists a unique $u \in H_0^1(0, L)$ such that (26) holds.

This is equivalent to the minimization problem: find $u \in H_0^1(0, L)$ such that

$$\mathcal{F}(u) = \min_{v \in H_0^1(0, L)} \mathcal{F}(v), \quad (27)$$

where

$$\mathcal{F}(v) = \frac{1}{2} \int_0^L (kv'(x)^2 + v(x)^2) \, dx - \int_0^L f(x)v(x) \, dx, \quad \forall v \in H_0^1(0, L).$$

Consider a partition of $[0, L]$ into $N + 1$ subintervals with nodes $0 = x_0 < x_1 < \dots < x_{N+1} = L$ and $h = \max_{0 \leq i \leq N} (x_{i+1} - x_i)$. The simplest choice is a uniform partition, i.e., $x_i = ih$ for $i = 0, 1, \dots, N + 1$. Now we can take a finite-dimensional subspace V_h of $H^1(0, L)$, more precisely, the space of piecewise linear functions on this partition, i.e.,

$$V_h = \{v \in H^1(0, L) : v|_{[x_i, x_{i+1}]} \text{ is linear for } i = 0, 1, \dots, N\}.$$

Then define

$$V_{0h} = \{v \in H_0^1(0, L) : v|_{[x_i, x_{i+1}]} \text{ is linear for } i = 0, 1, \dots, N; v(0) = v(L) = 0\}.$$

Note that $\dim V_{0h} = N$ and a basis of V_{0h} is given by $\{\phi_1, \phi_2, \dots, \phi_N\}$, where ϕ_i is the piecewise linear function such that

$$\phi_i(x) = \begin{cases} \frac{x - x_{i-1}}{x_i - x_{i-1}}, & x \in [x_{i-1}, x_i], \\ \frac{x_{i+1} - x}{x_{i+1} - x_i}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise,} \end{cases}$$

since every $v \in V_{0h}$ can be written as $v = \sum_{i=1}^N v(x_i) \phi_i$.

How to approximate the solution of (25) by elements in V_{0h} ? We can give a correspondence between a function $u_h \in V_{0h}$ and a vector $\bar{u}_h = (u_h^1, u_h^2, \dots, u_h^N)^T$ by $u_h = \sum_{i=1}^N u_h^i \phi_i$. Now by the method of (26), we consider the problem: find $u_h \in V_{0h}$ such that

$$a(u_h, v_h) = \ell(v_h), \quad \forall v_h \in V_{0h}. \quad (28)$$

It is equivalent to the following linear system: Find $\bar{u}_h \in \mathbb{R}^N$ such that

$$\mathbb{A} \bar{u}_h = \bar{\ell}, \quad (29)$$

where $\mathbb{A}$ is defined by $\mathbb{A}_{ij} = a(\phi_i, \phi_j)$ and $\bar{\ell}$ is the load vector defined by $\ell_i = \ell(\phi_i)$. More precisely, we have

$$\mathbb{A} = K_{0h} + M_{0h},$$

where

$$K_{0h} = \left(\int_0^L k \phi'_i(x) \phi'_j(x) dx \right)_{1 \leq i, j \leq N},$$

$$M_{0h} = \left(\int_0^L \phi_i(x) \phi_j(x) dx \right)_{1 \leq i, j \leq N}.$$

Note that we have

$$\int_0^L \phi'_j(x) \phi'_i(x) dx = \begin{cases} -\frac{1}{x_{j+1}-x_j}, & \text{if } j = i-1, \\ \frac{1}{x_j-x_{j-1}} + \frac{1}{x_{j+1}-x_j}, & \text{if } j = i, \\ -\frac{1}{x_j-x_{j-1}}, & \text{if } j = i+1, \\ 0, & \text{otherwise.} \end{cases}$$

and

$$\int_0^L \phi_j(x) \phi_i(x) dx = \begin{cases} \frac{x_{j+1}-x_j}{6}, & \text{if } j = i-1, \\ \frac{x_j-x_{j-1}}{3} + \frac{x_{j+1}-x_j}{3}, & \text{if } j = i, \\ \frac{x_j-x_{j-1}}{6}, & \text{if } j = i+1, \\ 0, & \text{otherwise.} \end{cases}$$

If we suppose the partition is uniform, then we can compute K_{0h} and M_{0h} explicitly as follows:

$$K_{0h} = \left(\int_0^L k \phi'_i(x) \phi'_j(x) dx \right)_{1 \leq i, j \leq N} = \frac{k}{h} \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 2 \end{pmatrix},$$

$$M_{0h} = \left(\int_0^L \phi_i(x) \phi_j(x) dx \right)_{1 \leq i, j \leq N} = \frac{h}{6} \begin{pmatrix} 4 & 1 & 0 & \cdots & 0 \\ 1 & 4 & 1 & \cdots & 0 \\ 0 & 1 & 4 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 4 \end{pmatrix}.$$

And the load vector is given by

$$\bar{\ell} = \left(\int_0^L f(x) \phi_i(x) dx \right)_{1 \leq i \leq N}.$$

For practical computations, we can also approximate $\bar{\ell}$ by $\bar{\ell}_h = M_h \bar{f}_h$, where $\bar{f}_h = (f(x_1), f(x_2), \dots, f(x_N))^T$ is the vector of the values of f at the nodes.

As another example, consider the following problem:

$$\begin{cases} -ku''(x) + u(x) = f(x), & \text{in } (0, L), \\ u'(0) = u'(L) = 0. \end{cases} \quad (30)$$

The variational formulation is: find $u \in H^1(0, L)$ such that

$$a(u, v) = \ell(v), \quad \forall v \in H^1(0, L),$$

where

$$a(u, v) = \int_0^L ku'(x)v'(x) dx + \int_0^L u(x)v(x) dx, \quad \forall u, v \in H^1(0, L),$$

$$\ell(v) = \int_0^L f(x)v(x) dx, \quad \forall v \in H^1(0, L).$$

Can we just copy the method of the first problem? The answer is no since $\dim V_h = N + 2$ and a basis of V_h is given by $\{\psi_0, \psi_1, \dots, \psi_{N+1}\}$, where $\psi_i = \phi_i$ for $i = 1, 2, \dots, N$ and ψ_0 and ψ_{N+1} are defined by

$$\psi_0(x) = \begin{cases} 1 - \frac{x}{x_1}, & x \in [0, x_1], \\ 0, & \text{otherwise,} \end{cases} \quad \psi_{N+1}(x) = \begin{cases} 1 - \frac{L-x}{L-x_N}, & x \in [x_N, L], \\ 0, & \text{otherwise.} \end{cases}$$

Note that we still have the correspondence between a function $u_h \in V_h$ and a vector $\bar{u}_h = (u_h^0, u_h^1, \dots, u_h^N, u_h^{N+1})^T$ by $u_h = \sum_{i=0}^{N+1} u_h^i \psi_i$. Hence, we can still consider the problem of linear algebra: find $\bar{u}_h \in \mathbb{R}^{N+2}$ such that

$$\mathbb{A} \bar{u}_h = \bar{\ell},$$

where

$$\mathbb{A} = K_h + M_h,$$

with

$$K_h = \left(\int_0^L k \psi'_i(x) \psi'_j(x) dx \right)_{0 \leq i, j \leq N+1},$$

$$M_h = \left(\int_0^L \psi_i(x) \psi_j(x) dx \right)_{0 \leq i, j \leq N+1}.$$

If we suppose the partition is uniform, then we can compute K_h and M_h explicitly as follows:

$$K_h = \frac{k}{h} \begin{pmatrix} 1 & -1 & \cdots & 0 & 0 \\ -1 & & & & 0 \\ \vdots & & K_{0h} & & \vdots \\ 0 & & & & -1 \\ 0 & \cdots & 0 & -1 & 1 \end{pmatrix},$$

$$M_h = \frac{h}{6} \begin{pmatrix} 2 & 1 & \cdots & 0 & 0 \\ 1 & & & & 0 \\ \vdots & & M_{0h} & & \vdots \\ 0 & & & & 1 \\ 0 & \cdots & 0 & 1 & 2 \end{pmatrix}.$$

Moreover, the load vector is given by

$$\bar{\ell} = \left(\int_0^L f(x) \psi_i(x) dx \right)_{0 \leq i \leq N+1}.$$

It is also possible to approximate $\bar{\ell}$ by $\bar{\ell}_h = M_h \bar{f}_h$, where $\bar{f}_h = (f(x_0), f(x_1), \dots, f(x_N), f(x_{N+1}))^T$ is the vector of the values of f at the nodes.

4.2 2D Case

Definition 4.1. Let Ω be a simply connected polygonal domain in $\mathbb{R}^2$. A triangular mesh of $\bar{\Omega}$ is a set of N_e triangles $\{T_k\}_{1 \leq k \leq N_e}$ such that:

- Each triangle T_k has non-empty interior (i.e., $\text{int}(T_k) \neq \emptyset$). We then define h_k as the radius of the circumscribed circle (circumradius) of the triangle T_k and $h = \max_{1 \leq k \leq N_e} h_k$.
- $\text{int}(T_k) \cap \text{int}(T_{k'}) = \emptyset$ if $k \neq k'$.
- $\bigcup_{k=1}^{N_e} T_k = \bar{\Omega}$.
- Every edge of a triangle is either an edge of another triangle or an edge lying on the boundary $\partial\Omega$.

Notations:

- N_s : Number of vertices
- N_b : Number of vertices on the boundary
- N_e : Number of triangles
- N_a : Number of edges

Proposition 4.2. Let $\{T_k\}_{1 \leq k \leq N_e}$ be a triangular mesh of $\bar{\Omega}$, where Ω is a simply connected polygonal domain. Then we have

$$N_e = 2N_s - N_b - 2,$$

$$N_a = 3N_s - N_b - 3.$$

Given a triangular mesh $\mathcal{T}_h$ of $\bar{\Omega}$, we can define the finite element space V_h as follows:

$$V_h = \{v \in C^0(\bar{\Omega}) : v|_{T_k} \text{ is linear for } k = 1, 2, \dots, N_e\}.$$

Note that $\dim V_h = N_s$ and a basis of V_h is given by $\{\phi_1, \phi_2, \dots, \phi_{N_s}\}$, where ϕ_i is the piecewise linear function such that $\phi_i(x_j) = \delta_{ij}$ for $i, j = 1, 2, \dots, N_s$ where x_j are the vertices of the mesh.

Consider the following variational formulation: find $u \in H^1(\Omega)$ such that

$$a(u, v) = \ell(v), \quad \forall v \in H^1(\Omega), \quad (31)$$

where

$$\begin{aligned} a(u, v) &= \int_{\Omega} (k \nabla u \cdot \nabla v + uv) \, dx + \int_{\partial\Omega} uv \, d\sigma, \quad \forall u, v \in H^1(\Omega), \\ \ell(v) &= \int_{\Omega} f v \, dx + \int_{\partial\Omega} g v \, d\sigma, \quad \forall v \in H^1(\Omega). \end{aligned}$$

We want to find an approximation of u by elements in V_h . Hence, we consider the following problem: find $u_h \in V_h$ such that

$$a(u_h, v_h) = \ell(v_h), \quad \forall v_h \in V_h. \quad (32)$$

Note that we have the correspondence between a function $u_h \in V_h$ and a vector $\bar{u}_h = (u_h^1, u_h^2, \dots, u_h^{N_s})^T$ by $u_h = \sum_{i=1}^{N_s} u_h^i \phi_i$. Hence, we can still consider the problem of linear algebra: find $\bar{u}_h \in \mathbb{R}^{N_s}$ such that

$$\mathbb{A} \bar{u}_h = \bar{\ell},$$

where $\mathbb{A}$ is defined by $\mathbb{A}_{ij} = a(\phi_i, \phi_j)$ and $\bar{\ell}$ is the load vector defined by $\ell_i = \ell(\phi_i)$ for $i = 1, 2, \dots, N_s$. More precisely, we have

$$\mathbb{A} = K_h + M_h + S_h,$$

where

$$\begin{aligned} K_h &= \left(\int_{\Omega} k \nabla \phi_i \cdot \nabla \phi_j \, dx \right)_{1 \leq i, j \leq N_s}, \\ M_h &= \left(\int_{\Omega} \phi_i \phi_j \, dx \right)_{1 \leq i, j \leq N_s}, \\ S_h &= \left(\int_{\partial\Omega} \phi_i \phi_j \, d\sigma \right)_{1 \leq i, j \leq N_s}. \end{aligned}$$

And the load vector is given by

$$\bar{\ell} = \left(\int_{\Omega} f \phi_i \, dx + \int_{\partial\Omega} g \phi_i \, d\sigma \right)_{1 \leq i \leq N_s}.$$

For practical computations, we can also approximate $\bar{\ell}$ by $\bar{\ell}_h = M_h \bar{f}_h + S_h \bar{g}_h$, where $\bar{f}_h = (f(x_1), f(x_2), \dots, f(x_{N_s}))^T$ and $\bar{g}_h = (g(x_k) \mathbf{1}_{x_k \in \partial\Omega})_{1 \leq k \leq N_s}^T$ are the vectors of the values of f and g at the vertices.

The entries of K_h , M_h and S_h can be computed by the following formulas:

$$\begin{aligned} \int_{\Omega} k \nabla \phi_i \cdot \nabla \phi_j \, dx &= \sum_{\ell=1}^{N_e} \int_{T_\ell} k \nabla \phi_i \cdot \nabla \phi_j \, dx, \\ \int_{\Omega} \phi_i \phi_j \, dx &= \sum_{\ell=1}^{N_e} \int_{T_\ell} \phi_i \phi_j \, dx, \\ \int_{\partial\Omega} \phi_i \phi_j \, d\sigma &= \sum_e \int_e \phi_i \phi_j \, d\sigma, \end{aligned}$$

where e is an edge of the mesh. Note that $\phi_i|_{T_k}$ is linear for $k = 1, 2, \dots, N_e$ and $\phi_i|_e$ is linear for $e = 1, 2, \dots, N_a$. Hence, we can compute these integrals by the following formulas:

$$\begin{aligned} \int_{T_\ell} k \nabla \phi_i \cdot \nabla \phi_j \, dx &= \begin{cases} \frac{k}{2} (\cot \theta_{v_1}^\ell + \cot \theta_{v_2}^\ell), & \text{if } i = j \text{ and } x_i, v_1, v_2 \text{ are vertices of } T_\ell, \\ -\frac{k}{2} \cot \theta_v^\ell, & \text{if } i \neq j \text{ and } x_i, x_j, v \text{ are vertices of } T_\ell, \\ 0, & \text{otherwise,} \end{cases} \\ \int_{T_\ell} \phi_i \phi_j \, dx &= \frac{1}{12} |T_\ell| (1 + \delta_{ij}), \text{ if } x_i, x_j \text{ are vertices of } T_\ell, \\ \int_e \phi_i \phi_j \, d\sigma &= \frac{1}{6} |e| (1 + \delta_{ij}), \text{ if } x_i, x_j \text{ are vertices of } e, \end{aligned}$$

where $|T_\ell|$ is the area of the triangle T_ℓ and $|e|$ is the length of the edge e and θ_v^ℓ is the angle at vertex v of triangle T_ℓ .

4.3 Convergence of the Finite Element Method

We now turn to the convergence of the finite element method. Here we consider the case where u is the solution of (26) and u_h is the solution of (28). By Lemma 1.23, we have

$$\|u - u_h\|_{H_0^1} \leq C \inf_{v_h \in V_{0h}} \|u - v_h\|_{H_0^1}.$$

By taking v_h as the piecewise linear interpolation of u on the partition, i.e.,

$$v_h = \Pi_h^1 u := \sum_{i=1}^N u(x_i) \phi_i,$$

we have

$$\begin{aligned} \|u - v_h\|_{H_0^1}^2 &= \int_0^L (u'(x) - v_h'(x))^2 dx \\ &= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} (u'(x) - v_h'(x))^2 dx \\ &= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(u'(x) - \frac{u(x_{i+1}) - u(x_i)}{x_{i+1} - x_i} \right)^2 dx \\ &= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(\frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} (u'(x) - u'(t)) dt \right)^2 dx \\ &\leq \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} (u'(x) - u'(t))^2 dt dx. \end{aligned}$$

Using the fact that $u \in H^2(0, L)$, the following estimate holds:

$$\begin{aligned} \|u - v_h\|_{H_0^1}^2 &\leq \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} (u'(x) - u'(t))^2 dt dx \\ &= \sum_{i=0}^N \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} \int_{x_i}^{x_{i+1}} \left(\int_t^x u''(s) ds \right)^2 dt dx \\ &\leq \sum_{i=0}^N \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} \int_{x_i}^{x_{i+1}} |x - t| \int_{x_i}^{x_{i+1}} (u''(s))^2 ds dt dx \\ &= \sum_{i=0}^N \frac{(x_{i+1} - x_i)^2}{3} \int_{x_i}^{x_{i+1}} (u''(s))^2 ds \\ &\leq \frac{h^2}{3} \|u''\|_{L^2(0, L)}^2. \end{aligned}$$

Hence, we can conclude that $\|u - u_h\|_{H_0^1} \leq Ch \|u''\|_{L^2(0, L)} \rightarrow 0$ as $h \rightarrow 0$.

Now we consider the case where u is the solution of (30), and u_h is still the solution given by the finite element method. Define Π_h^1 as the piecewise linear interpolation operator, i.e.,

$$\Pi_h^1 u = \sum_{i=0}^{N+1} u(x_i) \psi_i.$$

We can show that

$$\begin{aligned} \|u - \Pi_h^1 u\|_{H^1}^2 &= \int_0^L (u'(x) - (\Pi_h^1 u)'(x))^2 dx + \int_0^L (u(x) - \Pi_h^1 u(x))^2 dx \\ &= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} (u'(x) - (\Pi_h^1 u)'(x))^2 dx + \int_0^L (u(x) - \Pi_h^1 u(x))^2 dx. \end{aligned}$$

We now use the fact that $u \in H^2(0, L)$. For the first term, we can use the same method as the first problem to show that

$$\sum_{i=0}^N \int_{x_i}^{x_{i+1}} (u'(x) - (\Pi_h^1 u)'(x))^2 dx \leq \frac{h^2}{3} \|u''\|_{L^2(0, L)}^2.$$

For the second term, we have

$$\begin{aligned}
\int_0^L (u(x) - \Pi_h^1 u(x))^2 dx &= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} (u(x) - \Pi_h^1 u(x))^2 dx \\
&= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(u(x) - u(x_i) - \frac{u(x_{i+1}) - u(x_i)}{x_{i+1} - x_i} (x - x_i) \right)^2 dx \\
&= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(\int_{x_i}^x u'(t) dt - \frac{u(x_{i+1}) - u(x_i)}{x_{i+1} - x_i} (x - x_i) \right)^2 dx \\
&\leq \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(\int_{x_i}^x \left| u'(t) - \frac{u(x_{i+1}) - u(x_i)}{x_{i+1} - x_i} \right| dt \right)^2 dx \\
&= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(\int_{x_i}^x \frac{1}{x_{i+1} - x_i} \left| \int_{x_i}^{x_{i+1}} (u'(t) - u'(s)) ds \right| dt \right)^2 dx \\
&\leq \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(\int_{x_i}^x \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} |u'(t) - u'(s)| ds dt \right)^2 dx \\
&= \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(\int_{x_i}^x \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} \left| \int_s^t u''(\tau) d\tau \right| ds dt \right)^2 dx \\
&\leq \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \left(\int_{x_i}^x \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} \left(|t - s| \int_{x_i}^{x_{i+1}} |u''(\tau)|^2 d\tau \right)^{\frac{1}{2}} ds dt \right)^2 dx \\
&\leq \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \|u''\|_{L^2(x_i, x_{i+1})}^2 \left(\int_{x_i}^x \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} |t - s|^{\frac{1}{2}} ds dt \right)^2 dx \\
&\leq \sum_{i=0}^N \int_{x_i}^{x_{i+1}} \|u''\|_{L^2(x_i, x_{i+1})}^2 \left(\int_{x_i}^{x_{i+1}} \int_{x_i}^{x_{i+1}} \frac{1}{(x_{i+1} - x_i)^{\frac{1}{2}}} ds dt \right)^2 dx \\
&= \sum_{i=0}^N (x_{i+1} - x_i)^4 \|u''\|_{L^2(x_i, x_{i+1})}^2 \\
&\leq h^4 \|u''\|_{L^2(0, L)}^2.
\end{aligned}$$

Hence we have

$$\|u - \Pi_h^1 u\|_{H^1} \leq Ch \|u''\|_{L^2(0, L)}.$$

Since $u'' = \frac{1}{k}(u - f)$, we have

$$\begin{aligned}
\|u''\|_{L^2(0, L)} &\leq \frac{1}{k} (\|u\|_{L^2(0, L)} + \|f\|_{L^2(0, L)}) \\
&\leq \frac{1}{k} (\|u\|_{H^1(0, L)} + \|f\|_{L^2(0, L)}).
\end{aligned}$$

Note that by the variational formulation, we have

$$\begin{aligned}
\|f\|_{L^2(0, L)} \|u\|_{H^1(0, L)} &\geq \|f\|_{L^2(0, L)} \|u\|_{L^2(0, L)} \\
&\geq \ell(u) = a(u, u) \\
&\geq \min\{1, k\} \|u\|_{H^1(0, L)}^2.
\end{aligned}$$

Finally we have

$$\|u - \Pi_h^1 u\|_{H^1} \leq Ch \|f\|_{L^2(0, L)}.$$

5 Spectral Analysis

Consider the following problem:

$$\begin{cases} -\Delta u = \lambda u, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

The associated variational formulation is: find $u \in H_0^1(\Omega)$ and $\lambda \in \mathbb{R}$ such that

$$a(u, v) = \lambda(u, v), \quad \forall v \in H_0^1(\Omega),$$

where

$$\begin{aligned} a(u, v) &= \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad \forall u, v \in H_0^1(\Omega), \\ (u, v) &= \int_{\Omega} uv \, dx, \quad \forall u, v \in H_0^1(\Omega). \end{aligned}$$

By Riesz representation theorem, we can define the linear operator $A : H_0^1(\Omega) \rightarrow (H_0^1(\Omega))'$ by

$$\forall u, v \in H_0^1(\Omega), \quad a(u, v) = (Au, v).$$

Then we can rewrite the variational formulation as: find $u \in H_0^1(\Omega)$ and $\lambda \in \mathbb{R}$ such that

$$(Au, v) = \lambda(u, v), \quad \forall v \in H_0^1(\Omega).$$

Actually we can consider a more general problem: suppose H is a separable Hilbert space and V is a subspace of H such that V is dense in H and the embedding from V to H is compact. Let a be a symmetric continuous coercive bilinear form on V . We want to find $u \in V \setminus \{0\}$ and $\lambda \in \mathbb{R}$ such that

$$a(u, v) = \lambda(u, v)_H, \quad \forall v \in V. \quad (33)$$

Since $a(u, \cdot)$ is a linear form on V , by Riesz representation theorem, there exists a unique linear operator $A : V \rightarrow V'$ such that

$$\forall v \in V, \quad a(u, v) = (Au, v)_{V', V}.$$

Hence we can rewrite (33) as: find $u \in V \setminus \{0\}$ and $\lambda \in \mathbb{R}$ such that $(Au, v)_{V', V} = \lambda(u, v)_H, \forall v \in V$. Since $u \in V \subseteq H$ and $Au \in V' \subseteq H' = H$, this is also equivalent to find $u \in V \setminus \{0\}$ and $\lambda \in \mathbb{R}$ such that $Au = \lambda u$ in H .

Theorem 5.1. *Assuming the conditions above hold, then there exists a sequence of eigenvalues $\{\lambda_n\}_{n=1}^{\infty}$ such that $\lambda_n \rightarrow +\infty$ as $n \rightarrow +\infty$ and there exists an orthonormal basis $\{e_n\}_{n=1}^{\infty}$ of H such that $Ae_n = \lambda_n e_n$ for all $n \geq 1$.*

Proof. Since a is symmetric, continuous and coercive on V , the solution operator $T = A^{-1} : H \rightarrow H$ is well defined, self-adjoint and positive; moreover, because the embedding $V \hookrightarrow H$ is compact, T is compact. Applying the spectral theorem for compact self-adjoint operators to T yields an orthonormal basis $\{e_n\}_{n=1}^{\infty}$ of H consisting of eigenvectors, with $Te_n = \mu_n e_n$ and $\mu_n \rightarrow 0^+$. Setting $\lambda_n = 1/\mu_n$ gives $Ae_n = \lambda_n e_n$ with $\lambda_n \rightarrow +\infty$. $\square$

Example 5.2. $H = L^2(0, L)$, $V = H_0^1(0, L)$, $a(u, v) = \int_0^L u'(x)v'(x) \, dx$. Then

$$e_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, \dots$$

If H is finite-dimensional, then we can write A as a matrix. Since a is symmetric and coercive, we can show that A is a symmetric positive definite matrix. Moreover, we can define the Rayleigh quotient

$$R_A(u) = \frac{a(u, u)}{(u, u)_H}, \quad \forall u \in H \setminus \{0\}.$$

In the rest of this section, we assume H is finite-dimensional. We have the following min-max principle for the eigenvalues of A .

Theorem 5.3 (min-max principle). *Let $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ be the eigenvalues of A . Then we have*

$$\lambda_k = \min_{W \in S_k} \max_{u \in W \setminus \{0\}} R_A(u) = \max_{W \in S_{n-k+1}} \min_{u \in W \setminus \{0\}} R_A(u), \quad k = 1, 2, \dots, n,$$

where S_k is the set of all k -dimensional subspaces of V .

Proof. Let $\{u_1, \dots, u_n\}$ be an orthonormal basis of H consisting of eigenvectors of A : $Au_i = \lambda_i u_i$ with $\lambda_1 \leq \dots \leq \lambda_n$. Consider $W_k = \text{span}\{u_1, \dots, u_k\}$. For $u = \sum_{i=1}^k c_i u_i \in W_k$, we have

$$R_A(u) = \frac{\sum_{i=1}^k \lambda_i c_i^2}{\sum_{i=1}^k c_i^2} \leq \lambda_k,$$

hence $\max_{u \in W_k} R_A(u) = \lambda_k$. On the other hand, let W be any k -dimensional subspace. Since $\dim W = k$ and $\dim \text{span}\{u_k, \dots, u_n\} = n - k + 1$, there exists a non-zero $u \in W \cap \text{span}\{u_k, \dots, u_n\}$. Write $u = \sum_{i=k}^n c_i u_i$; then

$$R_A(u) = \frac{\sum_{i=k}^n \lambda_i c_i^2}{\sum_{i=k}^n c_i^2} \geq \lambda_k.$$

Consequently,

$$\max_{u \in W} R_A(u) \geq \lambda_k.$$

Taking the minimum over all k -dimensional subspaces W , we obtain

$$\min_{W \in S_k} \max_{u \in W} R_A(u) = \lambda_k.$$

The second equality follows by applying the same argument to the operator $-A$. $\square$

Given a finite-dimensional subspace V_h of V , we can consider the following problem: find $u_h \in V_h \setminus \{0\}$ and $\lambda_h \in \mathbb{R}$ such that

$$a(u_h, v_h) = \lambda_h (u_h, v_h)_H, \quad \forall v_h \in V_h.$$

Theorem 5.4. Suppose $\dim(V_h) = N$. Let $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ be the eigenvalues of A and $\lambda_{1,h} \leq \lambda_{2,h} \leq \dots \leq \lambda_{N,h}$ be the eigenvalues of the problem above. Then we have

$$0 < \lambda_k \leq \lambda_{k,h}, \quad k = 1, 2, \dots, N.$$

And there exists a basis $\{e_{k,h}\}_{k=1}^N$ of V_h , orthonormal for the inner product $(\cdot, \cdot)_H$, consisting of the eigenfunctions of the discrete problem, i.e., $a(e_{k,h}, v_h) = \lambda_{k,h} (e_{k,h}, v_h)_H$ for all $v_h \in V_h$ and $k = 1, 2, \dots, N$.

Proof. By the min-max principle applied to A on V and to its restriction on V_h , we have

$$\lambda_k = \min_{\substack{W \subset V \\ \dim W = k}} \max_{u \in W \setminus \{0\}} R_A(u), \quad \lambda_{k,h} = \min_{\substack{W \subset V_h \\ \dim W = k}} \max_{u \in W \setminus \{0\}} R_A(u).$$

Hence, we obtain $\lambda_{k,h} \geq \lambda_k$. The existence of an orthonormal basis of eigenvectors in V_h follows from the orthogonal diagonalization of symmetric matrices. $\square$

6 Exercises

We say that a bounded open set Ω is regular if its boundary $\partial\Omega$ is C^1 . And we denote by ν the unit outer normal vector at the boundary $\partial\Omega$.

Problem 1. Let Ω be a bounded domain in $\mathbb{R}^n$ with C^1 boundary. Let $f \in C^0(\overline{\Omega})$, $g \in C^0(\partial\Omega)$, $\alpha > 0$ and $k \in C^1(\overline{\Omega})$ be given with $k > 0$, we consider the following problem:

$$\begin{cases} -\operatorname{div}(k(x)\nabla u(x)) = f(x), & \text{in } \Omega, \\ k(x)\frac{\partial u}{\partial \nu}(x) + \alpha u(x) = g(x), & \text{on } \partial\Omega. \end{cases} \quad (\star)$$

Show that if $u \in C^1(\overline{\Omega})$ is a solution of $(\star)$, then u satisfies the variational formulation

$$\forall v \in C^1(\overline{\Omega}), \quad a(u, v) = \ell(v),$$

with

- $a : C^1(\overline{\Omega}) \times C^1(\overline{\Omega}) \rightarrow \mathbb{R}$ a symmetric and positive bilinear form;
- $\ell : C^1(\overline{\Omega}) \rightarrow \mathbb{R}$ a linear form on $C^1(\overline{\Omega})$.

Solution 1. Let $u \in C^1(\overline{\Omega})$ be a solution of $(\star)$. For any $v \in C^1(\overline{\Omega})$, we multiply the first equation of $(\star)$ by v and integrate over Ω :

$$\int_{\Omega} -\operatorname{div}(k(x)\nabla u(x))v(x) \, dx = \int_{\Omega} f(x)v(x) \, dx.$$

Using the third item of Corollary 1.4, we have

$$\int_{\Omega} k(x)\nabla u(x) \cdot \nabla v(x) \, dx - \int_{\partial\Omega} k(x)\frac{\partial u}{\partial \nu}(x)v(x) \, d\sigma = \int_{\Omega} f(x)v(x) \, dx.$$

Using the boundary condition, we get

$$\int_{\Omega} k(x)\nabla u(x) \cdot \nabla v(x) \, dx - \int_{\partial\Omega} (g(x) - \alpha u(x))v(x) \, d\sigma = \int_{\Omega} f(x)v(x) \, dx.$$

Hence, we can suppose

$$\begin{aligned} a(u, v) &= \int_{\Omega} k(x)\nabla u(x) \cdot \nabla v(x) \, dx + \alpha \int_{\partial\Omega} u(x)v(x) \, d\sigma, \\ \ell(v) &= \int_{\Omega} f(x)v(x) \, dx + \int_{\partial\Omega} g(x)v(x) \, d\sigma. \end{aligned}$$

Problem 2. (Quadratic Energies)

1. Let E be a real vector space. Consider a symmetric positive bilinear form a on E , a linear form ℓ on E , and the energy functional $F : E \rightarrow \mathbb{R}$ defined by

$$\forall x \in E, \quad F(x) = \frac{1}{2}a(x, x) - \ell(x).$$

Show that $x_* \in E$ is a minimizer of F if and only if

$$\forall y \in E, \quad a(x_*, y) = \ell(y).$$

2. Assume that $E = \mathbb{R}^d$. In this case, there exists a symmetric positive matrix $A \in M_d(\mathbb{R})$ and a vector $z \in \mathbb{R}^d$ such that

$$F(x) = \frac{1}{2}x^T A x - x^T z.$$

Determine the equation satisfied by the minimizers of F on $\mathbb{R}^d$. Under what condition is there a unique minimizer?

Solution 2. 1. Let $x_* \in E$ be a minimizer of F . For any $y \in E$ and $t \in \mathbb{R}$, we have

$$F(x_* + ty) \geq F(x_*).$$

Expanding both sides, we get

$$\frac{1}{2}a(x_* + ty, x_* + ty) - \ell(x_* + ty) \geq \frac{1}{2}a(x_*, x_*) - \ell(x_*).$$

Simplifying, we obtain

$$ta(x_*, y) + \frac{t^2}{2}a(y, y) - t\ell(y) \geq 0.$$

Dividing by $t > 0$ and letting $t \rightarrow 0^+$, we find

$$a(x_*, y) - \ell(y) \geq 0.$$

Dividing by $t < 0$ and letting $t \rightarrow 0^-$, we find

$$a(x_*, y) - \ell(y) \leq 0.$$

Thus, we conclude that

$$a(x_*, y) = \ell(y).$$

Conversely, suppose that $x_* \in E$ satisfies

$$a(x_*, y) = \ell(y), \quad \forall y \in E.$$

For any $x \in E$, we have

$$\begin{aligned} F(x) - F(x_*) &= \frac{1}{2}a(x, x) - \ell(x) - \left(\frac{1}{2}a(x_*, x_*) - \ell(x_*) \right) \\ &= \frac{1}{2}a(x - x_*, x - x_*) + a(x_*, x - x_*) - \ell(x - x_*) \\ &= \frac{1}{2}a(x - x_*, x - x_*). \end{aligned}$$

Since a is positive, we have $F(x) - F(x_*) \geq 0$, which shows that x_* is a minimizer of F .

- By identifying $a(x, y) = x^T A y$ and $\ell(y) = y^T z$, the equation satisfied by the minimizers of F on $\mathbb{R}^d$ is

$$Ax_* = z.$$

Since A is symmetric positive, it is invertible if and only if it is positive definite. Thus, there is a unique minimizer if and only if A is positive definite.

Problem 3 (Euler–Lagrange Equation). Let Ω be a bounded regular open subset of $\mathbb{R}^d$. Consider the space

$$E := \{u \in C^1(\bar{\Omega}) \mid u = 0 \text{ on } \partial\Omega\}.$$

Let $k \in C^1(\bar{\Omega})$ be a strictly positive function and let $f \in C^0(\bar{\Omega})$. Define the energy functional $F : E \rightarrow \mathbb{R}$ such that

$$\forall u \in E, \quad F(u) = \frac{1}{2} \int_{\Omega} k(x) |\nabla u(x)|^2 dx - \int_{\Omega} f(x) u(x) dx.$$

- Determine the variational problem $(\mathcal{P}_V)$ satisfied by the minimizers of F on E .
- Show that $(\mathcal{P}_V)$ has at most one solution; in other words, if u_* is a minimizer of F , then it is unique.
- Assume that there exists $u_* \in E$, solution of the variational problem $(\mathcal{P}_V)$, such that $k\nabla u_* \in C^1(\bar{\Omega})^d$. Determine the partial differential equation satisfied by u_* . This equation is called the **Euler–Lagrange equation** associated with the minimization of F on E .
- Conversely, assume that there exists $u_* \in E$, solution of the previous Euler–Lagrange equation, such that $k\nabla u_* \in C^1(\bar{\Omega})^d$. Prove that u_* is the minimizer of F .
- Given $u_0 \in C^1(\bar{\Omega})$, replace the space E by $\tilde{E} := \{u \in C^1(\bar{\Omega}) \mid u = u_0 \text{ on } \partial\Omega\}$. What changes must be made to the previous analysis?
- Let now K be a positive matrix in $\mathbb{R}^{2 \times 2}$, i.e., such that $Kx \cdot x \geq 0$ for every $x \in \mathbb{R}^2$. Define $G : E \rightarrow \mathbb{R}$ such that

$$\forall u \in E, \quad G(u) = \frac{1}{2} \int_{\Omega} K \nabla u(x) \cdot \nabla u(x) dx - \int_{\Omega} f(x) u(x) dx.$$

Write the variational problem satisfied by the minimizers of G on E .

Solution 3. 1. Using the conclusion from the previous problem, the variational problem $(\mathcal{P}_V)$ satisfied by the minimizers of F on E is given by

$$\forall v \in E, \quad \int_{\Omega} k(x) \nabla u_*(x) \cdot \nabla v(x) \, dx = \int_{\Omega} f(x) v(x) \, dx.$$

2. Suppose that there exist two minimizers $u_1, u_2 \in E$ of F . Then, for any $v \in E$, we have

$$\int_{\Omega} k(x) \nabla u_1(x) \cdot \nabla v(x) \, dx = \int_{\Omega} f(x) v(x) \, dx,$$

and

$$\int_{\Omega} k(x) \nabla u_2(x) \cdot \nabla v(x) \, dx = \int_{\Omega} f(x) v(x) \, dx.$$

Subtracting these two equations, we get

$$\int_{\Omega} k(x) \nabla(u_1 - u_2)(x) \cdot \nabla v(x) \, dx = 0.$$

Taking $v = u_1 - u_2$, we find

$$\int_{\Omega} k(x) |\nabla(u_1 - u_2)(x)|^2 \, dx = 0.$$

Since $k > 0$, this implies that $\nabla(u_1 - u_2) = 0$. Note that $u_1|_{\partial\Omega} = u_2|_{\partial\Omega} = 0$, i.e., $u_1 = u_2$. Hence, the minimizer is unique.

3. Applying the third item of Corollary 1.4, we have

$$\int_{\Omega} k(x) \nabla u_*(x) \cdot \nabla v(x) \, dx = - \int_{\Omega} \operatorname{div}(k(x) \nabla u_*(x)) v(x) \, dx + \int_{\partial\Omega} k(x) \frac{\partial u_*}{\partial \nu}(x) v(x) \, d\sigma.$$

Since $v \in E$ vanishes on $\partial\Omega$, the boundary term disappears. Thus, we have

$$\int_{\Omega} \operatorname{div}(k(x) \nabla u_*(x)) v(x) \, dx = - \int_{\Omega} k(x) \nabla u_*(x) \cdot \nabla v(x) \, dx = - \int_{\Omega} f(x) v(x) \, dx.$$

Since this holds for all $v \in E$, we conclude that

$$-\operatorname{div}(k(x) \nabla u_*(x)) = f(x) \text{ in } \Omega.$$

4. We can use the same calculation as in the previous item to show that u_* satisfies the variational problem $(\mathcal{P}_V)$. Since the minimizer is unique, u_* is the minimizer of F .

5. Replacing E by $\tilde{E}$, we can define a new function $w = u - u_0$ for $u \in \tilde{E}$. Then, $w \in E$ since $w = 0$ on $\partial\Omega$. The energy functional can be rewritten as

$$F(u) = \frac{1}{2} \int_{\Omega} k(x) |\nabla(w(x) + u_0(x))|^2 \, dx - \int_{\Omega} f(x) (w(x) + u_0(x)) \, dx.$$

The variational problem and the Euler–Lagrange equation can be derived similarly by considering w instead of u .

6. Note that

$$Kx \cdot x = x^\top Kx = (x^\top Kx)^\top = x^\top K^\top x = K^\top x \cdot x.$$

Thus, for every $x \in \mathbb{R}^2$, we have

$$\begin{aligned} G(u) &= \frac{1}{2} \int_{\Omega} K \nabla u(x) \cdot \nabla u(x) \, dx - \int_{\Omega} f(x) u(x) \, dx \\ &= \frac{1}{2} \int_{\Omega} K^\top \nabla u(x) \cdot \nabla u(x) \, dx - \int_{\Omega} f(x) u(x) \, dx \\ &= \frac{1}{2} \int_{\Omega} \left(\frac{K + K^\top}{2} \right) \nabla u(x) \cdot \nabla u(x) \, dx - \int_{\Omega} f(x) u(x) \, dx. \end{aligned}$$

Therefore, by the conclusion of the previous problem, the variational problem satisfied by the minimizers of G on E is given by

$$\forall v \in E, \quad \int_{\Omega} \left(\frac{K + K^\top}{2} \right) \nabla u_*(x) \cdot \nabla v(x) \, dx = \int_{\Omega} f(x) v(x) \, dx.$$

Problem 4. (Riesz Theorem in $L^2(\mathbb{R}^d)$) Let $K \in L^1(\mathbb{R}^d)$ and $f \in L^2(\mathbb{R}^d)$. Assume that K is even and $\|K\|_{L^1} < 1$. We seek $u \in L^2(\mathbb{R}^d)$ that minimizes the energy

$$E(u) := \frac{1}{2} \int_{\mathbb{R}^d \times \mathbb{R}^d} K(x-y)u(x)u(y) \, dx \, dy + \frac{1}{2} \int_{\mathbb{R}^d} u(x)^2 \, dx - \int_{\mathbb{R}^d} f(x)u(x) \, dx.$$

Show that the energy is well defined. Show that E admits a unique minimizer u_* and that moreover the map $f \mapsto u_*$ is a continuous linear map from $L^2(\mathbb{R}^d)$ to itself. Show that u_* is the unique solution of the functional equation

$$u_* + K * u_* = f, \quad \text{where} \quad (K * u)(x) := \int_{\mathbb{R}^d} K(x-y)u(y) \, dy.$$

Solution 4. Since $u \in L^2(\mathbb{R}^d)$ and $f \in L^2(\mathbb{R}^d)$, we only need to show that the first term of $E(u)$ is well defined. By Cauchy-Schwarz inequality, we have

$$\begin{aligned} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} K(x-y)u(x)u(y) \, dx \, dy \right)^2 &\leq \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} |K(x-y)||u(x)|^2 \, dx \, dy \right) \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} |K(x-y)||u(y)|^2 \, dx \, dy \right) \\ &\leq \|K\|_{L^1}^2 \cdot \|u\|_{L^2}^4 < \infty. \end{aligned}$$

Moreover, we have

$$E(u) = \frac{1}{2}a(u, u) - \ell(u),$$

where

$$\begin{aligned} a(u, v) &= \int_{\mathbb{R}^d \times \mathbb{R}^d} K(x-y)u(x)v(y) \, dx \, dy + \int_{\mathbb{R}^d} u(x)v(x) \, dx, \\ \ell(u) &= \int_{\mathbb{R}^d} f(x)u(x) \, dx. \end{aligned}$$

Since K is even, the bilinear form a is symmetric.

Existence of u_* : we suppose u_* is the unique solution of variational problem: find $u \in L^2(\mathbb{R}^d)$ such that

$$\forall v \in L^2(\mathbb{R}^d), \quad a(u, v) = \ell(v).$$

It is easy to check that a is coercive since $\|K\|_{L^1} < 1$. Hence, by the Lax-Milgram theorem, there exists a unique solution u_* of the variational problem above. Then for all $t \in \mathbb{R}$ and $v \in L^2(\mathbb{R}^d)$, we have

$$\begin{aligned} E(u_* + tv) &= \frac{1}{2}a(u_* + tv, u_* + tv) - \ell(u_* + tv) \\ &= \frac{1}{2}a(u_*, u_*) - \ell(u_*) + t(a(u_*, v) - \ell(v)) + \frac{t^2}{2}a(v, v) \\ &= E(u_*) + \frac{t^2}{2}a(v, v) \geq E(u_*). \end{aligned}$$

Hence u_* is a minimizer of E .

Uniqueness of the minimizer: if there exists another minimizer u'_* , then we have

$$\begin{aligned} E(u'_*) &= \frac{1}{2}a(u'_*, u'_*) - \ell(u'_*) \\ &= \frac{1}{2}a(u'_*, u'_*) - a(u'_*, u_*) + \frac{1}{2}a(u_*, u_*) + E(u_*) \\ &= \frac{1}{2}a(u'_* - u_*, u'_* - u_*) + E(u_*) \geq E(u_*). \end{aligned}$$

Since u'_* is also a minimizer, we have $E(u'_*) = E(u_*)$, which implies that $a(u'_* - u_*, u'_* - u_*) = 0$. Hence, we conclude that $u'_* = u_*$.

Suppose $u_{*,f}$ is the unique minimizer of E_f . Then for all $f, g \in L^2(\mathbb{R}^d)$, we have

$$\begin{aligned} a(u_{*,f}, v) &= \ell_f(v) = \int_{\mathbb{R}^d} f(x)v(x) \, dx, \\ a(u_{*,g}, v) &= \ell_g(v) = \int_{\mathbb{R}^d} g(x)v(x) \, dx. \end{aligned}$$

Hence, we have

$$\begin{aligned} a(u_{*,f} - u_{*,g}, v) &= \int_{\mathbb{R}^d} (f(x) - g(x))v(x) \, dx \\ &\leq \|f - g\|_{L^2} \cdot \|v\|_{L^2}. \end{aligned}$$

Since a is coercive, we have $\|u_{*,f} - u_{*,g}\|_{L^2} \leq C\|f - g\|_{L^2}$ for some constant $C > 0$. Hence, the map $f \mapsto u_{*,f}$ is continuous from $L^2(\mathbb{R}^d)$ to itself.

Finally, for all $v \in L^2(\mathbb{R}^d)$, we have

$$\begin{aligned} a(u_*, v) &= \int_{\mathbb{R}^d} u_*(x)v(x) \, dx + \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} K(x-y)u_*(y) \, dy \right) v(x) \, dx \\ &= \int_{\mathbb{R}^d} (u_*(x) + (K * u_*)(x)) v(x) \, dx \\ &= \ell(v) = \int_{\mathbb{R}^d} f(x)v(x) \, dx, \end{aligned}$$

which implies that $u_* + K * u_* = f$ in $L^2(\mathbb{R}^d)$.

Remark. Another method is to check that a defines a scalar product on $L^2(\mathbb{R}^d)$ which is equivalent to the standard scalar product, and then to apply Riesz representation theorem to conclude.

Problem 5. Let V be a real vector space, a a bilinear form from $V \times V$ to $\mathbb{R}$, and ℓ a linear form on V . We are interested in the variational problem

$$\text{Find } u \in V \text{ such that, } a(u, v) = \ell(v), \quad \forall v \in V. \quad (\dagger)$$

Finite Dimension.

1. Suppose in this question that $V = \mathbb{R}^d$. Show that if a is positive definite, then $(\dagger)$ has a unique solution. Prove that a is positive definite if and only if a is coercive.

Infinite Dimension.

Let $\Omega \subset \mathbb{R}^d$ be a bounded regular open set.

2. Assume $V = C^1(\overline{\Omega})$ and define a such that

$$a(u, v) = \int_{\Omega} (\nabla u \cdot \nabla v + uv) \, dx, \quad \forall u, v \in V.$$

Show that a defines an inner product on V . Deduce that problem $(\dagger)$ has at most one solution. Assume ℓ is continuous on V with respect to the norm associated with the inner product defined by a . Prove that V equipped with the norm associated with the inner product a is not complete, which prevents the use of the Lax-Milgram theorem to conclude the existence of a solution to $(\dagger)$.

From Infinite Dimension to Finite Dimension

3. Consider a linearly independent family of functions $(\phi_n)_{1 \leq n \leq N} \in C^1(\overline{\Omega})$ and let $V = \text{span}\{\phi_n\}_{1 \leq n \leq N}$. Show that solving $(\dagger)$ amounts to inverting a matrix, the coefficients of which should be given.

Solution 5. 1. Consider the matrix A associated with the bilinear form a in the canonical basis of $\mathbb{R}^d$. If A is positive definite, then $x^T A x \geq \lambda_1 \|x\|^2$, where λ_1 is the smallest eigenvalue of A . Therefore a is coercive. Conversely, if a is coercive, then there exists $\alpha > 0$ such that $x^T A x \geq \alpha \|x\|^2$ for all $x \in \mathbb{R}^d$, which implies that A is positive definite. By the Lax-Milgram theorem, there exists a unique solution $u \in \mathbb{R}^d$ of $(\dagger)$ when a is positive definite.

2. The form $a(u, v) = \int_{\Omega} (\nabla u \cdot \nabla v + uv) \, dx$ is symmetric, bilinear, and $a(u, u) = 0$ iff $u = 0$. Hence it defines an inner product on $C^1(\overline{\Omega})$. If a solution exists, it is unique because $a(u_1 - u_2, v) = 0$ for all $v \in V$ forces $u_1 = u_2$ since we can take $v = u_1 - u_2$. The space $C^1(\overline{\Omega})$ equipped with this norm is not complete in general. For example, take $\Omega = (-1, 1)$ and $u_n(x) = \sqrt{x^2 + 1/n}$. Each $u_n \in C^1([-1, 1])$, the sequence converges in H^1 to $u(x) = |x|$, which is not C^1 . Therefore V is not a Hilbert space.

3. Let $\{\phi_1, \dots, \phi_N\}$ be linearly independent and set $V_N = \text{span}\{\phi_n\}$. Write $u = \sum_{j=1}^N \alpha_j \phi_j$. The variational problem $a(u, \phi_i) = \ell(\phi_i)$ for $i = 1, \dots, N$ becomes

$$\sum_{j=1}^N \alpha_j a(\phi_j, \phi_i) = \ell(\phi_i), \quad i = 1, \dots, N.$$

This is a linear system $M\alpha = b$ with $M_{ij} = a(\phi_j, \phi_i)$ and $b_i = \ell(\phi_i)$. The matrix M is symmetric positive definite (because a is positive definite on V_N), hence invertible. Thus there exists a unique solution $u \in V_N$.

Problem 6. (Revisiting Coercivity) Let $\Omega \subset \mathbb{R}^d$ be a bounded regular open set. Consider the bilinear form

$$a(u, v) = \int_{\Omega} \alpha \nabla u \cdot \nabla v + \beta uv \, dx, \quad \forall u, v \in H^1(\Omega),$$

where α and β are strictly positive functions in $C^0(\overline{\Omega})$. We are interested in the variational problem

$$\text{Find } u \in H^1(\Omega) \text{ such that } a(u, v) = \ell(v), \quad \forall v \in H^1(\Omega). \quad (\diamond)$$

1. Show that for any linear continuous form ℓ on $H^1(\Omega)$ (equipped with its usual inner product), problem $(\diamond)$ has a unique solution that depends continuously on ℓ .
2. Now define the form $b(\cdot, \cdot)$ such that

$$b(u, v) = \int_{\Omega} uv \, dx, \quad \forall u, v \in H^1(\Omega).$$

Show that $b(\cdot, \cdot)$ is continuous and positive definite but not coercive on $H^1(\Omega)$.

Solution 6. 1. Since α and β are strictly positive and $\overline{\Omega}$ is compact, there exists $0 < m \leq M$ such that $m \leq \alpha(x), \beta(x) \leq M$ for all $x \in \overline{\Omega}$. Hence, we have

$$m\|u\|_{H^1}^2 \leq a(u, u) \leq M\|u\|_{H^1}^2,$$

i.e., a is coercive and continuous on $H^1(\Omega)$. By the Lax-Milgram theorem, there exists a unique solution $u_{\ell} \in H^1(\Omega)$ of $(\diamond)$ and $\|u_{\ell}\|_{H^1} \leq \frac{1}{m}\|\ell\|_{(H^1)'}.$

2. We take $u \in C_c^1(\mathbb{R}^d) \setminus \{0\}$. Then, we fix $x_0 \in \Omega$, and for every $n \in \mathbb{N}_{>0}$, we define

$$u_n(x) = n^{d/2-1}u(n(x - x_0)).$$

Then, we have

$$b(u_n, u_n) = \|u_n\|_{L^2}^2 = \int_{\mathbb{R}^d} n^{d-2}|u(n(x - x_0))|^2 \, dx = \frac{1}{n^2} \int_{\mathbb{R}^d} |u(y)|^2 \, dy.$$

However, we have

$$\begin{aligned} \|u_n\|_{H^1}^2 &= \int_{\mathbb{R}^d} n^{d-2}|u(n(x - x_0))|^2 \, dx + \int_{\mathbb{R}^d} n^d |\nabla u(n(x - x_0))|^2 \, dx \\ &= \frac{1}{n^2} \int_{\mathbb{R}^d} |u(y)|^2 \, dy + \int_{\mathbb{R}^d} |\nabla u(y)|^2 \, dy. \end{aligned}$$

Hence, $b(u_n, u_n)/\|u_n\|_{H^1}^2 \rightarrow 0$, which implies that b is not coercive on $H^1(\Omega)$. However, for every $u, v \in H^1(\Omega) \setminus \{0\}$, we have $b(u, u) = \|u\|_{L^2}^2 > 0$ and $b(u, v) \leq \|u\|_{L^2}\|v\|_{L^2} \leq \|u\|_{H^1}\|v\|_{H^1}$, which implies that b is continuous and positive definite on $H^1(\Omega)$.

Problem 7. (From Riesz to Lax-Milgram) Let $\Omega \subset \mathbb{R}^d$ be a bounded regular open set and $\underline{v} \in (C^1(\overline{\Omega}))^d$ a given divergence-free vector field such that $\underline{v} \cdot \underline{\nu} = 0$ on $\partial\Omega$. We are interested in the problem

$$\text{Find } u \in H^1(\Omega) \text{ such that } \begin{cases} -\Delta u + (\underline{v} \cdot \nabla u) + u = f & \text{in } \Omega, \\ \frac{\partial u}{\partial \underline{\nu}} = 0 & \text{on } \partial\Omega, \end{cases} \quad (\diamond\diamond)$$

where f is a source term belonging to $L^2(\Omega)$.

1. Write the variational formulation (VF) naturally associated with this problem in a space to be specified. Show that if u is a solution of (VF) belonging to $H^2(\Omega)$, then u satisfies $(\diamond\diamond)$.
2. Can we use the Riesz representation theorem to show that (VF) has a unique solution? What other tool should be used? Prove that the mapping

$$A : L^2(\Omega) \rightarrow H^1(\Omega), \quad f \mapsto u$$

where u is the solution of (VF), is linear and continuous.

3. Would we have obtained an identical result if the divergence of $\underline{v}$ admitted strictly positive values?

4. Let V_h be a finite-dimensional subspace of the space introduced in question 1. Show that there exists a unique solution to the problem

$$\text{Find } u_h \in V_h \text{ such that } \int_{\Omega} \nabla u_h \cdot \nabla w_h + (\underline{v} \cdot \nabla u_h) w_h + u_h w_h \, dx = \int_{\Omega} f w_h \, dx, \quad \forall w_h \in V_h.$$

Solution 7. 1. The variational formulation (VF) associated with problem ($\diamond$) is given by

$$\text{Find } u \in H^1(\Omega) \text{ such that } \int_{\Omega} \nabla u \cdot \nabla w + (\underline{v} \cdot \nabla u) w + u w \, dx = \int_{\Omega} f w \, dx, \quad \forall w \in H^1(\Omega).$$

If u is a solution of (VF) belonging to $H^2(\Omega)$, then for all $w \in H^1(\Omega)$, we have

$$\begin{aligned} \int_{\Omega} \nabla u \cdot \nabla w + (\underline{v} \cdot \nabla u) w + u w \, dx &= - \int_{\Omega} \Delta u w \, dx + \int_{\partial\Omega} \frac{\partial u}{\partial \nu} w |_{\partial\Omega} \, d\sigma + \int_{\Omega} (\underline{v} \cdot \nabla u) w \, dx + \int_{\Omega} u w \, dx \\ &= \int_{\Omega} (-\Delta u + (\underline{v} \cdot \nabla u) + u) w \, dx + \int_{\partial\Omega} \frac{\partial u}{\partial \nu} w |_{\partial\Omega} \, d\sigma. \end{aligned}$$

Taking $w \in C_c^\infty(\Omega)$, we obtain $-\Delta u + (\underline{v} \cdot \nabla u) + u = f$ in Ω . Then we can see

$$\int_{\partial\Omega} \frac{\partial u}{\partial \nu} \phi \, d\sigma = 0, \quad \forall \phi \in \text{Im}(\gamma).$$

Since $\text{Im}(\gamma)$ is dense in $L^2(\partial\Omega)$, we conclude that $\frac{\partial u}{\partial \nu} = 0$ almost everywhere on $\partial\Omega$.

2. The bilinear form a is not symmetric, so we cannot use the Riesz representation theorem to show that (VF) has a unique solution. Instead, we can use the Lax-Milgram theorem.

$$\begin{aligned} a(u, w) &= \int_{\Omega} \nabla u \cdot \nabla w + (\underline{v} \cdot \nabla u) w + u w \, dx \\ &\leq \|\nabla u\|_{L^2} \|\nabla w\|_{L^2} + \|\underline{v}\|_{L^\infty} \|\nabla u\|_{L^2} \|w\|_{L^2} + \|u\|_{L^2} \|w\|_{L^2} \\ &\leq C \|u\|_{H^1(\Omega)} \|w\|_{H^1(\Omega)}. \end{aligned}$$

So the bilinear form a is continuous. Moreover, we have

$$\begin{aligned} a(u, u) &= \int_{\Omega} |\nabla u|^2 + (\underline{v} \cdot \nabla u) u + u^2 \, dx \\ &= \|u\|_{H^1(\Omega)}^2 + \int_{\Omega} (\underline{v} \cdot \nabla u) u \, dx \end{aligned}$$

Note that

$$\int_{\Omega} (\underline{v} \cdot \nabla u) u \, dx = \frac{1}{2} \int_{\Omega} \underline{v} \cdot \nabla (u^2) \, dx = -\frac{1}{2} \int_{\Omega} (\nabla \cdot \underline{v}) u^2 \, dx + \frac{1}{2} \int_{\partial\Omega} (\underline{v} \cdot \underline{\nu}) u^2 \, d\sigma = 0.$$

Hence, we have $a(u, u) = \|u\|_{H^1(\Omega)}^2$. So the bilinear form a is coercive. It is trivial that ℓ is continuous. By the Lax-Milgram theorem, there exists a unique solution u of (VF). We can also check that the map A is linear and continuous by using the coercivity of a .

3. It depends on the value of $\nabla \cdot \underline{v}$. If $\text{esssup}(\nabla \cdot \underline{v})$ is strictly smaller than 2, we can still use the Lax-Milgram theorem to conclude the existence and uniqueness of the solution.
4. Since V_h is finite-dimensional, it is closed in $H^1(\Omega)$. Therefore, the subspace $(V_h, (\cdot, \cdot)_{H^1})$ is a Hilbert space. Note that the bilinear form a is continuous and coercive on V_h . Moreover, the linear form ℓ is continuous on V_h . Hence, we can still apply the Lax-Milgram theorem to conclude that there exists a unique solution u_h of the problem.

Problem 8. (Sobolev Space H^1 in Dimension 1)

1. Let $I = (-1, 1)$, show that the function u such that $u(x) = (x + |x|)/2$ belongs to $H^1(I)$. Determine its weak derivative. More generally, show that any continuous function on $\bar{I}$, piecewise continuously differentiable on $\bar{I}$, belongs to $H^1(I)$.

Remark: It can be proved that this result remains true in dimension $d \geq 2$ (here I is a domain in $\mathbb{R}^d$ with $d = 1$).

2. Let I be a bounded or unbounded interval. Show that every element $u \in H^1(I)$ is continuous and uniformly bounded.

Remark: Be careful, this result is not valid in dimension $d \geq 2$ (see Problem 9 below).

3. Using the results from questions 1 and 2, determine whether the functions u and v defined respectively by

$$u(x) = \begin{cases} a & \text{on } (0, 1/2) \\ b & \text{on } (1/2, 1) \end{cases}, \quad a, b \in \mathbb{R} \quad \text{and} \quad v(x) = x^\alpha, \quad \alpha \in \mathbb{R},$$

belong to $H^1(I)$ where $I = (0, 1)$.

4. Let I be an open interval of $\mathbb{R}$. Show that the functions in $H^1(I)$ are $1/2$ -Hölder continuous, i.e., for every $u \in H^1(I)$, there exists a constant $C = C(I)$ such that

$$\forall x, y \in I, \quad |u(x) - u(y)| \leq C|x - y|^{1/2}.$$

Assuming the Arzelà-Ascoli Theorem recalled below, deduce that if I is bounded then every bounded sequence in $H^1(I)$ admits a subsequence convergent in $C^0(\bar{I})$.

This result is false if I is unbounded (take a sequence of functions of the form $x \mapsto u(x - n)$).

Arzelà-Ascoli Theorem. Let K be a compact metric space and let $\mathcal{H}$ be a bounded subset of $C^0(K)$. Suppose that $\mathcal{H}$ is uniformly equicontinuous, i.e.,

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } \forall f \in \mathcal{H}, \quad d(x_1, x_2) < \delta \implies |f(x_1) - f(x_2)| < \epsilon.$$

Then $\mathcal{H}$ is relatively compact in $C^0(K)$; in other words, every sequence of elements of $\mathcal{H}$ admits a subsequence convergent in $C^0(K)$.

Solution 8. 1. Let $I = (-1, 1)$. One easily verifies that $u(x) = (x + |x|)/2$ belongs to $L^2(I)$. For any $\varphi \in C_c^\infty(I)$,

$$\int_I u(x) \varphi'(x) dx = \int_0^1 x \varphi'(x) dx = - \int_0^1 \varphi(x) dx.$$

Hence the weak derivative of u is

$$u'(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0, \end{cases}$$

which lies in $L^2(I)$.

More generally, let u be continuous on $\bar{I}$ and piecewise C^1 on $\bar{I}$. Then there exist points $a_0 = -1, a_1, \dots, a_N = 1$ with $a_i < a_{i+1}$ such that $u|_{[a_i, a_{i+1}]}$ is $C^1([a_i, a_{i+1}])$. For any $\varphi \in C_c^\infty(I)$,

$$\int_{-1}^1 u \varphi' dx = \sum_{i=0}^{N-1} \int_{a_i}^{a_{i+1}} u \varphi' dx = \sum_{i=0}^{N-1} \left(u(a_{i+1}) \varphi(a_{i+1}) - u(a_i) \varphi(a_i) \right) - \sum_{i=0}^{N-1} \int_{a_i}^{a_{i+1}} u' \varphi dx.$$

By continuity of u and φ at the interior points and because φ vanishes near ± 1 , the boundary terms cancel, leaving

$$\int_{-1}^1 u \varphi' dx = - \int_{-1}^1 u' \varphi dx.$$

Thus the classical derivative (defined almost everywhere) is also the weak derivative, and since it is piecewise continuous and bounded, it belongs to $L^2(I)$. Hence $u \in H^1(I)$.

2. **Case $I = \mathbb{R}$.** Let $u \in H^1(\mathbb{R})$. By density of $C_c^\infty(\mathbb{R})$ in $H^1(\mathbb{R})$, there exists a sequence $(u_n) \subset C_c^\infty(\mathbb{R})$ such that $\|u - u_n\|_{H^1} \rightarrow 0$. For any $v \in C_c^\infty(\mathbb{R})$ and any $x \in \mathbb{R}$,

$$v(x)^2 = 2 \int_{-\infty}^x v(t) v'(t) dt.$$

Cauchy-Schwarz gives $\|v\|_\infty^2 \leq 2\|v\|_{L^2}\|v'\|_{L^2} \leq \|v\|_{H^1}^2$. Hence (u_n) is a Cauchy sequence in $L^\infty(\mathbb{R})$ and converges uniformly to some $v \in C_b^0(\mathbb{R})$. For any $\varphi \in C_c^\infty(\mathbb{R})$, the H^1 convergence implies $\int u_n \varphi \rightarrow \int u \varphi$, while uniform convergence gives $\int u_n \varphi \rightarrow \int v \varphi$. Thus $\int (u - v) \varphi = 0$ for all such φ , so $u = v$ a.e. Therefore u has a continuous representative (still denoted u) and $\|u\|_\infty \leq \|u\|_{H^1}$.

Case I bounded. For any $x, y \in I$ and $v \in C^\infty(\bar{I})$,

$$v(x)^2 = v(y)^2 + 2 \int_x^y v(s) v'(s) ds.$$

Cauchy-Schwarz yields $|v(x)|^2 \leq |v(y)|^2 + 2\|v\|_{L^2}\|v'\|_{L^2} \leq |v(y)|^2 + \|v\|_{H^1}^2$. Integrating with respect to y over I gives

$$|I| |v(x)|^2 \leq \|v\|_{L^2}^2 + 2|I| \|v\|_{H^1}^2 \leq (1 + 2|I|) \|v\|_{H^1}^2.$$

By density of smooth functions, the same estimate holds for any $u \in H^1(I)$. Thus u is bounded and, using the same approximation argument as for $\mathbb{R}$, one shows that u is continuous, i.e., has a continuous representative.

3. For u defined as a on $(0, 1/2)$ and b on $(1/2, 1)$: if $a = b$, then u is constant and belongs to $H^1(I)$. If $a \neq b$, u is discontinuous at $1/2$ and by question 2 cannot belong to $H^1(I)$. A direct proof by contradiction: assume u has a weak derivative $\psi \in L^2(I)$. Then for any $\varphi \in C_c^\infty(I)$,

$$\int_0^1 u\varphi' = \int_0^{1/2} a\varphi' + \int_{1/2}^1 b\varphi' = (a - b)\varphi(1/2) = - \int_0^1 \psi\varphi.$$

Taking φ with support in $(0, 1/2) \cup (1/2, 1)$ forces $\psi = 0$ a.e., hence $(a - b)\varphi(1/2) = 0$ for all φ . It is impossible if $a \neq b$.

For $v(x) = x^\alpha$ on $I = (0, 1)$: $v \in L^2(I)$ iff $\int_0^1 x^{2\alpha} dx < \infty$, i.e., $\alpha > -1/2$. When $\alpha \neq 0$, the classical derivative $v'(x) = \alpha x^{\alpha-1}$ belongs to $L^2(I)$ iff $\int_0^1 x^{2(\alpha-1)} dx < \infty$, i.e., $\alpha > 1/2$. When $\alpha = 0$, $v \equiv 1$ is smooth. Therefore $v \in H^1(I)$ iff $\alpha \in \{0\} \cup (1/2, \infty)$.

4. Let $u \in C^\infty(I)$. For $x > y \in I$,

$$u(x) - u(y) = \int_y^x u'(t) dt,$$

so by Cauchy-Schwarz,

$$|u(x) - u(y)| \leq \left(\int_y^x dt \right)^{1/2} \left(\int_y^x |u'|^2 dt \right)^{1/2} \leq \|u\|_{H^1(I)} |x - y|^{1/2}.$$

By density of smooth functions and continuity of H^1 functions (question 2), this estimate holds for every $u \in H^1(I)$ with the same constant $C = 1$ (or any $C \geq 1$).

Now assume I bounded and let (u_n) be a bounded sequence in $H^1(I)$. The previous estimate shows that (u_n) is uniformly bounded (by $\|u_n\|_\infty \leq C\|u_n\|_{H^1}$) and equicontinuous: given $\varepsilon > 0$, take $\delta = \varepsilon^2/M^2$ where $M = \sup_n \|u_n\|_{H^1}$. Then $|x - y| < \delta$ implies $|u_n(x) - u_n(y)| \leq M|x - y|^{1/2} < \varepsilon$. By the Arzelà-Ascoli theorem, (u_n) has a subsequence converging uniformly on $\bar{I}$.

If I is unbounded, the result fails: take a nonzero $u \in H^1(\mathbb{R})$ and define $u_n(x) = u(x + n)$. Then $\|u_n\|_{H^1} = \|u\|_{H^1}$ is bounded, but no subsequence can converge uniformly because the supports move to infinity.

Problem 9. (Sobolev Space H^1 in Dimension $d \geq 2$) In this exercise, we establish that elements of H^1 in dimension $d \geq 2$ are generally not continuous.

- For $d = 2$, let $B = B(0, a)$ be the open ball of $\mathbb{R}^2$ of radius $a < 1$. Show that the function $u(x) = |\ln(|x|)|^\alpha$ belongs to $H^1(B)$ for $0 < \alpha < 1/2$ but is unbounded near the origin.
- For $d \geq 3$, let B be the open unit ball of $\mathbb{R}^d$. Show that the function u such that $u(x) = |x|^{-\beta}$ belongs to $H^1(B)$ for $0 < \beta < (d - 2)/2$ but is unbounded near the origin.

Solution 9. 1. **Case $d = 2$.** Define $u(x) = |\ln |x||^\alpha$ if $x \neq 0$ and $u(0) = 0$. Using polar coordinates,

$$\int_B |u|^2 dx = 2\pi \int_0^a |\ln r|^{2\alpha} r dr < \infty.$$

For $x \neq 0$, u is classically differentiable and

$$\nabla u(x) = -\alpha \frac{x}{|x|^2} |\ln |x||^{\alpha-1} =: \psi(x).$$

Then

$$\int_B |\psi|^2 dx = 2\pi\alpha^2 \int_0^a \frac{1}{r^2} |\ln r|^{2\alpha-2} r dr = 2\pi\alpha^2 \int_0^a \frac{1}{r} |\ln r|^{2\alpha-2} dr < \infty,$$

because $2\alpha - 2 < -1$ and the integral converges (via the substitution $t = -\ln r$). Hence $\psi \in L^2(B)^2$.

To prove that $\nabla u = \psi$ in the weak sense, take $\varphi \in C_c^\infty(B)^2$ and for $\varepsilon > 0$ small,

$$\int_B u \operatorname{div} \varphi dx = \int_{\varepsilon < |x| < a} u \operatorname{div} \varphi dx + \int_{|x| < \varepsilon} u \operatorname{div} \varphi dx.$$

On the annulus, integrate by parts:

$$\int_{\varepsilon < |x| < a} u \operatorname{div} \varphi = - \int_{\varepsilon < |x| < a} \psi \cdot \varphi \, dx + \int_{|x|=\varepsilon} u \varphi \cdot n \, ds,$$

where n is the unit outer normal to the circle of radius ε (pointing inward). The boundary term satisfies

$$\left| \int_{|x|=\varepsilon} u \varphi \cdot n \, ds \right| \leq |\ln \varepsilon|^\alpha \|\varphi\|_\infty 2\pi\varepsilon \rightarrow 0,$$

and the integral over the small ball tends to zero by the Cauchy–Schwarz inequality because its measure shrinks. Letting $\varepsilon \rightarrow 0$ gives

$$\int_B u \operatorname{div} \varphi = - \int_B \psi \cdot \varphi,$$

so $u \in H^1(B)$ and $\nabla u = \psi$. Clearly u is unbounded near the origin because $|\ln |x||^\alpha \rightarrow \infty$ as $x \rightarrow 0$.

2. **Case $d \geq 3$.** Let $0 < \beta < (d-2)/2$ and define $u(x) = |x|^{-\beta}$ on the unit ball B . Using spherical coordinates,

$$\int_B |u|^2 \, dx = |S^{d-1}| \int_0^1 r^{d-1} r^{-2\beta} \, dr = |S^{d-1}| \int_0^1 r^{d-1-2\beta} \, dr < \infty,$$

because $d-1-2\beta > -1$ (since $\beta < (d-2)/2$ implies $d-1-2\beta > 1$).

For $x \neq 0$, $\nabla u(x) = -\beta x |x|^{-(\beta+2)} =: \psi(x)$. Then

$$\int_B |\psi|^2 \, dx = \beta^2 \int_B |x|^{-2(\beta+1)} \, dx = \beta^2 |S^{d-1}| \int_0^1 r^{d-1} r^{-2(\beta+1)} \, dr = \beta^2 |S^{d-1}| \int_0^1 r^{-2\beta+d-3} \, dr,$$

which converges because $-2\beta + d - 3 > -1$ (again using $\beta < (d-2)/2$). An argument similar to the 2D case (splitting the ball into an annulus and a small ball, letting the inner radius tend to zero) shows that ψ is indeed the weak gradient of u . Hence $u \in H^1(B)$ and u is unbounded near the origin since $|x|^{-\beta} \rightarrow \infty$ as $x \rightarrow 0$.

Problem 10. (Mixed Dirichlet/Robin Conditions) Let $\Omega \subset \mathbb{R}^d$ be a bounded sufficiently regular open set. Let Γ_1, Γ_2 be two non-empty open sets such that $\partial\Omega = \overline{\Gamma}_1 \cup \overline{\Gamma}_2$ and $\Gamma_1 \cap \Gamma_2 = \emptyset$. We are interested in the problem

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = g, & \text{on } \Gamma_1, \\ \frac{\partial u}{\partial \nu} + \alpha u = h, & \text{on } \Gamma_2, \end{cases} \quad (\text{DR})$$

where $f \in L^2(\Omega)$, $g \in L^2(\Gamma_1)$, $h \in L^2(\Gamma_2)$, and $\alpha \in \mathbb{R}$ is strictly positive.

1. First consider the case $g = 0$. Write the variational formulation (VF) naturally associated with problem (DR) in a space to be specified. Show that if u is a solution of (VF) belonging to $H^2(\Omega)$, then u satisfies (DR).
2. Still in the case $g = 0$, show that (VF) has a unique solution.
3. Suppose that g admits a lift into $H^1(\Omega)$, i.e., there exists $u_g \in H^1(\Omega)$ such that $u_g|_{\Gamma_1} = g$. Write the corresponding variational formulation (VF $_g$) and show that it has a unique solution.
4. Let $k : \Omega \rightarrow \mathbb{R}$ be a continuous positive function and $A \in \mathcal{M}_2(\mathbb{R})$ a positive definite matrix. What would become the condition on Γ_2 in the case where the equation we wish to solve is

$$-\operatorname{div}(k\nabla u) = f \text{ in } \Omega \quad \text{or} \quad -\operatorname{div}(A\nabla u) = f \text{ in } \Omega \quad ?$$

Solution 10. 1. As with the trace on $\partial\Omega$, one can consider “partial” traces defined on $C^1(\overline{\Omega})$ with values in $L^2(\Gamma_i)$. These maps are linear and one can show the existence of constants C_i such that

$$\|u\|_{L^2(\Gamma_i)} \leq C_i \|u\|_{H^1(\Omega)} \quad \text{for all } u \in C^1(\overline{\Omega}).$$

These traces can therefore be uniquely extended to continuous linear maps

$$\gamma_i : H^1(\Omega) \rightarrow L^2(\Gamma_i), \quad i = 1, 2.$$

One then defines the following subspace of $H^1(\Omega)$:

$$H_{0,\Gamma_1}^1(\Omega) = \{v \in H^1(\Omega) \mid \gamma_1(v) = 0\}.$$

In the following, for $i = 1, 2$, on Γ_i , we will abusively write v instead of $\gamma_i v$. We look for u in $H_{0,\Gamma_1}^1(\Omega)$ and consider a test function $v \in H_{0,\Gamma_1}^1(\Omega)$. We multiply the equation by v and integrate over Ω to obtain

$$\int_{\Omega} -\Delta u(x)v(x) \, dx = \int_{\Omega} f(x)v(x) \, dx.$$

Integrating by parts and using the fact that $v = 0$ on Γ_1 , we find

$$\int_{\Omega} \nabla u(x) \cdot \nabla v(x) \, dx - \int_{\Gamma_2} \frac{\partial u}{\partial \nu}(x)v(x) \, d\sigma = \int_{\Omega} f(x)v(x) \, dx.$$

Using the boundary condition on Γ_2 , we deduce

$$\int_{\Omega} \nabla u(x) \cdot \nabla v(x) \, dx - \int_{\Gamma_2} (h(x) - \alpha u(x))v(x) \, d\sigma = \int_{\Omega} f(x)v(x) \, dx.$$

Finally, the variational formulation associated with problem (DR) is written as

$$\text{Find } u \in H_{0,\Gamma_1}^1(\Omega) \text{ such that } \forall v \in H_{0,\Gamma_1}^1(\Omega), \quad a(u, v) = \ell(v) \quad (\text{VF})$$

where

$$a(u, v) = \int_{\Omega} \nabla u(x) \cdot \nabla v(x) \, dx + \alpha \int_{\Gamma_2} u(x)v(x) \, d\sigma$$

and

$$\ell(v) = \int_{\Omega} f(x)v(x) \, dx + \int_{\Gamma_2} h(x)v(x) \, d\sigma.$$

It is then straightforward to check that if u is a solution of (VF) belonging to $H^2(\Omega)$, then u satisfies (DR).

2. The bilinear form $a(\cdot, \cdot)$ is continuous on $H_{0,\Gamma_1}^1(\Omega) \times H_{0,\Gamma_1}^1(\Omega)$ since

$$\begin{aligned} |a(v, w)| &\leq \|\nabla v\|_{(L^2(\Omega))^n} \|\nabla w\|_{(L^2(\Omega))^n} + \alpha \|v\|_{L^2(\Gamma_2)} \|w\|_{L^2(\Gamma_2)} \\ &\leq \|v\|_{H^1(\Omega)} \|w\|_{H^1(\Omega)} + \alpha C_2^2 \|v\|_{H^1(\Omega)} \|w\|_{H^1(\Omega)} \\ &\leq (1 + \alpha C_2^2) \|v\|_{H^1(\Omega)} \|w\|_{H^1(\Omega)}. \end{aligned}$$

Then we show that the bilinear form $a(\cdot, \cdot)$ is also coercive on $H_{0,\Gamma_1}^1(\Omega)$. We will prove the following inequality, which is a variant of the Poincaré inequality: there exists a constant $C > 0$ such that for all $v \in H_{0,\Gamma_1}^1(\Omega)$,

$$\|v\|_{L^2(\Omega)} \leq C [\|\nabla v\|_{(L^2(\Omega))^n} + \|\gamma_2(v)\|_{L^2(\Gamma_2)}]. \quad (*)$$

It implies

$$\begin{aligned} a(v, v) &= \|\nabla v\|_{(L^2(\Omega))^n}^2 + \alpha \|v\|_{L^2(\Gamma_2)}^2 \\ &\geq \min(1, \alpha) [\|\nabla v\|_{(L^2(\Omega))^n}^2 + \|v\|_{L^2(\Gamma_2)}^2] \\ &\geq \frac{\min(1, \alpha)}{2} [\|\nabla v\|_{(L^2(\Omega))^n} + \|v\|_{L^2(\Gamma_2)}]^2 \\ &\geq \frac{\min(1, \alpha)}{2(C+1)^2} \|v\|_{H^1(\Omega)}^2. \end{aligned}$$

Now we prove (*) by contradiction. Suppose that for every integer $n \geq 1$, there exists $v_n \in H_{0,\Gamma_1}^1(\Omega)$ such that

$$\|v_n\|_{L^2(\Omega)} > n [\|\nabla v_n\|_{(L^2(\Omega))^n} + \|\gamma_2(v_n)\|_{L^2(\Gamma_2)}].$$

By replacing v_n by $v_n/\|v_n\|_{L^2(\Omega)}$, we can assume that $\|v_n\|_{L^2(\Omega)} = 1$ for all n . Then we have

$$\|\nabla v_n\|_{(L^2(\Omega))^n} + \|\gamma_2(v_n)\|_{L^2(\Gamma_2)} < \frac{1}{n}.$$

The sequence $(v_n)_{n \geq 1}$ is bounded in $H^1(\Omega)$. By the Rellich-Kondrachov theorem, we can suppose that there exists a subsequence (still denoted $(v_n)_{n \geq 1}$) that converges strongly in $L^2(\Omega)$ to v . Since $\|v_n\|_{L^2(\Omega)} =$

1 for all n , we have $\|v\|_{L^2(\Omega)} = 1$. In addition, $(v_n)_{n \geq 1}$ is a Cauchy sequence in $L^2(\Omega)$. Moreover, $(\nabla v_n)_{n \geq 1}$ is a Cauchy sequence in $(L^2(\Omega))^n$ since

$$\|\nabla v_n - \nabla v_m\|_{(L^2(\Omega))^n} \leq \|\nabla v_n\|_{(L^2(\Omega))^n} + \|\nabla v_m\|_{(L^2(\Omega))^n} < \frac{1}{n} + \frac{1}{m}.$$

We deduce that $((v_n)_{n \geq 1})$ is a Cauchy sequence in $H_{0,\Gamma_1}^1(\Omega)$ and therefore converges strongly in $H_{0,\Gamma_1}^1(\Omega)$ to v . Thus, we have $\|\nabla v\|_{(L^2(\Omega))^n} = 0$, which implies that v is constant in each connected component of Ω . Since $v \in H_{0,\Gamma_1}^1(\Omega)$, we have $v = 0$ on Γ_1 . By the continuity of $\gamma_2 : H^1(\Omega) \rightarrow L^2(\Gamma_2)$, we have $v = 0$ on Γ_2 . Hence v is equal to zero on $\partial\Omega$, which implies that $v = 0$ in Ω since v is constant on the connected components. This contradicts the fact that $\|v\|_{L^2(\Omega)} = 1$. This concludes the proof of $(*)$.

The linear form $\ell(\cdot)$ is continuous on $H_{0,\Gamma_1}^1(\Omega)$ since

$$\begin{aligned} |\ell(v)| &\leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|h\|_{L^2(\Gamma_2)} \|v\|_{L^2(\Gamma_2)} \\ &\leq \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)} + C_2 \|h\|_{L^2(\Gamma_2)} \|v\|_{H^1(\Omega)} \\ &\leq (\|f\|_{L^2(\Omega)} + C_2 \|h\|_{L^2(\Gamma_2)}) \|v\|_{H^1(\Omega)}. \end{aligned}$$

By the Lax-Milgram theorem, we deduce that (VF) has a unique solution.

3. The variational formulation (VFg) is to look for u in the affine space $V = \{v \in H^1(\Omega) \mid \gamma_1(v) = g\}$ such that $a(u, v) = \ell(v)$. However, it is more convenient to look for $u_0 = u - u_g$ in the linear space $H_{0,\Gamma_1}^1(\Omega)$ with $u_g \in H^1(\Omega)$ a lift of g . We look for $u_0 \in H_{0,\Gamma_1}^1(\Omega)$ such that $u = u_0 + u_g$. Then u_0 satisfies the variational formulation:

$$\text{Find } u_0 \in H_{0,\Gamma_1}^1(\Omega) \text{ such that } \forall v \in H_{0,\Gamma_1}^1(\Omega), \quad a(u_0, v) = \ell_g(v) \quad (\text{VFg}') \quad (1)$$

where

$$\ell_g(v) = \ell(v) - a(u_g, v) = \int_{\Omega} f(x)v(x) dx + \int_{\Gamma_2} h(x)v(x) d\sigma - \int_{\Omega} \nabla u_g(x) \cdot \nabla v(x) dx - \alpha \int_{\Gamma_2} u_g(x)v(x) d\sigma.$$

Using the same arguments as in question 2, we show that (VFg') has a unique solution. Since the lift u_g is not unique, we have to check that the solution u of (VFg) does not depend on the choice of the lift. Let $u_g^1, u_g^2 \in H^1(\Omega)$ be two lifts of g and let $u_0^1, u_0^2 \in H_{0,\Gamma_1}^1(\Omega)$ be the corresponding solutions of (VFg') . We consider $u^1 = u_0^1 + u_g^1$ and $u^2 = u_0^2 + u_g^2$. We have

$$a(u^1, v) = a(u^2, v), \quad \forall v \in H_{0,\Gamma_1}^1(\Omega),$$

which implies that $\nabla(u^1 - u^2) = 0$ in Ω and $u^1 - u^2 = 0$ on Γ_2 . By definition, we have also $u^1 - u^2 = 0$ on Γ_1 . Thus, we have $u^1 = u^2$ in Ω .

4. WLOG, we can suppose $g = 0$. For the equation $-\text{div}(k\nabla u) = f$ in Ω , by multiplying the equation by a test function $v \in H_{0,\Gamma_1}^1(\Omega)$ and integrating over Ω , we can see that the boundary condition on Γ_2 must have a connection with $k \frac{\partial u}{\partial \nu}$. For example, it could be

$$k \frac{\partial u}{\partial \nu} + \alpha u = h \text{ on } \Gamma_2.$$

For the equation $-\text{div}(A\nabla u) = f$ in Ω , by multiplying the equation by a test function $v \in H_{0,\Gamma_1}^1(\Omega)$ and integrating over Ω , we can see that the boundary condition on Γ_2 must have a connection with $(A\nabla u) \cdot \underline{\nu}$. For example, it could be

$$(A\nabla u) \cdot \underline{\nu} + \alpha u = h \text{ on } \Gamma_2.$$

Problem 11. (Laplacian with Neumann Boundary Conditions) Let $\Omega \subset \mathbb{R}^d$ be a bounded sufficiently regular connected open set. We are interested in the Laplace problem with Neumann boundary conditions:

$$\text{Find } u \text{ such that } -\Delta u = f \text{ in } \Omega \text{ and } \frac{\partial u}{\partial \nu} = g \text{ on } \partial\Omega, \quad (\text{LN}) \quad (2)$$

where f, g denote source terms belonging respectively to $L^2(\Omega)$ and $L^2(\partial\Omega)$.

1. Can we hope to prove a uniqueness result for the solution of problem (LN) ? Why?

2. Give a necessary condition (called a compatibility condition) that f and g must satisfy for (LN) to admit a solution.

Define the space of functions with zero mean:

$$H_{\#}^1(\Omega) := \left\{ \phi \in H^1(\Omega) \mid \int_{\Omega} \phi(x) \, dx = 0 \right\}.$$

3. Show that if u satisfies problem (LN), then there exists a constant α such that $w = u - \alpha$ is a solution of the problem

$$\text{Find } w \in H_{\#}^1(\Omega) \text{ such that } a(w, v) = \ell(v), \quad \forall v \in H_{\#}^1(\Omega). \quad (\text{LNV})$$

Give the expressions for $a(\cdot, \cdot)$ and $\ell(\cdot)$.

Conversely, prove that if w is a solution of (LNV) belonging to $H^2(\Omega)$, then w satisfies (LN) provided that f, g satisfy the compatibility relation obtained in question 2.

4. Show that $H_{\#}^1(\Omega)$ equipped with the inner product of $H^1(\Omega)$ is a Hilbert space.
5. Show that $H_{\#}^1(\Omega)$ equipped with the inner product $(\phi, \psi) \mapsto \int_{\Omega} \nabla \phi(x) \cdot \nabla \psi(x) \, dx$ is also a Hilbert space.
6. Show that (LNV) admits a unique solution. Is it necessary to assume that f, g satisfy the compatibility condition from question 2? Establish that the mapping

$$A : L^2(\Omega) \times L^2(\partial\Omega) \rightarrow H_{\#}^1(\Omega), \quad (f, g) \mapsto w,$$

where w is the solution of (LNV), is linear and continuous.

Solution 11. We have discussed this problem in the main text in Section 2.3. Here are more details.

1. No. If u is a solution of (LN), then for every constant $\alpha \in \mathbb{R}$, $u + \alpha$ is also a solution of (LN).
2. The compatibility condition is

$$\int_{\Omega} f(x) \, dx + \int_{\partial\Omega} g(x) \, d\sigma = 0.$$

3. We can take

$$\alpha = \frac{1}{|\Omega|} \int_{\Omega} u(x) \, dx.$$

The bilinear form $a(\cdot, \cdot)$ and the linear form $\ell(\cdot)$ are defined by

$$a(w, v) = \int_{\Omega} \nabla w(x) \cdot \nabla v(x) \, dx$$

and

$$\ell(v) = \int_{\Omega} f(x)v(x) \, dx + \int_{\partial\Omega} g(x)v(x) \, d\sigma.$$

If w is a solution of (LNV) belonging to $H^2(\Omega)$, then by taking $v \in C_c^\infty(\Omega)$ in (LNV), we obtain $-\Delta w = f$ in Ω . By taking $v \in H^1(\Omega)$ in (LNV) and integrating by parts, we obtain $\frac{\partial w}{\partial \nu} = g$ on $\partial\Omega$. Then by Green formula, f, g satisfy the compatibility relation obtained in question 2.

4. Note that

$$H^1(\Omega) \rightarrow \mathbb{R}, \quad \phi \mapsto \int_{\Omega} \phi(x) \, dx$$

is a continuous linear form since Ω is bounded. Thus, $H_{\#}^1(\Omega)$ is a closed subspace of the Hilbert space $H^1(\Omega)$ since it is the kernel of a continuous linear form. Hence it is a Hilbert space.

5. Under this inner product, every Cauchy sequence $(\phi_n)_{n \geq 1}$ in $H_{\#}^1(\Omega)$ satisfies that $(\nabla \phi_n)_{n \geq 1}$ is a Cauchy sequence in $(L^2(\Omega))^d$. By Poincaré-Wirtinger inequality, there exists a constant $C > 0$ such that for every $w \in H_{\#}^1(\Omega)$, we have

$$\|w\|_{L^2(\Omega)} \leq C \|\nabla w\|_{(L^2(\Omega))^d}.$$

Therefore $(\phi_n)_{n \geq 1}$ is a Cauchy sequence in $H_{\#}^1(\Omega)$ equipped with the H^1 -norm. Since $H_{\#}^1(\Omega)$ equipped with the H^1 -norm is a Hilbert space (see question 4), there exists $\phi \in H_{\#}^1(\Omega)$ such that $\phi_n \rightarrow \phi$ in $H_{\#}^1(\Omega)$ equipped with the H^1 -norm. In particular, we have $\nabla \phi_n \rightarrow \nabla \phi$ in $(L^2(\Omega))^d$. This concludes the proof.

6. By the Lax-Milgram theorem, (LNV) admits a unique solution since the bilinear form $a(\cdot, \cdot)$ is continuous and coercive on $H_{\#}^1(\Omega)$ and the linear form $\ell(\cdot)$ is continuous on $H_{\#}^1(\Omega)$. It is not necessary to assume that f, g satisfy the compatibility condition from question 2 since we are looking for a solution in $H_{\#}^1(\Omega)$. The mapping A is linear since the problem is linear. Moreover, we have

$$\begin{aligned} \|w\|_{H^1(\Omega)} &\leq C \|\nabla w\|_{(L^2(\Omega))^d} \\ &\leq C \left| \int_{\partial\Omega} w \frac{\partial w}{\partial \nu} d\sigma - \int_{\Omega} \Delta w w dx \right|^{\frac{1}{2}} \\ &\leq C \left(\|w\|_{\partial\Omega} \|g\|_{L^2(\partial\Omega)} + \|w\|_{L^2(\Omega)} \|f\|_{L^2(\Omega)} \right)^{\frac{1}{2}} \\ &\leq C \left(C_{tr} \|w\|_{H^1(\Omega)} \|g\|_{L^2(\partial\Omega)} + \|w\|_{H^1(\Omega)} \|f\|_{L^2(\Omega)} \right)^{\frac{1}{2}}. \end{aligned}$$

Therefore, there exists a constant $C > 0$ such that

$$\|w\|_{H^1(\Omega)} \leq C \left(\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \right).$$

Problem 12. (Penalization Methods for the Neumann Problem) To compute an approximation of the solution of a Neumann-type problem like (LNV), one must avoid having to work with spaces of functions with zero mean.

Let $\Omega \subset \mathbb{R}^d$ be a bounded regular connected open set. For $\kappa > 0$, introduce the problem

$$\text{Find } w_{\kappa} \in H^1(\Omega) \text{ such that : } a_{\kappa}(w_{\kappa}, v) = \ell(v), \quad \forall v \in H^1(\Omega), \quad (\mathcal{P}_v^{\kappa})$$

where

$$a_{\kappa}(w_{\kappa}, v) = a(w_{\kappa}, v) + \kappa^{-1} \left(\int_{\Omega} w_{\kappa}(x) dx \right) \left(\int_{\Omega} v(x) dx \right),$$

and where $\ell(\cdot)$ and $a(\cdot, \cdot)$ are defined as in (LNV).

1. Prove that for every $\kappa > 0$, $(\mathcal{P}_v^{\kappa})$ has a unique solution. Establish that if $\ell(1) = 0$, then $w_{\kappa} = w$, the solution of (LNV). In the general case, by testing with $v = \int_{\Omega} w_{\kappa}(x) dx$ in $(\mathcal{P}_v^{\kappa})$, show that there exists a constant $C > 0$ independent of $\kappa \in (0, \kappa_0]$ such that

$$\left| \int_{\Omega} w_{\kappa}(x) dx \right| \leq C \kappa \left(\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \right).$$

Here $\kappa_0 > 0$ is fixed once and for all.

2. Observing that there exists an α independent of $\kappa \in (0, \kappa_0]$ such that

$$\alpha \|w_{\kappa} - w\|_{H^1(\Omega)}^2 \leq a_{\kappa}(w_{\kappa} - w, w_{\kappa} - w),$$

and by expanding the right-hand side of this inequality, show that we have

$$\|w_{\kappa} - w\|_{H^1(\Omega)}^2 \leq C \kappa^{-1} \left| \int_{\Omega} w_{\kappa}(x) dx \right|^2, \quad \forall \kappa \in (0, \kappa_0].$$

Deduce the error estimate

$$\|w_{\kappa} - w\|_{H^1(\Omega)} \leq C \sqrt{\kappa} \left(\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \right),$$

where $C > 0$ is a constant independent of $\kappa \in (0, \kappa_0]$.

We now propose an alternative to the previous penalized problem $(\mathcal{P}_v^{\kappa})$:

$$\text{Find } w_{\epsilon} \in H^1(\Omega) \text{ such that : } a_{\epsilon}(w_{\epsilon}, v) = \ell(v), \quad \forall v \in H^1(\Omega), \quad (P_v^{\epsilon})$$

where

$$a_{\epsilon}(w_{\epsilon}, v) = a(w_{\epsilon}, v) + \epsilon \int_{\Omega} w_{\epsilon}(x) v(x) dx,$$

and where $\ell(\cdot)$ and $a(\cdot, \cdot)$ are defined as in (LNV).

3. Prove that this problem is well-posed. Assume $\ell(1) = 0$. Show that

$$w_{\epsilon} \in H_{\#}^1(\Omega) := \left\{ \phi \in H^1(\Omega) \mid \int_{\Omega} \phi(x) dx = 0 \right\},$$

and that there exists C such that for every $\epsilon > 0$

$$\|w_{\epsilon}\|_{H^1(\Omega)} \leq C \left(\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \right).$$

4. We remain in the setting $\ell(1) = 0$. Prove that there exists C such that for every $\epsilon > 0$

$$\|w - w_\epsilon\|_{H^1(\Omega)} \leq C\epsilon (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)}).$$

Solution 12. 1. By the Lax-Milgram theorem, $(\mathcal{P}_v^\kappa)$ has a unique solution w_κ since the bilinear form $a_\kappa(\cdot, \cdot)$ is continuous and coercive on $H^1(\Omega)$ and the linear form $\ell(\cdot)$ is continuous on $H^1(\Omega)$. If $\ell(1) = 0$, we have

$$\begin{aligned} a(w, v) &= a\left(w, v - \frac{1}{|\Omega|} \int_\Omega v \, dx\right) + \left(\frac{1}{|\Omega|} \int_\Omega v \, dx\right) a(w, 1) \\ &= \ell\left(v - \frac{1}{|\Omega|} \int_\Omega v \, dx\right) = \ell(v) - \left(\frac{1}{|\Omega|} \int_\Omega v \, dx\right) \ell(1) = \ell(v). \end{aligned}$$

Since $w \in H_\#^1(\Omega)$, we have $a_\kappa(w, v) = a(w, v)$. Hence $w_\kappa = w$. For the general case, by testing with $v = c_\kappa := \int_\Omega w_\kappa(x) \, dx$ in $(\mathcal{P}_v^\kappa)$, we have

$$\begin{aligned} a_\kappa(w_\kappa, c_\kappa) &= a(w_\kappa, c_\kappa) + \kappa^{-1} |\Omega| c_\kappa^2 = \kappa^{-1} |\Omega| c_\kappa^2 \\ &= \ell(c_\kappa) \leq C (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)}) \|c_\kappa\|_{H^1(\Omega)} \\ &= C (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)}) |c_\kappa| \sqrt{|\Omega|}. \end{aligned}$$

Hence, $|c_\kappa| \leq C' \kappa (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)})$.

2. We have

$$\begin{aligned} a_\kappa(w_\kappa - w, w_\kappa - w) &= a(w_\kappa - w, w_\kappa - w) + \kappa^{-1} \left| \int_\Omega w_\kappa(x) \, dx \right|^2 \\ &= \|\nabla(w_\kappa - w)\|_{(L^2(\Omega))^d}^2 + \kappa^{-1} \left| \int_\Omega (w_\kappa(x) - w(x)) \, dx \right|^2 \\ &\geq \min\left(1, \frac{1}{\kappa_0}\right) \left(\|\nabla(w_\kappa - w)\|_{(L^2(\Omega))^d}^2 + \left| \int_\Omega (w_\kappa - w) \, dx \right|^2 \right) \\ &\geq \frac{\min\left(1, \frac{1}{\kappa_0}\right)}{C_P + 1} \|w_\kappa - w\|_{H^1(\Omega)}^2, \end{aligned}$$

where $\kappa \leq \kappa_0$ and C_P is the (generalized Poincaré–Wirtinger) constant such that $\|\phi\|_{L^2(\Omega)}^2 \leq C_P (\|\nabla \phi\|_{(L^2(\Omega))^d}^2 + |\int_\Omega \phi \, dx|^2)$ for all $\phi \in H^1(\Omega)$. We assume $v = w - w_\kappa + c_\kappa/|\Omega|$, then $v \in H_\#^1(\Omega)$. Note that

$$\begin{aligned} a_\kappa(w_\kappa - w, w_\kappa - w) &= a_\kappa(w_\kappa - w, w_\kappa - w + v) = \frac{c_\kappa}{|\Omega|} a_\kappa(w_\kappa - w, 1) \\ &= \frac{c_\kappa}{|\Omega|} a(w_\kappa - w, 1) + c_\kappa \kappa^{-1} \left(\int_\Omega w_\kappa(x) \, dx \right) \\ &= \kappa^{-1} c_\kappa^2. \end{aligned}$$

Hence, we have

$$\|w_\kappa - w\|_{H^1(\Omega)}^2 \leq \frac{\kappa^{-1}}{\alpha} \left| \int_\Omega w_\kappa(x) \, dx \right|^2 \stackrel{\text{Question 1}}{\leq} C \kappa (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)})^2.$$

3. By the Lax-Milgram theorem, $(\mathcal{P}_v^\epsilon)$ admits a unique solution w_ϵ since the bilinear form $a_\epsilon(\cdot, \cdot)$ is continuous and coercive on $H^1(\Omega)$ and the linear form $\ell(\cdot)$ is continuous on $H^1(\Omega)$. Assume $\ell(1) = 0$. We have

$$0 = \ell(1) = a_\epsilon(w_\epsilon, 1) = a(w_\epsilon, 1) + \epsilon \int_\Omega w_\epsilon(x) \, dx = \epsilon \int_\Omega w_\epsilon(x) \, dx.$$

Hence, $w_\epsilon \in H_\#^1(\Omega)$. Moreover, we have

$$\begin{aligned} a(w, w_\epsilon) &= \ell(w_\epsilon) = a_\epsilon(w_\epsilon, w_\epsilon) \\ &= \int_\Omega |\nabla w_\epsilon(x)|^2 \, dx + \epsilon \int_\Omega w_\epsilon(x)^2 \, dx \\ &\geq \int_\Omega |\nabla w_\epsilon(x)|^2 \, dx. \end{aligned}$$

By Cauchy-Schwarz inequality, we have

$$\|\nabla w_\epsilon\|_{(L^2(\Omega))^d}^2 \leq a(w, w_\epsilon) \leq \|\nabla w\|_{(L^2(\Omega))^d} \cdot \|\nabla w_\epsilon\|_{(L^2(\Omega))^d}.$$

Hence, by Poincaré-Wirtinger inequality, we have

$$\|w_\epsilon\|_{H^1(\Omega)} \leq C_P \|\nabla w_\epsilon\|_{(L^2(\Omega))^d} \leq C_P \|\nabla w\|_{(L^2(\Omega))^d} \stackrel{\text{Problem 11}}{\leq} C (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)}).$$

4. By question 3, $w_\epsilon \in H_{\#}^1(\Omega)$. By Poincaré-Wirtinger inequality, we only need to prove that

$$\|\nabla(w - w_\epsilon)\|_{(L^2(\Omega))^d} \leq C\epsilon (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)}).$$

Note that

$$\begin{aligned} \|\nabla(w - w_\epsilon)\|_{L^2(\Omega)}^2 &= a(w - w_\epsilon, w - w_\epsilon) \\ &= a(w, w - w_\epsilon) - a(w_\epsilon, w - w_\epsilon) \\ &= \ell(w - w_\epsilon) - a_\epsilon(w_\epsilon, w - w_\epsilon) + \epsilon \int_{\Omega} w_\epsilon(x)(w(x) - w_\epsilon(x)) \, dx \\ &\leq \epsilon \|w_\epsilon\|_{L^2(\Omega)} \|w - w_\epsilon\|_{L^2(\Omega)} \\ &\leq C_P \epsilon \|w_\epsilon\|_{H^1(\Omega)} \|\nabla(w - w_\epsilon)\|_{(L^2(\Omega))^d}. \end{aligned}$$

Hence, we have

$$\|\nabla(w - w_\epsilon)\|_{(L^2(\Omega))^d} \leq C_P \epsilon \|w_\epsilon\|_{H^1(\Omega)} \stackrel{\text{Question 3}}{\leq} C\epsilon (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)}).$$

Problem 13. (Error estimation in 1D) For $\Omega = (0, 1)$ and $f \in L^2(\Omega)$, consider the problem: Find $u \in H^1(\Omega)$ such that

$$\begin{cases} -u'' = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases} \quad (\text{P1})$$

1. Write the variational formulation associated with this problem. We know that the hypotheses of the Lax-Milgram theorem are satisfied for this formulation, which guarantees that it has a unique solution $u \in H_0^1(\Omega)$.
2. Let u_h denote the approximation of the solution of (P1) by $\mathbb{P}^1$ Lagrange finite elements. Specify this definition, in particular by giving a meaning to h .
3. Establish the error estimate

$$\|u - u_h\|_{H_0^1(\Omega)} \leq Ch \|f\|_{L^2(\Omega)},$$

for some constant C that does not depend on f and h . This is a convergence result in the $H_0^1(\Omega)$ norm of order 1 in h .

4. Show that in the $L^2(\Omega)$ norm, we have the error estimate of order 2 (hence better)

$$\|u - u_h\|_{L^2(\Omega)} \leq Ch^2 \|f\|_{L^2(\Omega)},$$

where the constant C does not depend on f and h . This last result is called the Aubin-Nitsche lemma.

Solution 13. 1. The variational formulation is: Find $u \in H_0^1(\Omega)$ such that for every $v \in H_0^1(\Omega)$,

$$\int_{\Omega} u'(x)v'(x) \, dx = \int_{\Omega} f(x)v(x) \, dx.$$

2. Suppose $h = \frac{1}{N}$, where N is the number of subintervals in the partition of Ω . We define the finite element space

$$V_h = \{\phi \in C^0(\bar{\Omega}) \mid \phi|_{[x_i, x_{i+1}]} \in \mathbb{P}^1([x_i, x_{i+1}]), \forall i = 0, \dots, N-1; \phi(0) = \phi(1) = 0\}.$$

The approximation u_h of the solution of (P1) by $\mathbb{P}^1$ Lagrange finite elements is defined as the unique element of V_h such that for every $v_h \in V_h$,

$$\int_{\Omega} u_h'(x)v_h'(x) \, dx = \int_{\Omega} f(x)v_h(x) \, dx.$$

The Lax-Milgram theorem guarantees that u_h is well-defined.

3. Suppose

$$\begin{aligned} a(u, v) &= \int_{\Omega} u'(x) v'(x) \, dx, \\ \ell(v) &= \int_{\Omega} f(x) v(x) \, dx. \end{aligned}$$

Suppose M, ν the continuity and coercivity constants of $a(\cdot, \cdot)$, respectively. Then we have

$$\begin{aligned} \nu \|u - u_h\|_{H_0^1(\Omega)}^2 &\leq a(u - u_h, u - u_h) \\ &= a(u - u_h, u - v_h) \\ &\leq M \|u - u_h\|_{H_0^1(\Omega)} \|u - v_h\|_{H_0^1(\Omega)}, \quad \forall v_h \in V_h. \end{aligned}$$

So we have

$$\|u - u_h\|_{H_0^1(\Omega)} \leq \frac{M}{\nu} \inf_{v_h \in V_h} \|u - v_h\|_{H_0^1(\Omega)}.$$

Taking v_h as the piecewise linear interpolant of u on the partition of Ω into subintervals, we have

$$v_h(x) = \Pi_h^1 u := \sum_{i=0}^{N-1} u(x_i) \phi_i(x),$$

where ϕ_i is the hat function associated with the node x_i . Using the fact that $u \in H^2(\Omega)$, we have

$$\begin{aligned} \|u - \Pi_h^1 u\|_{H_0^1(\Omega)}^2 &= \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} \left(u'(x) - \frac{u(x_{i+1}) - u(x_i)}{x_{i+1} - x_i} \right)^2 \, dx \\ &= \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} \left(\frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} (u'(x) - u'(t)) \, dt \right)^2 \, dx \\ &\leq \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} (u'(x) - u'(t))^2 \, dt \, dx \\ &\leq \sum_{i=0}^{N-1} \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} \int_{x_i}^{x_{i+1}} |x - t| \int_{x_i}^{x_{i+1}} (u''(s))^2 \, ds \, dt \, dx \\ &\leq \sum_{i=0}^{N-1} \frac{(x_{i+1} - x_i)^2}{3} \int_{x_i}^{x_{i+1}} (u''(s))^2 \, ds \\ &= \frac{h^2}{3} \|u''\|_{L^2(\Omega)}^2 = \frac{h^2}{3} \|f\|_{L^2(\Omega)}^2. \end{aligned}$$

Hence, we have

$$\|u - u_h\|_{H_0^1(\Omega)} \leq \frac{M}{\nu} \|u - \Pi_h^1 u\|_{H_0^1(\Omega)} \leq Ch \|f\|_{L^2(\Omega)},$$

where $C = \frac{M}{\nu\sqrt{3}}$ is a constant independent of f and h .

4. Consider the following problem: Find $w \in H_0^1(\Omega)$ such that

$$\begin{cases} -w'' = u - u_h & \text{in } \Omega, \\ w = 0 & \text{on } \partial\Omega. \end{cases}$$

By the variational formulation of this problem, we have for every $v \in H_0^1(\Omega)$,

$$\int_{\Omega} w'(x) v'(x) \, dx = \int_{\Omega} (u(x) - u_h(x)) v(x) \, dx.$$

Taking $v = u - u_h$ in the equation above, we have

$$\begin{aligned} \|u - u_h\|_{L^2(\Omega)}^2 &= \int_{\Omega} w'(x) (u(x) - u_h(x))' \, dx \\ &= a(u - u_h, w) \\ &= a(u - u_h, w - \Pi_h^1 w) \\ &\leq \|(u - u_h)'\|_{L^2(\Omega)} \cdot \|(w - \Pi_h^1 w)'\|_{L^2(\Omega)}. \end{aligned}$$

By the previous error estimate, we have

$$\begin{aligned}\|(w - \Pi_h^1 w)'\|_{L^2(\Omega)} &\leq \|w - \Pi_h^1 w\|_{H_0^1(\Omega)} \leq Ch \|w''\|_{L^2(\Omega)} = Ch \|u - u_h\|_{L^2(\Omega)}. \\ \|(u - u_h)'\|_{L^2(\Omega)} &\leq \|u - u_h\|_{H_0^1(\Omega)} \leq Ch \|f\|_{L^2(\Omega)}.\end{aligned}$$

Hence, we have

$$\|u - u_h\|_{L^2(\Omega)} \leq Ch^2 \|f\|_{L^2(\Omega)},$$

where C is a constant independent of f and h .

Problem 14. (Bramble-Hilbert theorem and finite elements) Let $\Omega \subset \mathbb{R}^2$ be a bounded connected domain. Define the set $F := \{u \in H^2(\Omega) \mid |u|_{H^2(\Omega)} = 0\}$ where

$$|u|_{H^2(\Omega)} = \left(\int_{\Omega} |\partial_x^2 u|^2 + |\partial_y^2 u|^2 + 2|\partial_x \partial_y u|^2 \, dx \, dy \right)^{1/2}.$$

1. Show that F is a closed vector subspace of $H^2(\Omega)$. More precisely, establish that the functions in F are $\mathbb{P}^1$ on Ω so that F is finite-dimensional.

Let $\Pi_F : H^2(\Omega) \rightarrow F$ be the projector such that for every $u \in H^2(\Omega)$, $\Pi_F(u)$ is the element of F satisfying

$$(\Pi_F(u), f)_{H^2(\Omega)} = (u, f)_{H^2(\Omega)}, \quad \forall f \in F.$$

2. Show that Π_F is a bounded operator and that $\|\Pi_F(u)\|_{H^2(\Omega)} \leq \|u\|_{H^2(\Omega)}$ for all $u \in H^2(\Omega)$.
3. Prove the existence of a constant $C > 0$ such that

$$\|u\|_{H^1(\Omega)} \leq C (|u|_{H^2(\Omega)} + \|\Pi_F(u)\|_{H^1(\Omega)}), \quad \forall u \in H^2(\Omega).$$

Observe the analogy with the Poincaré-Wirtinger inequality

$$\|u\|_{L^2(\Omega)} \leq C (|u|_{H^1(\Omega)} + |m(u)|), \quad \forall u \in H^1(\Omega),$$

for connected Ω with $m(u) = |\Omega|^{-1} \int_{\Omega} u \, d\Omega$.

4. Deduce the Bramble-Hilbert theorem: there exists a constant $C > 0$ such that

$$\|u - \Pi_F(u)\|_{H^1(\Omega)} \leq C |u|_{H^2(\Omega)}, \quad \forall u \in H^2(\Omega).$$

This result is used in the proof of finite element error estimates. Note that this inequality is similar to, again for connected Ω ,

$$\|u - m(u)\|_{L^2(\Omega)} \leq C |u|_{H^1(\Omega)}, \quad \forall u \in H^1(\Omega).$$

Solution 14. 1. The map $\rho : u \mapsto |u|_{H^2(\Omega)}$ is continuous on $H^2(\Omega)$ since

$$|\rho(u) - \rho(v)| \leq \|u - v\|_{H^2(\Omega)}, \quad \forall u, v \in H^2(\Omega).$$

Therefore, $F = \ker(\rho)$ is a closed vector subspace of $H^2(\Omega)$. Moreover, if $u \in F$, then $\partial_x^2 u = \partial_y^2 u = \partial_x \partial_y u = 0$ in Ω . Since Ω is connected, $\partial_x u$ and $\partial_y u$ must be constant, and thus u is a linear function, i.e., $u \in \mathbb{P}^1(\Omega)$. Moreover, the dimension of $\mathbb{P}^1(\Omega)$ is 3, so F is finite-dimensional.

2. Note that for every $u \in H^2(\Omega)$, we have

$$\|\Pi_F(u)\|_{H^2(\Omega)}^2 = (\Pi_F(u), \Pi_F(u))_{H^2(\Omega)} = (u, \Pi_F(u))_{H^2(\Omega)} \leq \|u\|_{H^2(\Omega)} \|\Pi_F(u)\|_{H^2(\Omega)}.$$

Hence, we have

$$\|\Pi_F(u)\|_{H^2(\Omega)} \leq \|u\|_{H^2(\Omega)}, \quad \forall u \in H^2(\Omega),$$

which implies that Π_F is a bounded operator.

3. We argue by contradiction. Assume that for every $n \in \mathbb{N}_*$, there exists $u_n \in H^2(\Omega)$ such that

$$\|u_n\|_{H^1(\Omega)} > n (|u_n|_{H^2(\Omega)} + \|\Pi_F(u_n)\|_{H^1(\Omega)}).$$

Define $v_n = u_n / \|u_n\|_{H^1(\Omega)}$. Then we have

$$\|v_n\|_{H^1(\Omega)} = 1, \quad |v_n|_{H^2(\Omega)} < \frac{1}{n}, \quad \|\Pi_F(v_n)\|_{H^1(\Omega)} < \frac{1}{n}.$$

It follows that (∇v_n) is bounded in $H^1(\Omega)^2$. By the Rellich-Kondrachov compactness theorem, there exists a subsequence of (v_n) (still denoted by (v_n)) such that ∇v_n converges in $L^2(\Omega)^2$, which implies that (∇v_n) is a Cauchy sequence in $L^2(\Omega)^2$. Since (v_n) is bounded in $H^1(\Omega)$, by Rellich-Kondrachov compactness theorem again, there exists a subsequence of (v_n) (still denoted by (v_n)) such that (v_n) converges in $L^2(\Omega)$, which implies that (v_n) is a Cauchy sequence in $L^2(\Omega)$. Hence, (v_n) is a Cauchy sequence in $H^1(\Omega)$. Since $|v_n|_{H^2(\Omega)} < \frac{1}{n}$, we conclude that (v_n) is a Cauchy sequence in $H^2(\Omega)$. Therefore, there exists $v \in H^2(\Omega)$ such that v_n converges to v in $H^2(\Omega)$, which implies that $\|v\|_{H^1(\Omega)} = 1$, $|v|_{H^2(\Omega)} = 0$ and $\|\Pi_F(v)\|_{H^1(\Omega)} = 0$. Since $|v|_{H^2(\Omega)} = 0$, we have $v \in F$. Since $\Pi_F(v) = v$, we have $\|\Pi_F(v)\|_{H^1(\Omega)} = \|v\|_{H^1(\Omega)} = 1$, which is a contradiction.

4. Suppose $u \in H^2(\Omega)$, and $v = u - \Pi_F(u)$. By the previous question, we have

$$\|v\|_{H^1(\Omega)} \leq C(|v|_{H^2(\Omega)} + \|\Pi_F(v)\|_{H^1(\Omega)}).$$

Since $\Pi_F(v) = 0$, we have

$$\|u - \Pi_F(u)\|_{H^1(\Omega)} = \|v\|_{H^1(\Omega)} \leq C|v|_{H^2(\Omega)} = C|u|_{H^2(\Omega)}.$$

Problem 15. (Inverse inequality for finite elements) Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain. Let $\mathcal{T}_h = (T_\ell)_{1 \leq \ell \leq N}$ be a regular triangulation of Ω with mesh size

$$h := \max_{T_\ell \in \mathcal{T}_h} h_\ell \quad \text{where} \quad h_\ell = \text{diam}(T_\ell).$$

Consider the $\mathbb{P}^1$ finite element space

$$V_h := \{v_h \in C^0(\bar{\Omega}) \mid v_h|_{T_\ell} \in \mathbb{P}_1(T_\ell), \forall T_\ell \in \mathcal{T}_h\}.$$

Let $\hat{T}$ denote the reference simplex.

1. **Inverse inequality on the reference triangle $\hat{T}$.** Show that there exists a constant $C > 0$, independent of h , such that for every $\hat{v} \in \mathbb{P}^1(\hat{T})$

$$\|\nabla \hat{v}\|_{L^2(\hat{T})^2} \leq C\|\hat{v}\|_{L^2(\hat{T})}.$$

2. **Mapping to the reference element.** Let $\phi_\ell : \hat{T} \rightarrow T_\ell$ be the affine map that takes the reference triangle $\hat{T}$ to T_ℓ : $x = \phi_\ell(\hat{x}) = F_\ell \hat{x} + t_\ell$, where F_ℓ and t_ℓ are a matrix and a vector whose values depend only on the vertices of triangle T_ℓ . Show that for every $v_h \in \mathbb{P}^1(T_\ell)$,

$$\|v_h\|_{L^2(T_\ell)}^2 = |\det F_\ell| \cdot \|\hat{v}_h\|_{L^2(\hat{T})}^2 \quad \text{and} \quad \|\nabla v_h\|_{L^2(T_\ell)^2}^2 = |\det F_\ell| \cdot \|F_\ell^{-T} \nabla \hat{v}_h\|_{L^2(\hat{T})^2}^2,$$

where $\hat{v}_h = v_h \circ \phi_\ell$.

3. Show that there exists a constant $C > 0$, independent of h and T_ℓ , such that for every $v_h \in V_h$ and every $T_\ell \in \mathcal{T}_h$,

$$\|\nabla v_h\|_{L^2(T_\ell)^2} \leq Ch_\ell^{-1} \|v_h\|_{L^2(T_\ell)}.$$

4. **Global inverse inequality.** Formulate a condition on the mesh such that there exists $C > 0$, independent of h , with

$$\|\nabla v_h\|_{L^2(\Omega)^2} \leq Ch^{-1} \|v_h\|_{L^2(\Omega)}, \quad \forall v_h \in V_h.$$

Solution 15. 1. Since V_h is finite-dimensional, all norms are equivalent on V_h . Therefore, we have

$$\|\hat{v}\|_{H^1(\hat{T})} \leq C\|\hat{v}\|_{L^2(\hat{T})}, \quad \forall \hat{v} \in \mathbb{P}^1(\hat{T}),$$

which implies that

$$\|\nabla \hat{v}\|_{L^2(\hat{T})^2} \leq C\|\hat{v}\|_{L^2(\hat{T})}, \quad \forall \hat{v} \in \mathbb{P}^1(\hat{T}).$$

2. By the change of variable formula, we have

$$\begin{aligned} \|v_h\|_{L^2(T_\ell)}^2 &= \int_{T_\ell} |v_h(x)|^2 dx = \int_{\hat{T}} |v_h(\phi_\ell(\hat{x}))|^2 |\det F_\ell| d\hat{x} = |\det F_\ell| \cdot \|\hat{v}_h\|_{L^2(\hat{T})}^2, \\ \|\nabla v_h\|_{L^2(T_\ell)^2}^2 &= \int_{T_\ell} |\nabla v_h(x)|^2 dx = \int_{\hat{T}} |F_\ell^{-T} \nabla \hat{v}_h(\hat{x})|^2 |\det F_\ell| d\hat{x} = |\det F_\ell| \cdot \|F_\ell^{-T} \nabla \hat{v}_h\|_{L^2(\hat{T})^2}^2. \end{aligned}$$

3. By the previous questions, we have

$$\begin{aligned}\|\nabla v_h\|_{L^2(T_\ell)^2}^2 &= |\det F_\ell| \cdot \|F_\ell^{-T} \nabla \hat{v}_h\|_{L^2(\hat{T})^2}^2 \\ &\leq |\det F_\ell| \cdot \|F_\ell^{-T}\|_2^2 \cdot \|\nabla \hat{v}_h\|_{L^2(\hat{T})^2}^2.\end{aligned}$$

where $\|F_\ell^{-T}\|_2$ is the operator norm of F_ℓ^{-T} .

Note that

$$\|F_\ell^{-T}\|_2 = \sup_{x \in \mathbb{R}^2 \setminus \{0\}} \frac{\|F_\ell^{-T} x\|_2}{\|x\|_2} \leq \frac{C}{h_\ell},$$

where C is a constant independent of h and T_ℓ .

Hence, we have

$$\|\nabla v_h\|_{L^2(T_\ell)^2} \leq |\det F_\ell|^{1/2} \cdot \frac{C}{h_\ell} \cdot \|\nabla \hat{v}_h\|_{L^2(\hat{T})^2}.$$

Since $\|\nabla \hat{v}_h\|_{L^2(\hat{T})^2} \leq \hat{C} \|\hat{v}_h\|_{L^2(\hat{T})}$ by the norm equivalence on the reference element $\hat{T}$, and $\|v_h\|_{L^2(T_\ell)} = |\det F_\ell|^{1/2} \|\hat{v}_h\|_{L^2(\hat{T})}$, the factors $|\det F_\ell|^{1/2}$ cancel and we have

$$\|\nabla v_h\|_{L^2(T_\ell)^2} \leq C h_\ell^{-1} \|v_h\|_{L^2(T_\ell)},$$

where C is a constant independent of h_ℓ and T_ℓ .

4. Note that

$$\|\nabla v_h\|_{L^2(\Omega)^2}^2 = \sum_{T_\ell \in \mathcal{T}_h} \|\nabla v_h\|_{L^2(T_\ell)^2}^2 \text{ and } \|v_h\|_{L^2(\Omega)}^2 = \sum_{T_\ell \in \mathcal{T}_h} \|v_h\|_{L^2(T_\ell)}^2.$$

We only need to ensure that $h_i \geq ch$ for some constant $c > 0$ for all i to get the global inverse inequality.

Problem 16. (Weak convergence not strong) For $a < b$ in $[-\infty, +\infty]$, the space $L^2(a, b)$ is equipped with its usual inner product. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be a non-zero smooth function with compact support.

1. **(Vanishing)** Define u_n by $u_n(x) = \phi(x - n)$. Show that (u_n) converges weakly to 0 in $L^2(\mathbb{R})$ but not strongly.
2. **(Concentration)** Define v_n by $v_n(x) = \sqrt{n} \phi(nx)$. Show that (v_n) converges weakly to 0 in $L^2(-1, 1)$ but not strongly.
3. **(Oscillations)** For $w : \mathbb{R} \rightarrow \mathbb{R}$ a 2π -periodic function, define w_n by $w_n(x) = w(nx)$. Show that (w_n) converges weakly to the mean value of w in $L^2((0, 2\pi))$ but not strongly in general.

Solution 16. 1. It suffices to show that

$$\int_{\mathbb{R}} u_n(x) \psi(x) dx \rightarrow 0, \quad \forall \psi \in L^2(\mathbb{R}).$$

Since $C_c^\infty(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, there exists $\tilde{\psi} \in C_c^\infty(\mathbb{R})$ such that $\|\tilde{\psi} - \psi\|_{L^2(\mathbb{R})} < \varepsilon$. Then we have

$$\begin{aligned}\left| \int_{\mathbb{R}} u_n(x) \psi(x) dx \right| &\leq \left| \int_{\mathbb{R}} u_n(x) (\psi(x) - \tilde{\psi}(x)) dx \right| + \left| \int_{\mathbb{R}} u_n(x) \tilde{\psi}(x) dx \right| \\ &\leq \|\phi\|_{L^2(\mathbb{R})} \|\psi - \tilde{\psi}\|_{L^2(\mathbb{R})} + \left| \int_{\mathbb{R}} u_n(x) \tilde{\psi}(x) dx \right|.\end{aligned}$$

As $n \rightarrow \infty$, we have $\text{supp}(u_n) \cap \text{supp}(\tilde{\psi}) = \emptyset$ for large n , hence

$$\int_{\mathbb{R}} u_n(x) \tilde{\psi}(x) dx \rightarrow 0.$$

Hence we have

$$\limsup_{n \rightarrow \infty} \left| \int_{\mathbb{R}} u_n(x) \psi(x) dx \right| \leq \varepsilon \|\phi\|_{L^2(\mathbb{R})}.$$

Since ε is arbitrary, we have

$$\int_{\mathbb{R}} u_n(x) \psi(x) dx \rightarrow 0, \quad \forall \psi \in L^2(\mathbb{R}).$$

However, we have $\|u_n\|_{L^2(\mathbb{R})} = \|\phi\|_{L^2(\mathbb{R})} > 0$ for all n , so (u_n) does not converge strongly to 0.

2. It suffices to show that

$$\int_{-1}^1 v_n(x)\psi(x) dx \rightarrow 0, \quad \forall \psi \in L^2(-1, 1).$$

Since $C_c^\infty(-1, 1)$ is dense in $L^2(-1, 1)$, there exists $\tilde{\psi} \in C_c^\infty(-1, 1)$ such that $\|\tilde{\psi} - \psi\|_{L^2(-1, 1)} < \varepsilon$. Then we have

$$\begin{aligned} \left| \int_{-1}^1 v_n(x)\psi(x) dx \right| &\leq \left| \int_{-1}^1 v_n(x)(\psi(x) - \tilde{\psi}(x)) dx \right| + \left| \int_{-1}^1 v_n(x)\tilde{\psi}(x) dx \right| \\ &\leq \|v_n\|_{L^2(-1, 1)} \|\psi - \tilde{\psi}\|_{L^2(-1, 1)} + \left| \int_{-1}^1 v_n(x)\tilde{\psi}(x) dx \right|. \end{aligned}$$

Note that

$$\|v_n\|_{L^2(-1, 1)}^2 = \int_{-1}^1 n|\phi(nx)|^2 dx = \int_{-n}^n |\phi(y)|^2 dy \rightarrow \|\phi\|_{L^2(\mathbb{R})}^2,$$

and $\text{supp}(v_n) = \frac{1}{n} \text{supp}(\phi)$. Since $\tilde{\psi}$ has compact support, we obtain $|\text{supp}(v_n) \cap \text{supp}(\tilde{\psi})| \rightarrow 0$. It follows that

$$\limsup_{n \rightarrow \infty} \left| \int_{-1}^1 v_n(x)\psi(x) dx \right| \leq \varepsilon \limsup_{n \rightarrow \infty} \|v_n\|_{L^2(-1, 1)} = \varepsilon \|\phi\|_{L^2(\mathbb{R})}.$$

Since ε is arbitrary, we have

$$\int_{-1}^1 v_n(x)\psi(x) dx \rightarrow 0, \quad \forall \psi \in L^2(-1, 1).$$

However, we have $\|v_n\|_{L^2(-1, 1)} \rightarrow \|\phi\|_{L^2(\mathbb{R})} > 0$ as $n \rightarrow \infty$, so (v_n) does not converge strongly to 0.

3. Suppose w has mean value $m(w)$, where

$$m(w) = \frac{1}{2\pi} \int_0^{2\pi} w(x) dx.$$

It suffices to show that

$$\int_0^{2\pi} w_n(x)\psi(x) dx \rightarrow m(w) \int_0^{2\pi} \psi(x) dx, \quad \forall \psi \in L^2((0, 2\pi)).$$

Since $C_c^\infty((0, 2\pi))$ is dense in $L^2((0, 2\pi))$, there exists $\tilde{\psi} \in C_c^\infty((0, 2\pi))$ such that $\|\tilde{\psi} - \psi\|_{L^2((0, 2\pi))} < \varepsilon$. Then we have

$$\begin{aligned} &\left| \int_0^{2\pi} w_n(x)\psi(x) dx - m(w) \int_0^{2\pi} \psi(x) dx \right| \\ &\leq \left| \int_0^{2\pi} w_n(x)(\psi(x) - \tilde{\psi}(x)) dx \right| + \left| \int_0^{2\pi} w_n(x)\tilde{\psi}(x) dx - m(w) \int_0^{2\pi} \tilde{\psi}(x) dx \right| + m(w) \left| \int_0^{2\pi} (\tilde{\psi}(x) - \psi(x)) dx \right| \\ &\leq \|w_n\|_{L^2((0, 2\pi))} \|\psi - \tilde{\psi}\|_{L^2((0, 2\pi))} + \left| \int_0^{2\pi} w_n(x)\tilde{\psi}(x) dx - m(w) \int_0^{2\pi} \tilde{\psi}(x) dx \right| + m(w) \|\tilde{\psi} - \psi\|_{L^1((0, 2\pi))} \\ &= \varepsilon (\|w\|_{L^2((0, 2\pi))} + m(w)) + \left| \int_0^{2\pi} (w_n(x) - m(w))\tilde{\psi}(x) dx \right|. \end{aligned}$$

Suppose $v(x) = w(x) - m(w)$. Then v has mean value 0, and we have

$$\int_0^{2\pi} (w_n(x) - m(w))\tilde{\psi}(x) dx = \int_0^{2\pi} v(nx)\tilde{\psi}(x) dx = -\frac{1}{n} \int_0^{2\pi} \mu(nx)\psi'(x) dx,$$

where $\mu(x) = \int_0^x v(y) dy$. Since μ is bounded, we have

$$\left| \int_0^{2\pi} (w_n(x) - m(w))\tilde{\psi}(x) dx \right| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Hence we have

$$\limsup_{n \rightarrow \infty} \left| \int_0^{2\pi} w_n(x)\psi(x) dx - m(w) \int_0^{2\pi} \psi(x) dx \right| \leq \varepsilon (\|w\|_{L^2((0, 2\pi))} + m(w)).$$

Since ε is arbitrary, we have

$$\int_0^{2\pi} w_n(x)\psi(x) dx \rightarrow m(w) \int_0^{2\pi} \psi(x) dx, \quad \forall \psi \in L^2((0, 2\pi)).$$

Problem 17. (Weak convergence and convexity) Let V be a Hilbert space.

1. Show that a set that is closed for weak convergence is closed for strong convergence. The converse is generally false. Nevertheless, prove that if K is convex, then K closed for strong convergence implies K closed for weak convergence.
2. Prove the following result: Let $F : V \rightarrow \mathbb{R}$ be convex, continuous, and let (u_n) be a sequence in V that converges weakly to u . Then

$$F(u) \leq \liminf_n F(u_n).$$

Which inequality does this theorem generalize?

3. Let K be a non-empty closed convex subset of V and $J : V \rightarrow \mathbb{R}$ a convex, continuous, and coercive (i.e., $J(v) \rightarrow +\infty$ as $\|v\| \rightarrow \infty$) functional on K . Show that there exists $u \in K$ such that $J(u) = \inf_{v \in K} J(v)$.

Solution 17. 1. The first result is straightforward. Now we give a counterexample to show that the converse is generally false. We consider the unit sphere S of space $\ell^2 := \{(x_n)_{n \in \mathbb{N}} \mid \sum_{n=0}^{\infty} |x_n|^2 < \infty\}$, the sequence of standard basis vectors e_n satisfies $\|e_n\| = 1$ but $e_n \rightarrow 0$ weakly, and $0 \notin S$.

Let K be a convex and strongly closed subset of V . We show that K is weakly closed. By the Hahn-Banach separation theorem, any closed convex set in a locally convex space can be represented as the intersection of all closed half-spaces containing it. More precisely,

$$K = \bigcap_{f \in V^*, \alpha \in \mathbb{R}} \{x \in V : f(x) \leq \alpha\}$$

where the intersection is taken over all pairs (f, α) such that $K \subset \{x : f(x) \leq \alpha\}$. Each closed half-space $\{x : f(x) \leq \alpha\}$ is weakly closed because f is weakly continuous (linear functionals are continuous for the weak topology). Therefore, the intersection of weakly closed sets is weakly closed, so K is weakly closed. (Note: This characterization holds because K is convex and closed.)

2. Consider the epigraph of F :

$$\text{Epi}(F) := \{(v, \lambda) \in V \times \mathbb{R} : F(v) \leq \lambda\}.$$

Since F is convex, $\text{Epi}(F)$ is convex. Moreover, because F is continuous, $\text{Epi}(F)$ is closed: if $(v_n, \lambda_n) \in \text{Epi}(F)$ converges to (v, λ) in $V \times \mathbb{R}$ (strongly in V and in $\mathbb{R}$), then $F(v_n) \leq \lambda_n$ and by continuity $F(v) = \lim F(v_n) \leq \lim \lambda_n = \lambda$, so $(v, \lambda) \in \text{Epi}(F)$. Hence $\text{Epi}(F)$ is a closed convex set. By the previous question, it is also weakly closed in $V \times \mathbb{R}$ (where the weak topology on the product is the product of the weak topology on V and the usual topology on $\mathbb{R}$, since $\mathbb{R}$ is finite-dimensional).

Now let $u_n \rightharpoonup u$ weakly in V . Set $L = \liminf_n F(u_n)$. Then there exists a subsequence (u_{n_k}) such that $F(u_{n_k}) \rightarrow L$ as $k \rightarrow \infty$. Since $u_{n_k} \rightharpoonup u$ weakly, we have $(u_{n_k}, F(u_{n_k})) \rightharpoonup (u, L)$ weakly in $V \times \mathbb{R}$. As $\text{Epi}(F)$ is weakly closed and $(u_{n_k}, F(u_{n_k})) \in \text{Epi}(F)$ (because $F(u_{n_k}) \leq F(u_{n_k})$ trivially), we obtain $(u, L) \in \text{Epi}(F)$, i.e., $F(u) \leq L$. Therefore,

$$F(u) \leq \liminf_n F(u_n).$$

Generalization. This theorem generalizes Fatou's lemma.

3. Define $m = \inf_{v \in K} J(v)$. Since J is coercive, m is finite. Let $(u_n) \subset K$ be a minimizing sequence, i.e., $J(u_n) \rightarrow m$ as $n \rightarrow \infty$. We claim that (u_n) is bounded. Indeed, if it were unbounded, there would exist a subsequence (u_{n_k}) with $\|u_{n_k}\| \rightarrow \infty$, and by coercivity $J(u_{n_k}) \rightarrow \infty$, contradicting $J(u_n) \rightarrow m < \infty$. Hence (u_n) is bounded.

In a Hilbert space, every bounded sequence has a weakly convergent subsequence. Thus we can extract a subsequence, still denoted (u_{n_k}) , such that $u_{n_k} \rightharpoonup u$ weakly for some $u \in V$. Since K is convex and closed, it is weakly closed by the first question, so $u \in K$.

Now apply the result of the previous question to the convex continuous function J (restricted to K but it is defined on all V). We obtain

$$J(u) \leq \liminf_{k \rightarrow \infty} J(u_{n_k}) = m.$$

But m is the infimum, so $J(u) \geq m$. Hence $J(u) = m$, and u is a minimizer.

Problem 18. Let V be a Hilbert space. Consider $J : V \rightarrow \mathbb{R}_+$ a strictly convex, continuous, coercive functional, and $\phi : V \rightarrow \mathbb{R}_+$ a convex continuous function. We wish to minimize J under the constraint that the minimizer makes ϕ vanish.

1. Show that $K := \{v \in V \mid \phi(v) = 0\}$ is a convex subset of V . In Problem 17, we showed that J admits a minimum. What can be said about uniqueness?
2. Taking the constraint into account in the search for the minimum of J is often difficult to implement. We therefore propose to penalize it. This leads us to define, for $n \in \mathbb{N}$, the functional J_n by $J_n := J + n\phi$. Show that there exists a unique $u_n \in V$ such that

$$J_n(u_n) = \min_V J_n.$$

3. Verify that $J(u_n) \leq \min_K J$. Deduce that there exists a subsequence $(u_{\psi(n)})$ of (u_n) such that $(u_{\psi(n)})$ converges weakly to some u in V .
4. Show that $u \in K$.
5. Prove that $u = \operatorname{argmin}_K J$. Deduce that the whole sequence (u_n) converges weakly in V .
6. **Special case of a quadratic J .** Now suppose J is of the form

$$J(v) = \frac{1}{2}a(v, v) - \ell(v),$$

where $a(\cdot, \cdot)$ is a symmetric coercive bilinear form, $\ell(\cdot)$ is a linear form. Establish that the convergence of (u_n) to u is strong.

7. Suppose moreover that K is a vector subspace of V defined as the kernel of a continuous symmetric positive bilinear form

$$K = \{v \in V \mid \forall w \in V, b(v, w) = 0\} = \{v \in V \mid \phi(v) := \frac{1}{2}b(v, v) = 0\}.$$

Show that $(v \mapsto nb(u_n, v))$ converges to some $\lambda \in V'$ which we will express in terms of u .

Solution 18. 1. Suppose $v_1, v_2 \in K$ and $t \in [0, 1]$. Then we have

$$\phi(tv_1 + (1-t)v_2) \leq t\phi(v_1) + (1-t)\phi(v_2) = 0,$$

since ϕ is convex and $\phi(v_1) = \phi(v_2) = 0$. Moreover, since ϕ is non-negative, we have $\phi(tv_1 + (1-t)v_2) = 0$, so $tv_1 + (1-t)v_2 \in K$. Hence K is convex. Since J is strictly convex, if there were two distinct minimizers $u_1, u_2 \in K$ such that $J(u_1) = J(u_2) = \inf_K J$, then for any $t \in (0, 1)$, we would have

$$J(tu_1 + (1-t)u_2) < tJ(u_1) + (1-t)J(u_2) = \inf_K J,$$

contradicting the definition of $\inf_K J$. Hence the minimizer is unique.

2. Since J_n is strictly convex, continuous, and coercive, there exists a unique $u_n \in V$ such that $J_n(u_n) = \min_V J_n$.
3. Suppose $\bar{u}$ is the unique minimizer of J on K . Then we have

$$J(u_n) \leq J(u_n) + n\phi(u_n) = J_n(u_n) \leq J_n(\bar{u}) = J(\bar{u}) = \min_K J,$$

Hence $(J(u_n))$ is bounded. Since J is coercive, (u_n) is bounded in V . Therefore, there exists a subsequence $(u_{\psi(n)})$ that converges weakly to some $u \in V$.

4. Since K is closed and convex, it is weakly closed. Therefore, the weak limit u of the subsequence $(u_{\psi(n)})$ belongs to K by Problem 17.
5. By Problem 17 and question 3, we have

$$J(u) \leq \liminf_{n \rightarrow \infty} J(u_{\psi(n)}) \leq \min_K J.$$

Since $u \in K$, u is the minimizer of J on K . Since the minimizer is unique, $u = \operatorname{argmin}_K J$. Moreover, since every subsequence of (u_n) has a subsequence that converges weakly to u , the whole sequence (u_n) converges weakly to u . By question 4, we have

$$\limsup J(u_n) \leq \min_K J = J(u) \leq \liminf J(u_n),$$

hence $J(u_n) \rightarrow J(u)$.

6. We have

$$J(v) \geq \frac{\alpha}{2} \|v\|^2 - \|\ell\| \|v\| \geq -\frac{\|\ell\|^2}{2\alpha}.$$

Suppose $M = \frac{\|\ell\|^2}{2\alpha}$. Then $J(v) \geq -M$ for all $v \in V$. Suppose $\tilde{J} = J + M \geq 0$, then we have

$$\begin{aligned} \|u - u_n\|_V^2 &\leq \frac{1}{\alpha} a(u - u_n, u - u_n) = \frac{1}{\alpha} (a(u, u) - 2a(u, u_n) + a(u_n, u_n)) \\ &= \frac{1}{\alpha} (2\tilde{J}(u_n) - 2M + 2\ell(u_n) - 2a(u, u_n) + a(u, u)). \end{aligned}$$

By question 5, we have $\tilde{J}(u_n) \rightarrow \tilde{J}(u)$. Moreover, since $u_n \rightharpoonup u$ weakly, we have $\ell(u_n) \rightarrow \ell(u)$ (by Riesz representation theorem) and $a(u, u_n) \rightarrow a(u, u)$. Hence $\|u - u_n\|_V^2 \rightarrow 0$, i.e., u_n converges strongly to u .

7. Here J_n is differentiable. So for the minimizer, it suffices to consider the zero of the derivative of J_n . Suppose u_n is the minimizer of J_n , then we have

$$\forall v \in V, \quad a(u_n, v) + nb(u_n, v) = \ell(v).$$

Therefore, we have

$$nb(u_n, \cdot) \rightarrow \ell(\cdot) - a(u, \cdot) \text{ in } V'.$$

Problem 19. (The Stokes system) We consider the Stokes problem in a bounded regular connected domain Ω where for $\mu > 0$ and $\underline{f} \in L^2(\Omega)^d$, we seek $(\underline{u}, p) \in H^1(\Omega)^d \times L^2(\Omega)$ such that

$$\begin{cases} -\mu \Delta \underline{u} + \nabla p = \underline{f} & \text{in } \Omega, \\ \operatorname{div} \underline{u} = 0 & \text{in } \Omega, \\ \underline{u} = 0 & \text{on } \partial\Omega, \end{cases}$$

where the vector Laplacian is the Laplacian applied componentwise.

1. Recall

$$\nabla \underline{u} = \left(\frac{\partial u_i}{\partial x_j} \right)_{1 \leq i, j \leq d} = (\nabla u_i^T)_{1 \leq i \leq d} \quad \text{and} \quad \operatorname{div} \underline{u} = \left(\sum_{j=1}^d \frac{\partial u_{ij}}{\partial x_j} \right)_{1 \leq i \leq d}.$$

Write a variational formulation of this problem involving only the variable $\underline{u}$.

2. Show that this variational formulation admits a unique solution.

3. Prove the equivalence between the variational formulation and the strong formulation.

4. Show that the variational formulation derives from the minimization of a functional J under constraint.

5. Define the functional J_ε such that for every $\underline{v} \in V = H_0^1(\Omega)^d$,

$$J_\varepsilon(\underline{v}) := J(\underline{v}) + \frac{1}{\varepsilon} \int_{\Omega} |\operatorname{div} \underline{v}|^2 \, d\Omega.$$

Write a variational formulation satisfied by $(\underline{u}_\varepsilon, p_\varepsilon)$ associated with the minimization.

Solution 19. 1. Let $\underline{v} \in H_0^1(\Omega)^d$. Then we have

$$\begin{aligned} \int_{\Omega} \mu \nabla \underline{u} : \nabla \underline{v} \, d\Omega &= \sum_{i,j=1}^d \int_{\Omega} \mu \frac{\partial u_i}{\partial x_j} \frac{\partial v_i}{\partial x_j} \, d\Omega \\ &= - \sum_{i,j=1}^d \int_{\Omega} \mu v_i \frac{\partial^2 u_i}{\partial x_j^2} \, d\Omega + \sum_{i,j=1}^d \int_{\partial\Omega} \mu \frac{\partial u_i}{\partial x_j} v_i \nu_j \, dS \\ &= - \sum_{i=1}^d \int_{\Omega} \mu v_i \Delta u_i \, d\Omega = \int_{\Omega} \underline{v} \cdot (-\mu \Delta \underline{u}) \, d\Omega \\ &= \int_{\Omega} \underline{v} \cdot (\underline{f} - \nabla p) \, d\Omega = \int_{\Omega} \underline{v} \cdot \underline{f} \, d\Omega - \int_{\Omega} \underline{v} \cdot \nabla p \, d\Omega \\ &= \int_{\Omega} \underline{v} \cdot \underline{f} \, d\Omega + \int_{\Omega} p \operatorname{div} \underline{v} \, d\Omega. \end{aligned}$$

By considering $\underline{v} \in V := \{\underline{v} \in H_0^1(\Omega)^d : \operatorname{div} \underline{v} = 0\}$, we obtain the variational formulation involving only the variable $\underline{u}$, i.e., find $\underline{u} \in V$ such that

$$\int_{\Omega} \mu \nabla \underline{u} : \nabla \underline{v} \, d\Omega = \int_{\Omega} \underline{v} \cdot \underline{f} \, d\Omega, \quad \forall \underline{v} \in V.$$

2. By Poincaré inequality, we have

$$\int_{\Omega} \mu \nabla \underline{v} : \nabla \underline{v} \, d\Omega \geq C \|\underline{v}\|_{H^1(\Omega)^d}^2, \quad \forall \underline{v} \in V,$$

Hence we can apply the Lax-Milgram theorem to obtain the existence and uniqueness of the solution.

3. Suppose

$$\ell(\underline{v}) = \int_{\Omega} \mu \nabla \underline{u} : \nabla \underline{v} \, d\Omega - \int_{\Omega} \underline{v} \cdot \underline{f} \, d\Omega$$

is a continuous linear form on $H_0^1(\Omega)^d$ that vanishes on V . By de Rham's theorem, there exists a function $p \in L^2(\Omega)$ such that

$$\forall \underline{v} \in H_0^1(\Omega)^d, \quad \ell(\underline{v}) = \int_{\Omega} p \operatorname{div} \underline{v} \, d\Omega.$$

Suppose $\underline{\sigma} = \mu \nabla \underline{u} - pI$, then we have

$$\left| \int_{\Omega} \underline{\sigma} : \nabla \underline{v} \right| = \left| \int_{\Omega} \underline{v} \cdot \underline{f} \, d\Omega \right| \leq C \|\underline{v}\|_{H^1(\Omega)^d}.$$

This implies that $\underline{\sigma}$ has a weak divergence, and we have $\operatorname{div} \underline{\sigma} = -\underline{f}$ in the weak sense. Hence, we have

$$-\mu \Delta \underline{u} + \nabla p = \underline{f} \text{ in } \Omega \text{ a.e.}$$

Since $\underline{u} \in V$, we have $\operatorname{div} \underline{u} = 0$ in Ω a.e. and $\underline{u} = 0$ on $\partial\Omega$. Therefore, the variational formulation implies the strong formulation.

4. The energy functional is given by

$$J(\underline{v}) = \frac{\mu}{2} \int_{\Omega} |\nabla \underline{v}|^2 \, d\Omega - \int_{\Omega} \underline{v} \cdot \underline{f} \, d\Omega.$$

And the variational formulation can be derived from the minimization of J on V .

5. The variational formulation associated with the minimization of J_{ε} on V is given by: find $\underline{u}_{\varepsilon} \in H_0^1(\Omega)^d$ such that

$$\int_{\Omega} \mu \nabla \underline{u}_{\varepsilon} : \nabla \underline{v} \, d\Omega + \frac{1}{\varepsilon} \int_{\Omega} \operatorname{div} \underline{u}_{\varepsilon} \operatorname{div} \underline{v} \, d\Omega = \int_{\Omega} \underline{v} \cdot \underline{f} \, d\Omega, \quad \forall \underline{v} \in H_0^1(\Omega)^d.$$

(We can suppose $p_{\varepsilon} = -\frac{1}{\varepsilon} \operatorname{div} \underline{u}_{\varepsilon}$, then we can also write the variational formulation in the form of the Stokes system.)

Problem 20. (Poincaré Inequality and Laplacian Eigenvalues) Let $\Omega \subset \mathbb{R}^d$, $d \geq 1$, be a bounded domain with Lipschitz boundary. According to the Poincaré inequality, there exists a constant $C > 0$ such that

$$\|v\|_{L^2(\Omega)} \leq C \|\nabla v\|_{(L^2(\Omega))^d}, \quad \forall v \in H_0^1(\Omega). \quad (\text{P})$$

In this exercise, we show that one can take a smaller constant C in (P) and express its value in terms of the first eigenvalue of the Dirichlet Laplacian.

For this, we define the set $K := \{v \in H_0^1(\Omega) : \|v\|_{L^2(\Omega)} = 1\}$ and consider the minimization problem

$$\inf_{v \in K} \left\{ J(v) = \int_{\Omega} |\nabla v|^2 \, dx \right\}.$$

1. Show that J attains a minimum λ on K , achieved by at least one function $u \in K$.
2. Find a functional $\tilde{J} : H_0^1(\Omega) \rightarrow \mathbb{R}$ such that if u solves the minimization problem $\min_{v \in K} J(v)$, then u solves the minimization problem $\min_{v \in H_0^1(\Omega)} \tilde{J}(v)$.
3. Write the equation satisfied by the minimizers of J on K . Deduce that the minimizers of J on K are exactly the eigenfunctions associated with the first eigenvalue of the Dirichlet Laplacian problem.

4. Deduce the following estimate:

$$\|v\|_{L^2(\Omega)} \leq \frac{1}{\sqrt{\lambda}} \|\nabla v\|_{(L^2(\Omega))^d}, \quad \forall v \in H_0^1(\Omega),$$

where λ denotes the first eigenvalue of the Dirichlet Laplacian. Explain why the constant C in (P) cannot be chosen smaller than $1/\sqrt{\lambda}$.

5. Assuming further that Ω is connected, similarly characterize the smallest constant $C_W > 0$ in the Poincaré-Wirtinger inequality

$$\|v\|_{L^2(\Omega)} \leq C_W \|\nabla v\|_{(L^2(\Omega))^d}, \quad \forall v \in H_{\#}^1(\Omega) := \{\varphi \in H^1(\Omega) : \int_{\Omega} \varphi \, dx = 0\}.$$

Solution 20. 1. Let $(v_n)_{n \geq 1}$ be a minimizing sequence of J on K . Then we have

$$\int_{\Omega} |\nabla v_n|^2 \, dx \rightarrow \lambda = \inf_{v \in K} J(v).$$

The sequence $(v_n)_{n \geq 1}$ is bounded in $H_0^1(\Omega)$. So we can extract a subsequence (still denoted $(v_n)_{n \geq 1}$) such that $v_n \rightharpoonup u$ weakly in $H_0^1(\Omega)$. Then by the Rellich-Kondrachov compactness theorem, there exists a subsequence (still denoted $(v_n)_{n \geq 1}$) such that $v_n \rightarrow v$ strongly in $L^2(\Omega)$.

We need to verify that $v = u \in H_0^1(\Omega)$. For all $\phi \in L^2(\Omega)$, we define the linear form on $H_0^1(\Omega)$ by

$$\ell_{\phi}(w) = \int_{\Omega} \phi(x) w(x) \, dx.$$

Since $H_0^1(\Omega)$ is a Hilbert space, by Riesz representation theorem, there exists a unique element $z \in H_0^1(\Omega)$ such that for every $w \in H_0^1(\Omega)$, we have

$$\ell_{\phi}(w) = (z, w)_{H_0^1(\Omega)}.$$

Since $v_n \rightarrow u$ weakly in $H_0^1(\Omega)$, we have

$$\ell_{\phi}(v_n) = (z, v_n)_{H_0^1(\Omega)} \rightarrow (z, u)_{H_0^1(\Omega)} = \ell_{\phi}(u).$$

On the other hand, since $v_n \rightarrow v$ strongly in $L^2(\Omega)$, we have

$$\ell_{\phi}(v_n) = \int_{\Omega} \phi(x) v_n(x) \, dx \rightarrow \int_{\Omega} \phi(x) v(x) \, dx = \ell_{\phi}(v).$$

Therefore, we have $\ell_{\phi}(u) = \ell_{\phi}(v)$ for every $\phi \in L^2(\Omega)$. This implies that $u = v$ in $L^2(\Omega)$.

It is easy to check that $u \in K$ since we have

$$\|u\|_{L^2(\Omega)} = \|v\|_{L^2(\Omega)} = \lim_{n \rightarrow +\infty} \|v_n\|_{L^2(\Omega)} = 1,$$

Moreover, we have

$$J(u) = \int_{\Omega} |\nabla u|^2 \, dx \leq \liminf_{n \rightarrow +\infty} \int_{\Omega} |\nabla v_n|^2 \, dx = \lambda.$$

This concludes the proof.

2. We define the functional $\tilde{J} : H_0^1(\Omega) \rightarrow \mathbb{R}$ by

$$\tilde{J}(v) = \int_{\Omega} |\nabla v|^2 \, dx - \lambda \int_{\Omega} |v(x)|^2 \, dx.$$

If $v \in L^2(\Omega)$ satisfies $\|v\|_{L^2(\Omega)} = a$, then we write $v = a\tilde{v}$ where $\tilde{v} \in K$. We can check that

$$\tilde{J}(v) = a^2 J(\tilde{v}) - \lambda a^2 \geq a^2 \lambda - \lambda a^2 = 0.$$

Moreover, we have $\tilde{J}(u) = 0$ if u is a minimizer of J on K . This implies that u is a minimizer of $\tilde{J}$ on $H_0^1(\Omega)$.

3. Suppose u is a minimizer of J on K . Then by the previous question, for every $v \in H_0^1(\Omega)$, we have

$$\tilde{J}(u) \leq \tilde{J}(v),$$

i.e.,

$$\int_{\Omega} |\nabla u|^2 dx - \lambda \int_{\Omega} |u(x)|^2 dx \leq \int_{\Omega} |\nabla v|^2 dx - \lambda \int_{\Omega} |v(x)|^2 dx.$$

By taking $v = u + t\phi$ where $t \in \mathbb{R}$ and $\phi \in H_0^1(\Omega)$, we have

$$\int_{\Omega} |\nabla u|^2 dx - \lambda \int_{\Omega} |u(x)|^2 dx \leq \int_{\Omega} |\nabla u + t\nabla\phi|^2 dx - \lambda \int_{\Omega} |u(x) + t\phi(x)|^2 dx.$$

By expanding the right-hand side of this inequality, we have

$$0 \leq 2t \left(\int_{\Omega} \nabla u(x) \cdot \nabla \phi(x) dx - \lambda \int_{\Omega} u(x)\phi(x) dx \right) + t^2 \left(\int_{\Omega} |\nabla \phi|^2 dx - \lambda \int_{\Omega} |\phi(x)|^2 dx \right).$$

By dividing by t and taking $t \rightarrow 0^+$ and $t \rightarrow 0^-$, we have

$$\begin{aligned} 0 &= \int_{\Omega} \nabla u(x) \cdot \nabla \phi(x) dx - \lambda \int_{\Omega} u(x)\phi(x) dx \\ &= \int_{\Omega} (-\Delta u(x) - \lambda u(x))\phi(x) dx. \end{aligned}$$

Hence, we have $-\Delta u = \lambda u$ a.e. in Ω . This implies that u is an eigenfunction associated with the eigenvalue λ of the Dirichlet Laplacian. Finally, we show that λ is the first eigenvalue of the Dirichlet Laplacian problem. Suppose $\tilde{u}$ is an eigenfunction in $H_0^1(\Omega)$ associated with $\mu < \lambda$. By replacing $\tilde{u}$ by $\frac{\tilde{u}}{\|\tilde{u}\|_{L^2(\Omega)}}$, we have $\tilde{u} \in K$ and

$$J(\tilde{u}) = \int_{\Omega} |\nabla \tilde{u}|^2 dx = \mu < \lambda,$$

which contradicts the definition of λ . Hence, λ is the first eigenvalue of the Dirichlet Laplacian problem.

4. Since $\tilde{J}(v) \geq 0$, we have

$$\int_{\Omega} |\nabla v|^2 dx \geq \lambda \int_{\Omega} |v(x)|^2 dx.$$

This implies that

$$\|v\|_{L^2(\Omega)} \leq \frac{1}{\sqrt{\lambda}} \|\nabla v\|_{(L^2(\Omega))^d}, \quad \forall v \in H_0^1(\Omega).$$

Suppose u is the eigenfunction associated with the first eigenvalue of the Dirichlet Laplacian such that $\|u\|_{L^2(\Omega)} = 1$. Then we have

$$J(u) = \int_{\Omega} |\nabla u|^2 dx = \lambda.$$

Hence, we have

$$\|u\|_{L^2(\Omega)} = 1 = \frac{1}{\sqrt{\lambda}} \|\nabla u\|_{(L^2(\Omega))^d}.$$

5. Suppose J has a minimum λ_W on the set $K_W = \{v \in H_{\#}^1(\Omega) : \|v\|_{L^2(\Omega)} = 1\}$. Then we consider the corresponding variational problem: find $u \in H_{\#}^1(\Omega)$ such that

$$\int_{\Omega} \nabla u(x) \cdot \nabla v(x) dx = \lambda_W \int_{\Omega} u(x)v(x) dx, \quad \forall v \in H_{\#}^1(\Omega).$$

And we can check that u is an eigenfunction associated with the eigenvalue λ_W of the Neumann Laplacian problem, i.e., u satisfies

$$\begin{cases} -\Delta u = \lambda u & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial\Omega. \end{cases}$$

Problem 21. (Dirichlet Laplacian Eigenvalues) Let $\Omega \subset \mathbb{R}^2$ be a bounded open set with Lipschitz boundary. In this exercise, we consider the eigenvalue problem: find $(\lambda, u) \in \mathbb{R} \times H^1(\Omega) \setminus \{0\}$ such that

$$\begin{cases} -\Delta u = \lambda u & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The equivalent variational formulation of this problem is: find $(\lambda, u) \in \mathbb{R} \times H_0^1(\Omega) \setminus \{0\}$ such that

$$a(u, v) = \lambda b(u, v), \quad \forall v \in H_0^1(\Omega), \quad (\text{DL})$$

with

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, d\Omega \quad \text{and} \quad b(u, v) = \int_{\Omega} uv \, d\Omega.$$

Let $(\mathcal{T}_h)$ be a family of regular triangulations of Ω such that if $h' < h$, $\mathcal{T}_{h'}$ can be obtained by refining $\mathcal{T}_h$. Denote by V_h the subspace of $H_0^1(\Omega)$ constructed from the $\mathbb{P}^1$ finite element on $\mathcal{T}_h$ and consider the discretized problem: find $(\lambda_h, u_h) \in \mathbb{R} \times V_h \setminus \{0\}$ such that

$$a(u_h, v_h) = \lambda_h b(u_h, v_h), \quad \forall v_h \in V_h. \quad (\text{DL h})$$

1. Numerical approximation of eigenvalues

Denote by λ_k (resp. $\lambda_{k,h}$) the k -th smallest eigenvalue of (DL) (resp. (DL h)). Note that λ_k^h is well-defined for h sufficiently small. Using the min-max principle, show that

$$\lambda_k \leq \lambda_{k,h},$$

and that $\lambda_{k,h'} \leq \lambda_{k,h}$ for $h' \leq h$. Verify these properties numerically. For this, work in $\Omega := (0, 1) \times (0, 1)$. For this particular geometry, give the exact expression of the eigenvalues and eigenfunctions of problem (DL).

2. Problem with a source term

For given $\omega \in \mathbb{R}$ and $f \in L^2(\Omega)$, consider now the problem with a source term: find $v \in H^1(\Omega)$ such that

$$\begin{cases} -\Delta v - \omega v = f & \text{in } \Omega \\ v = 0 & \text{on } \partial\Omega \end{cases}$$

whose variational formulation is: find $v \in H_0^1(\Omega)$ such that

$$a(v, \tilde{v}) - \omega b(v, \tilde{v}) = \ell(\tilde{v}), \quad \forall \tilde{v} \in H_0^1(\Omega), \quad (\text{DLS})$$

with $\ell(\tilde{v}) = \int_{\Omega} f \tilde{v} \, d\Omega$.

- (i) Show that (DLS) admits a unique solution if and only if ω is not an eigenvalue of problem (DL). To establish this result, use the fact that $L^2(\Omega)$ has a Hilbert basis consisting of eigenfunctions associated with the eigenvalues of problem (DL). When ω is an eigenvalue of (DL), discuss the characteristics of problem (DLS).
- (ii) Let λ be an eigenvalue of (DL) and u an associated eigenfunction. Consider (ω_n) a sequence of real numbers converging to λ , with $\omega_n \neq \lambda$ for all $n \in \mathbb{N}$. Show that

$$\lim_{n \rightarrow +\infty} \|v(\omega_n)\|_{H^1(\Omega)} = +\infty \quad \text{when } f \text{ is such that } \int_{\Omega} f u \, d\Omega \neq 0.$$

Solution 21. 1. By the min-max principle, we have

$$\lambda_k = \min_{\substack{S \subset H_0^1(\Omega) \\ \dim(S)=k}} \max_{v \in S \setminus \{0\}} \frac{a(v, v)}{\|v\|_{L^2(\Omega)}^2} \leq \min_{\substack{S \subset V_h \\ \dim(S)=k}} \max_{v \in S \setminus \{0\}} \frac{a(v, v)}{\|v\|_{L^2(\Omega)}^2} = \lambda_{k,h}.$$

Since $V_h \subset V_{h'}$ for $h' < h$, we have also $\lambda_{k,h'} \leq \lambda_{k,h}$. Now we consider the case where $\Omega = (0, 1) \times (0, 1)$. It is easy to check that the eigenvalues of problem (DL) are given by

$$\lambda_{m,n} = \pi^2(m^2 + n^2), \quad m, n \in \mathbb{N}^*,$$

and the corresponding eigenfunctions are given by

$$u_{m,n}(x, y) = \sin(m\pi x) \sin(n\pi y), \quad m, n \in \mathbb{N}^*.$$

since $(u_{m,n})$ forms an orthogonal basis of $L^2(\Omega)$ (with $\|u_{m,n}\|_{L^2(\Omega)} = 1/2$).

2. (i) Since $L^2(\Omega)$ has a Hilbert basis $(e_k)_{k \in \mathbb{N}^*}$ consisting of eigenfunctions associated with the eigenvalues of problem (DL), we can write

$$v = \sum_{k=1}^{+\infty} (v, e_k)_{L^2(\Omega)} e_k, \quad \forall v \in H_0^1(\Omega).$$

If (DLS) has a solution v , then we have

$$\begin{aligned}
a(v, e_{k'}) - \omega b(v, e_{k'}) &= \sum_{k=1}^{+\infty} (v, e_k)_{L^2(\Omega)} (a(e_k, e_{k'}) - \omega b(e_k, e_{k'})) \\
&= \sum_{k=1}^{+\infty} (\lambda_k - \omega) (v, e_k)_{L^2(\Omega)} \delta_{k,k'} \\
&= (\lambda_{k'} - \omega) (v, e_{k'})_{L^2(\Omega)} \\
&= \ell(e_{k'}) = (f, e_{k'})_{L^2(\Omega)}, \quad \forall k \in \mathbb{N}^*,
\end{aligned}$$

which implies that

$$(\lambda_k - \omega) (v, e_k)_{L^2(\Omega)} = (f, e_k)_{L^2(\Omega)}, \quad \forall k \in \mathbb{N}^*.$$

Therefore, there are two cases:

- If ω is not an eigenvalue of problem (DL), then we have

$$(v, e_k)_{L^2(\Omega)} = \frac{(f, e_k)_{L^2(\Omega)}}{\lambda_k - \omega}, \quad \forall k \in \mathbb{N}^*.$$

Since $f \in L^2(\Omega)$, the series

$$\sum_{k=1}^{+\infty} \frac{(f, e_k)_{L^2(\Omega)}}{\lambda_k - \omega} e_k.$$

converges in $H_0^1(\Omega)$. And it is easy to check that it is a solution of problem (DLS). Moreover, the solution is unique since we can consider the projection on every e_k .

- If ω is an eigenvalue of problem (DL), let $I_0 = \{i \in \mathbb{N}^* : \lambda_i = \omega\}$ be the (finite) set of indices of the corresponding eigenspace. Taking $k \in I_0$ in the identity above forces $(f, e_k)_{L^2(\Omega)} = 0$. Hence there are two cases:
 - If $(f, e_k)_{L^2(\Omega)} \neq 0$ for some $k \in I_0$, i.e., $f \notin \text{Span}\{e_k : k \in I_0\}^\perp$, then problem (DLS) has no solution.
 - If $(f, e_k)_{L^2(\Omega)} = 0$ for all $k \in I_0$, i.e., $f \in \text{Span}\{e_k : k \in I_0\}^\perp$, then problem (DLS) has infinitely many solutions: for any family $(\alpha_k)_{k \in I_0} \subset \mathbb{R}$,

$$\sum_{i \notin I_0} \frac{(f, e_i)_{L^2(\Omega)}}{\lambda_i - \omega} e_i + \sum_{k \in I_0} \alpha_k e_k$$

is a solution of problem (DLS).

(ii) Since $\omega_n \rightarrow \lambda$ and $\int_\Omega f u \, d\Omega \neq 0$, we have

$$(v(\omega_n), u)_{L^2(\Omega)} = \frac{(f, u)_{L^2(\Omega)}}{\lambda - \omega_n} \rightarrow +\infty.$$

WLOG, we can suppose $\|u\|_{L^2(\Omega)} = 1$. Hence we have

$$\|v(\omega_n)\|_{H^1(\Omega)} \geq \|v(\omega_n)\|_{L^2(\Omega)} \geq \frac{|(v(\omega_n), u)_{L^2(\Omega)}|}{\|u\|_{L^2(\Omega)}} \rightarrow +\infty.$$

References

- [1] C. Amrouche, P.G. Ciarlet, and C. Mardare, “On a lemma of Jacques-Louis Lions and its relation to other fundamental results,” *Journal de Mathématiques Pures et Appliquées*, vol. 104, no. 2, pp. 207–226, 2015.
- [2] R.G. Durán, “An elementary proof of the continuity of the Bogovskii operator,” arXiv:1103.3718v1, 2011.
- [3] L.C. Evans, *Partial Differential Equations*, 2nd ed., American Mathematical Society, 2010.